MULTI-TEMPORAL LAND USE AND LAND COVER CLASSIFICATION AND INTERACTIVE CHANGE VISUALIZATION OF THE CHENNAI URBAN STUDY AREA (2010–2025)

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Name with Affiliation :

Date :

Project *Viva voce* held on _____

Examiner 1 **Examiner 2**

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Harish Ravichandran

Abstract

This project set out to map land use and land cover (LULC) in Chennai at five-year intervals from 2005 to 2025 with one supervised classifier, Gaussian maximum likelihood classification (MLC), and to present the results in an interactive GIS application. A fixed 529.45 km² Chennai Urban Study Area was classified from full-year Landsat Collection 2 median composites in Google Earth Engine, using nine predictors, 600 balanced training pixels and 300 independent validation points per year. The 2025, 2020 and 2015 maps reached five-class overall accuracies of 83.67%, 86.33% and 85.67%, but required extensive manual interpretation of training data. Automated transfer of 2015 training data to 2010 yielded only 527 of 600 required pixels and was rejected; manually interpreted sites and a Landsat 5 + 7 stack gave a provisional 2010 map (69.33%).

The provisional 2010 map implied a 78.94 km² decline in Built-up area by 2015. GAIA and GHSL external products, dated historical imagery, a reconstruction of both classifiers and the validation errors indicated classification disagreement rather than urban loss, and side-by-side reading of all four maps then revealed systematic Built-up omission in 2015, 2020 and 2025. A revised 2015 training set, neighbourhood and SNIC segment context, and a Random Forest trained with GAIA-derived labels each failed to recover the omission reliably; the last raised 2015 Built-up recall but added commission errors and fell to 79.00% accuracy in 2020. The locked maps were therefore corrected by rule-based post-classification fusion with year-specific evidence (GAIA with WSF 2015 and the corrected 2020 map, ESA WorldCover 2020, and Dynamic World 2025). On reused validation points this recovered 4–9 Built-up points per year without new false positives (2015 and 2020: 88.67%), giving mapped Built-up of 279.82, 292.24, 296.74 and 342.60 km² for 2010–2025. A 2010 two-way fusion (304.72 km²; 71.33%) is retained alongside the original 2010 map and reverses the 2010–2015 interval, so only the overall increase is robust.

Persistent confusion between Agricultural/Open and Barren/Fallow land led to four-class reporting by post-classification aggregation (81.00–96.67% overall accuracy, an increase by construction), and 2005 was removed from scope. All rasters were verified after export, and a local FastAPI and React/Leaflet application presents the original series with official statistics and caveats. The project's contribution is a documented, critically evaluated multi-temporal workflow whose limitations were investigated, and partly corrected, rather than concealed. A separate 2020 geographical benchmark then applied the frozen five-class Chennai model to a manually interpreted Thanjavur pilot. Its area-adjusted accuracy was 23.19%; source-month restriction gave 20.66% and bounded unlabeled quantile alignment 35.94%. Additional coastal-city screening measured automated proxy agreement only, qualifying geographical transfer claims.

Keywords: land use / land cover; Landsat; Google Earth Engine; Gaussian maximum likelihood classification; random forest; accuracy assessment; GAIA; GHSL; post-classification fusion; built-up omission; post-classification change; Chennai; web GIS; Thanjavur; geographical generalisability; frozen-model transfer

Table of Contents

Title	Page No.
Bonafide Certificate	ii
Acknowledgements	iii
Abstract	v
List of Figures and Tables	vii
Abbreviations	ix
1. Introduction	1
1.1 The Original Project Idea	1
1.2 How the Project Evolved	1
1.3 Objectives, Scope and Conventions	2
2. Related Work	4
2.1 Landsat Time Series and Cloud Processing	4
2.2 Indices and Parametric Classification	4
2.3 Accuracy, Area and Change	4
2.4 Related Regional Study	5
2.5 Position of This Project	5
3. Study Area, Data and Methodology	6
3.1 Study Area	6
3.2 Landsat Data and Annual Composites	6
3.3 Pre-processing and Predictor Stack	7
3.4 Processing Workflow	9
3.5 Land-cover Classes	10
3.6 Training and Validation Data	10
3.7 Gaussian Maximum Likelihood Classifier	11
3.8 Accuracy Assessment	11
3.9 Areas, Aggregation and Transitions	11
3.10 Post-classification Built-up Fusion	13
4. Implementation Journey and Experimental Development	16
4.1 Developing the Initial Classification Workflow	16
4.2 Establishing the 2025 Baseline	17
4.3 Extending Classification to 2020 and 2015	17
4.4 The Practical Limitations of Manual Training	17
4.5 Attempting Automated Historical Training	17
4.6 Rebuilding the 2010 Reference Dataset and Predictor Stack	19
4.7 Evaluating Random Forest As an Alternative	20
4.8 Refining the Reporting Classification and Scope	24
4.9 Completing and Verifying the Final Maps	24
5. Classification and Validation Results	25
5.1 Final Land-cover Maps	25
5.2 Overall Accuracy	26
5.3 Class-level Errors and the Merge	26
5.4 What the Four-class Accuracy Means	27
5.5 Reliability of the 2010 Map	27
5.6 What Overall Accuracy Does Not Show	28
6. Multi-temporal Land-cover Change	29
6.1 Mapped Class Areas	29
6.2 Transitions Between Years	29
6.3 Where the Change Occurs	30
6.4 Net Versus Gross Change	31

7. Investigating the 2010–2015 Built-up Anomaly 32
7.1 The Unexpected Built-up Decline 32
7.2 A First Hypothesis: Too Few Urban Examples in 2015 32
7.3 External Cross-checks with GAIA and GHSL 33
7.4 Spatial Disagreement Inside the Disputed Pixels 34
7.5 Historical Imagery At Audited Locations 35
7.6 Reconstructing the Classifiers 36
7.7 Why the 2015 Classifier Preferred Open Land 36
7.8 Training Representativeness and Validation Errors 37
7.9 What Could and Could Not Be Concluded 38
8. Correcting Built-up Omission Across the Series 39
8.1 From a 2010 Anomaly to a Series-wide Omission Problem 39
8.2 Adding Spatial Context to the Predictors 40
8.3 Segment-based Context and Its Reproducibility 42
8.4 Random Forest with GAIA-assisted Training 43
8.5 From Retraining to Evidence-based Fusion 46
8.6 Validation of the Retained Candidates 49
8.7 The 2010 Sensitivity Test 50
8.8 Fusion-adjusted Built-up Areas and Transitions 52
8.9 What the Correction Does and Does Not Show 58
9. System Architecture and Interactive GIS Application 60
9.1 Purpose and Architecture 60
9.2 Interface and Timelapse Modes 61
9.3 Completed and Planned Features 62
10. Geographical Generalisability of the Chennai Gaussian MLC 63
10.1 Why an External Benchmark Was Needed 63
10.2 What Was Held Fixed 63
10.3 The Thanjavur Pilot and the First Transfer Result 64
10.4 Manual References and the Area Adjusted Estimate 65
10.5 Diagnosing the Failed Transfer Before Changing the Inputs 66
10.6 Experiment B Source Month Restriction 68
10.7 Experiment C Bounded Unlabeled Quantile Alignment 68
10.8 What the A B C Comparison Showed 69
10.9 Extending Screening to Three Coastal Cities 72
10.10 Three City Results and the Water Effect 74
10.11 What Can Be Concluded and What Remains Uncertain 76
10.12 Final Decision and Reproducibility 76
11. Discussion, Limitations and Conclusion 78
11.1 From the Original Plan to the Final Outcome 78
11.2 Limitations 79
11.3 Conclusion 80
11.4 Future Work 81
References 82
Appendix A — Reproducibility and Evidence 85
Appendix B — Confusion Matrices, Areas and Transitions 87
Appendix C — Geographical Transfer Evidence 93

List of Figures and Tables

Figures

Figure	Title	Page
3.1	Chennai Urban Study Area on the 2025 annual Landsat composite	6
3.2	Annual median true-colour composites for the four study years	7
3.3	NDVI, NDWI and NDBI for 2010–2025, each on one fixed scale	9
3.4	Implemented Earth Engine processing chain	10
3.5	Post-classification comparison of two maps on the shared 30 m grid: transition coding, matrix accumulation and class gains and losses	13
4.1	Development path and key decisions	16
4.2	Overall and class-level results of the Random Forest comparison	22
4.3	Paired five-class confusion matrices, Gaussian MLC and Random Forest, 2010 and 2020	23
5.1	Four-class LULC maps of the Chennai Urban Study Area, 2010–2025	25
6.1	Mapped class areas by year	29
6.2	Four-class transition matrices between adjacent years (km²)	30
6.3	Mapped class transitions in each period	30
7.1	External cross-checks of the 2010–2015 mapped Built-up decline	34
7.2	Audit evidence at disputed locations and controls	35
8.1	How the context predictors of experiments X1–X3 were computed: (a) 3 × 3 window, (b) 5 × 5 window and (c) SNIC segment	41
8.2	Automatic generation of the GAIA-assisted training set for the 2015 Random Forest (stage 08G-3)	44
8.3	How post-classification Built-up fusion works	46
8.4	Two-stage dependency of the retained 2020 and 2015 fusion candidates	48
8.5	The sequence of 2015 experiments on the same validation points	49
8.6	Overall accuracy and Built-up producer's accuracy, original and fusion-adjusted maps	50
8.7	Mapped Built-up area: original MLC maps and the fusion-adjusted reported series	53
8.8	Fusion-adjusted LULC composition and area trends across Chennai, 2010–2025	54
8.9	Reported-series LULC maps: original provisional 2010 MLC and retained 2015–2025 fusion classifications.	57
8.10	Original 2010 provisional MLC and reconstructed two-way fusion sensitivity alternative.	58
9.1	Implemented and planned system architecture	60
9.2	The local GIS application	61
10.1	Frozen-model transfer and the two distinct reference pathways.	64

Figure	Title	Page
10.2	Thanjavur 2020 classification maps under direct transfer (A), seasonal matching (B) and spectral harmonisation (C).	65
10.3	Monthly availability of clear observations at the sampled locations.	67
10.4	Area-adjusted Thanjavur accuracy under three frozen-model input protocols.	70
10.5	Manual-reference confusion matrices for A, B and C.	71
10.6	Mapped class areas for all three Thanjavur experiments on the common valid footprint.	71
10.7	Direct frozen-model classifications for the three coastal pilots, 2020.	73
10.8	Three-city cross-model agreement with and without Water.	74
10.9	Classwise matches against the automated proxy, 40 points per class and city.	75

Tables

Table	Title	Page
3.1	Sensors and full-year scene inventories	7
3.2	External evidence used in the fusion experiments	14
3.3	Three kinds of classified output reported in this study	15
4.1	Outcome of the automated 2010 training selection	18
4.2	Jeffries–Matusita distances between the fitted 2010 class distributions (Landsat 5 stack)	19
4.3	Site-held-out cross-validation of the 2010 training data (training data only)	20
4.4	Gaussian MLC and Random Forest on identical validation points	21
5.1	Overall accuracy and kappa of the five-class and four-class maps	26
5.2	Producer's (PA) and user's (UA) accuracy of the five-class maps (%)	26
6.1	Mapped class areas, km² (% of mapped area)	29
6.2	Gross and net Built-up flows between adjacent years (km²)	30
7.1	Five-class transitions of 2010 Built-up pixels, 2010 → 2015	32
7.2	Revised 2015 training experiment V2A against the locked 2015 model (same nine predictors, same 300 validation points)	33
7.3	Archived external cross-check areas within the Chennai Urban Study Area (km²)	34
7.4	GAIA impervious coverage inside the disputed MLC transitions (archived stage 07W)	35
7.5	Historical audit locations	36
7.6	Likelihood and index diagnostics for 340 sampled pixels by original transition	37
8.1	GAIA disagreement with the original MLC open-land classes (stage 08G-1, read-only)	40
8.2	Context and segmentation experiments on the 2015 validation set (11-band Gaussian MLC unless stated)	42
8.3	The four groups of GAIA-assisted pseudo-labelled pixels (stage 08G-3)	44
8.4	Composition of the combined 2015 Random Forest training set	45

Table	Title	Page
8.5	Fusion variants tested, with paired results on each year's validation points	48
8.6	Paired validation of the retained fusion candidates against the original MLC maps	50
8.7	Same-grid 2010 rasters scored on the 300 original 2010 validation points (stage 08G-12)	51
8.8	The 2010 sensitivity test on the 300 original 2010 validation points (stage 08G-15)	51
8.9	Five-class mapped areas of the reported series (km²; stage 08G-13)	52
8.10	Built-up change in the reported series (km²)	53
8.11	Gross gain, gross loss and net change of all five classes in the fusion-adjusted series (km²; stage 08G-13)	54
9.1	Application features by status	62
10.1	Original map strata and the correct predictions retained for every comparison.	66
10.2	Mean predictor differences at Built-up reference locations.	67
10.3	Applied band corrections from the original exported parameter CSV.	69
10.4	Thanjavur results on the same 112 points and original A area weights.	69
10.5	Correct predictions by actual manually interpreted reference class.	71
10.6	Automated consensus definitions and minimum Dynamic World mean probabilities.	73
10.7	Agreement on the automated consensus subset and its spatial coverage.	74
10.8	Exported mapped class areas (km²) across all six experiments.	76
A.1	Final Earth Engine assets and exported rasters	85
A.2	Experimental assets of the Built-up correction (stages 07U2, 08F, 08G)	85
A.3	Principal results and their evidence	85
B.1	Gaussian MLC five-class confusion matrices of the official maps (rows = reference, columns = map; bold = correct; *2010 provisional; four-class matrices follow by summing rows and columns 4 and 5)	87
B.2	Five-class mapped areas before aggregation (km²)	88
B.3	Random Forest five-class confusion matrices from the exploratory comparison (same layout; †reconstructed paired 2025 samples)	88
B.4	Context, segmentation and GAIA-assisted RF matrices on the 2015 validation set (rows = reference, columns = map; bold = correct)	89
B.5	Fusion matrices (rows = reference, columns = map; bold = correct)	90
B.6	Five-class transition matrices of the reported series (km²; rows = earlier year, columns = later year; stage 08G-13)	91

Abbreviations

BSI	Bare soil index
BU	Built-up
CV	Cross-validation
DW	Dynamic World
EE / GEE	Google Earth Engine
ETM+	Enhanced Thematic Mapper Plus (Landsat 7)
FP	False positive (commission)

GAIA	Global Artificial Impervious Area
GHSL	Global Human Settlement Layer
JM	Jeffries–Matusita distance
L5, L7, L8, L9	Landsat 5, 7, 8, 9
LULC	Land use / land cover
MLC	Maximum likelihood classification
NDBI	Normalized Difference Built-up Index
NDVI	Normalized Difference Vegetation Index
NDWI	Normalized Difference Water Index (McFeeters)
NIR, SWIR	Near-infrared; shortwave infrared
OA, PA, UA	Overall, producer's and user's accuracy
OLI	Operational Land Imager (Landsat 8/9)
PCC	Post-classification comparison
QA	Quality assessment
RF	Random Forest
ROI	Region of interest (the study area)
RGB	Red–green–blue true-colour composite
SLC	Scan line corrector (Landsat 7)
SNIC	Simple non-iterative clustering (segmentation)
SR	Surface reflectance
TM	Thematic Mapper (Landsat 5)
UTM	Universal Transverse Mercator
WSF	World Settlement Footprint
κ	Cohen's kappa
pp	Percentage points

CHAPTER 1
INTRODUCTION

1.1 THE ORIGINAL PROJECT IDEA

Chennai is one of India's largest coastal metropolitan regions. Over the past two decades its built-up fabric has spread into agricultural land, lakes, wetlands and vacant plots, changes that matter for urban planning, drainage and flood exposure. Land use / land cover (LULC) maps derived from satellite imagery are the usual way to describe such change consistently across a whole city, and the Landsat archive, processed in Google Earth Engine (GEE), makes it possible to map the same area repeatedly at 30 m resolution [2], [3].

The project began with a practical and deliberately simple plan: classify Chennai's land use and land cover at five-year intervals — **2005, 2010, 2015, 2020 and 2025** — with one supervised classifier, Gaussian maximum likelihood classification (MLC), measure the historical change recorded by the maps, and present the results in an interactive GIS application. The aim was historical mapping, not forecasting. The working assumption was that once a reliable workflow had been established for one year, it could be repeated for the others.

Comparing maps across years requires that they differ only because the land changed, not because the method did. The proposal therefore fixed a single study area, a single 30 m grid, one compositing method, one set of predictors and one classifier. The first boundary considered was the Greater Chennai Corporation area (about 426 km²); it was replaced by the 529.45 km² **Chennai Urban Study Area**, which also covers the urbanised south-west, and that boundary was then used unchanged for every year.

1.2 HOW THE PROJECT EVOLVED

The work did not proceed as planned. The three recent years were classified successfully, but only through extensive manual interpretation of training data. An attempt to automate training-data preparation for 2010 did not produce a usable training set, and manual historical interpretation followed. The 2010 map that resulted, although provisionally acceptable, implied a large and implausible loss of built-up land between 2010 and 2015, which had to be investigated before any change could be reported. Persistent confusion between two open-land classes led to a revised reporting scheme, a Random Forest classifier was tested as an alternative, and the time all of this required led to 2005 being removed from the completed scope. Four classified maps (2010–2025), their accuracy

assessments and change statistics, a documented investigation of the 2010–2015 anomaly and a working local GIS application were then completed.

A final phase followed. Reading the four maps side by side against their own Landsat composites showed that the anomaly was not confined to 2010: dense built-up neighbourhoods were often mapped as Agricultural/Open or Barren/Fallow in 2015, 2020 and 2025. Revised training data, spatial-context predictors, image segmentation and a Random Forest trained with automatically generated labels were tested in turn, and each either failed to recover the missing Built-up land or recovered it at the cost of new errors. The project therefore moved to **post-classification fusion**: correcting the locked MLC maps only where year-specific external evidence agreed. This produced fusion-adjusted Built-up candidates for 2015, 2020 and 2025, and two retained outcomes for 2010.

Several questions therefore became central only as the work progressed: whether historical training data could be transferred automatically; why the 2010 map generalised poorly; whether the mapped 2010–2015 decline was real; whether a different classifier would help; how confused classes should be reported; and, finally, whether systematic Built-up omission could be corrected without forcing the series towards an expected answer. These questions are answered in the order in which they arose, and the report is organised around that chronology rather than around a set of questions fixed in advance.

1.3 OBJECTIVES, SCOPE AND CONVENTIONS

The objectives, as finally achieved, were to (i) fix one study boundary and one 30 m analysis grid; (ii) build cloud-masked annual Landsat composites and a nine-band predictor stack for each year; (iii) collect balanced training data and independent validation data; (iv) classify each year with an explicitly implemented Gaussian MLC; (v) assess accuracy from confusion matrices; (vi) compute mapped class areas and transitions from saved rasters; (vii) investigate, and where external evidence allowed correct, systematic Built-up omission without retraining the locked maps; and (viii) present the results in a local interactive viewer. 2005 was prepared but not classified, and no 2005 result is reported.

Four conventions are used throughout. Mapped change means a difference between two classified maps, not change verified on the ground. The 2010 map is provisional (Section 5.5). Five-class results are the classifier's own output; four-class results come from merging two classes after classification and are always reported alongside the five-class results. Fusion-adjusted maps are experimental post-classification corrections of the locked MLC maps (Section 3.10), reported separately from them. Chapters 2–3 cover background and

methods, Chapters 4–8 the development, results and corrections, Chapter 9 the application, Chapter 10 the external benchmark and Chapter 11 the conclusions.

CHAPTER 2
RELATED WORK

2.1 LANDSAT TIME SERIES AND CLOUD PROCESSING

Landsat provides the longest continuous record of 30 m multispectral observations of the land surface. Collection 2 reprocessed the archive for geometric and radiometric consistency and introduced Level-2 surface reflectance for all sensors [3]; Level-2 values are converted to reflectance with a fixed scale and offset [4], and the QA_PIXEL cloud and shadow flags derive from the CFMask family of algorithms [5]. Two sensor issues matter for a 2010–2025 series. The scan line corrector of Landsat 7 failed in 2003, so later ETM+ scenes miss about 22% of each scene in wedge-shaped gaps [7]. Reflectance and vegetation indices also differ between TM, ETM+ and OLI because their spectral response functions differ [6], [8]. Google Earth Engine makes the whole archive available for server-side compositing and classification [2].

2.2 INDICES AND PARAMETRIC CLASSIFICATION

Normalised-difference indices are routinely added to reflectance bands: NDVI for vegetation [9], McFeeters' NDWI for open water [10] and NDBI for built-up land [11]. They are deterministic transformations of the bands, so they add contrast but not independent information. The Gaussian MLC models each class as a multivariate normal distribution and assigns each pixel to the class with the highest likelihood [12]. It is transparent and its likelihood terms can be inspected, but it assumes unimodal class distributions and depends heavily on how well the training pixels represent each class. Class separability can be summarised by the Jeffries–Matusita (JM) distance, from 0 (complete overlap) to 2 (complete separation) [12]. The most widely used non-parametric alternative is Random Forest, an ensemble of decision trees grown on bootstrap samples with randomised splits [22], which has become a standard land-cover classifier because it needs no distributional assumptions and handles non-linear class boundaries [23].

2.3 ACCURACY, AREA AND CHANGE

The confusion matrix, with overall, producer's and user's accuracy, is the standard summary of thematic accuracy [13], [14]. Kappa is widely reported but is a weak basis for comparing maps [16], and differences between two classifiers evaluated on the same points are properly tested with paired methods such as McNemar's test [24]. Olofsson et al. showed that pixel counts are biased area estimates and that unbiased areas with confidence intervals require a probability-based reference sample [15]. Post-classification comparison, the most

common LULC change method, gives a full from–to matrix without radiometric normalisation, but errors in either map pass directly into the change map [17].

2.4 RELATED REGIONAL STUDY

Nandi et al. [1] mapped LULC in Berhampore, West Bengal, from 2001 to 2023 and simulated land use for 2033 with an artificial neural network–cellular automata model. Like this project, they analysed a multi-temporal Landsat series for an Indian city. Unlike that study, the present work has no forecasting component, uses an explicitly implemented parametric classifier, and concentrates on whether each map and each mapped change can be trusted.

2.5 POSITION OF THIS PROJECT

City-scale LULC studies typically report a sequence of maps, an accuracy figure per map and a table of change. The literature above also warns that post-classification change inherits the errors of both maps [17] and that mapped areas are not unbiased estimates [15]. Those warnings turned out to be directly relevant here: the present report documents not only the final maps but also the training difficulties, the classifier comparison and the investigation of an implausible change that shaped how the results can be used.

CHAPTER 3
STUDY AREA, DATA AND METHODOLOGY

3.1 STUDY AREA

The study area covers Greater Chennai and adjoining urbanised areas to the south-west, including Chrompet, West Tambaram and Gerugambakkam (Fig. 3.1). Its vector area is **529.45 km²**. A municipal boundary of about 426 km² was considered first, and several intermediate boundaries were tested, but all were replaced by this cleaned boundary, which is used unchanged for every year. All processing uses WGS 84 / UTM zone 44N (EPSG:32644) on one locked 30 m grid of 951 × 1,412 cells, of which 592,039 fall inside the study area.

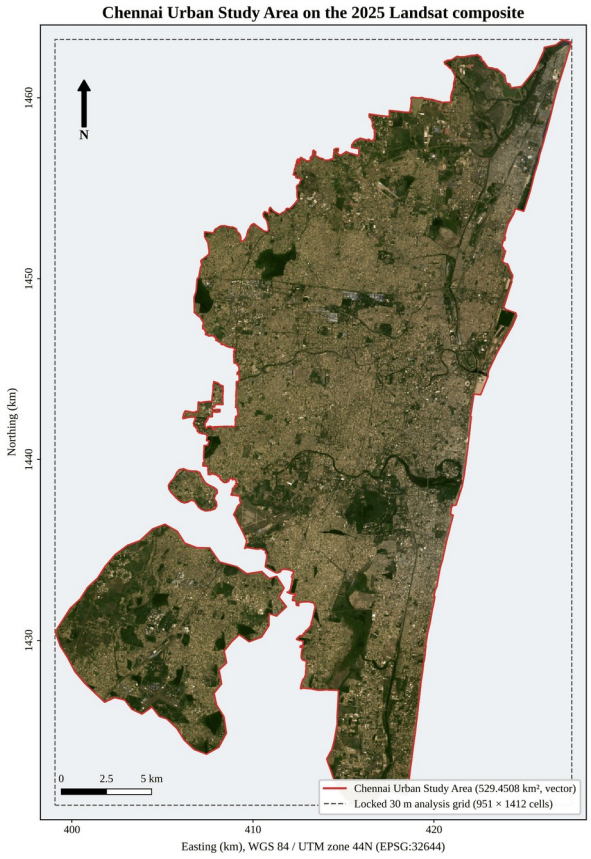


Fig. 3.1 Chennai Urban Study Area on the 2025 annual Landsat composite.

Boundary from the project's study-area asset; background is the project's 2025 Landsat 8+9 annual median true-colour composite. Dashed rectangle: locked raster grid. Coordinates in km, EPSG:32644.

3.2 LANDSAT DATA AND ANNUAL COMPOSITES

All imagery is Landsat Collection 2 Tier 1 Level-2 surface reflectance (Table 3.1). The original design used a common dry-season window (January–April) to limit seasonal differences. The scene inventory showed that this window contained only three usable Landsat 5 scenes for 2010, and an extended window was still insufficient. The design was

therefore changed to a **full-calendar-year per-pixel median composite** for every year. This keeps the method identical across years at the cost of mixing seasons within each composite, which particularly affects water and vegetation (Section 10.2). For 2010, Landsat 7 ETM+ observations were later added to Landsat 5 TM (Section 4.6); the per-pixel median of the combined set fills the SLC-off gaps from other dates.

Table 3.1 Sensors and full-year scene inventories.

Year	Sensor(s)	Scenes	Clear observations per pixel (min / median / max)	Status
2010	Landsat 5 TM + Landsat 7 ETM+	7 (L5)†	0 / 2 / 7 (L5)	Provisional
2015	Landsat 8 OLI	19	2 / 7 / 13	Accepted
2020	Landsat 8 OLI	21	3 / 8–9 / 14	Accepted
2025	Landsat 8 OLI + Landsat 9 OLI-2	40	5 / 20 / 27	Accepted

†Landsat 5 inventory. The number of Landsat 7 scenes in the final 2010 stack is not recorded in the available evidence (Appendix A).

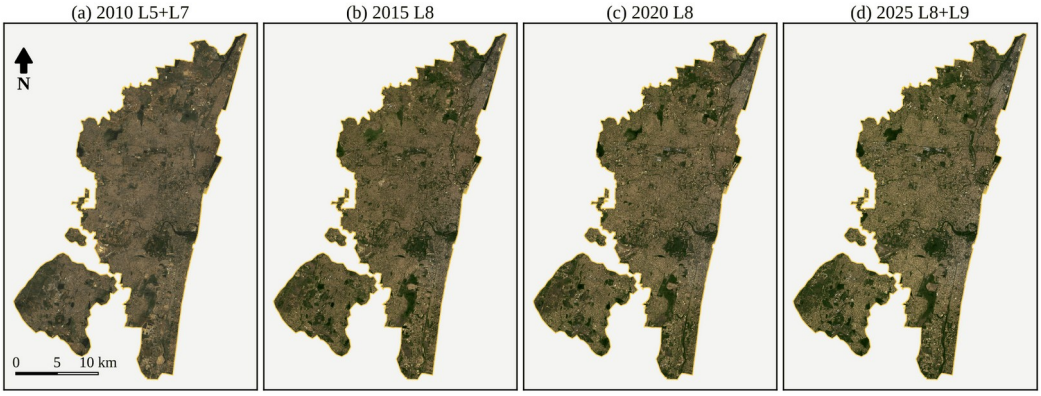


Fig. 3.2 Annual median true-colour composites for the four study years.

Project RGB GeoTIFFs exported from Earth Engine on the locked grid (QA-masked, scaled, annual median). Shown for visual context; they are not the classifier input.

3.3 PRE-PROCESSING AND PREDICTOR STACK

In each scene, pixels flagged in QA_PIXEL as fill, dilated cloud, cirrus, cloud, cloud shadow or snow, and radiometrically saturated pixels (QA_RADSAT ≠ 0), were masked. Digital numbers were converted to surface reflectance ρ with the Collection 2 scaling [4], and sensor-specific bands were renamed to a common set (blue, green, red, nir, swir1, swir2):

$$\rho = DN \times 0.0000275 - 0.2 \tag{3.1}$$

Three indices were computed from the annual composite by explicit arithmetic:

$$NDVI = (NIR - Red) / (NIR + Red) \tag{3.2}$$

$$\mathrm{NDWI} = (\mathrm{Green} - \mathrm{NIR}) / (\mathrm{Green} + \mathrm{NIR}) \quad (3.3)$$

$$\mathrm{NDBI} = (\mathrm{SWIR1} - \mathrm{NIR}) / (\mathrm{SWIR1} + \mathrm{NIR}) \quad (3.4)$$

The classifier's predictor vector has nine elements: the six reflectance bands, NDVI, NDWI and NDBI. Fig. 3.3 shows the three indices for all four years on fixed colour scales. The 2010 index distributions are compressed relative to later years (the 98th-percentile NDVI is 0.61 in 2010 against 0.70–0.74 in 2015–2025), a sensor and compositing effect that limits direct numerical comparison across the 2010/2015 sensor break. The median NDWI of pixels mapped as Water is also negative in every year (−0.05 to −0.30), consistent with seasonal tanks. Because the indices are classifier inputs, agreement between index patterns and mapped classes is expected and is not independent evidence that a class or transition is correct.

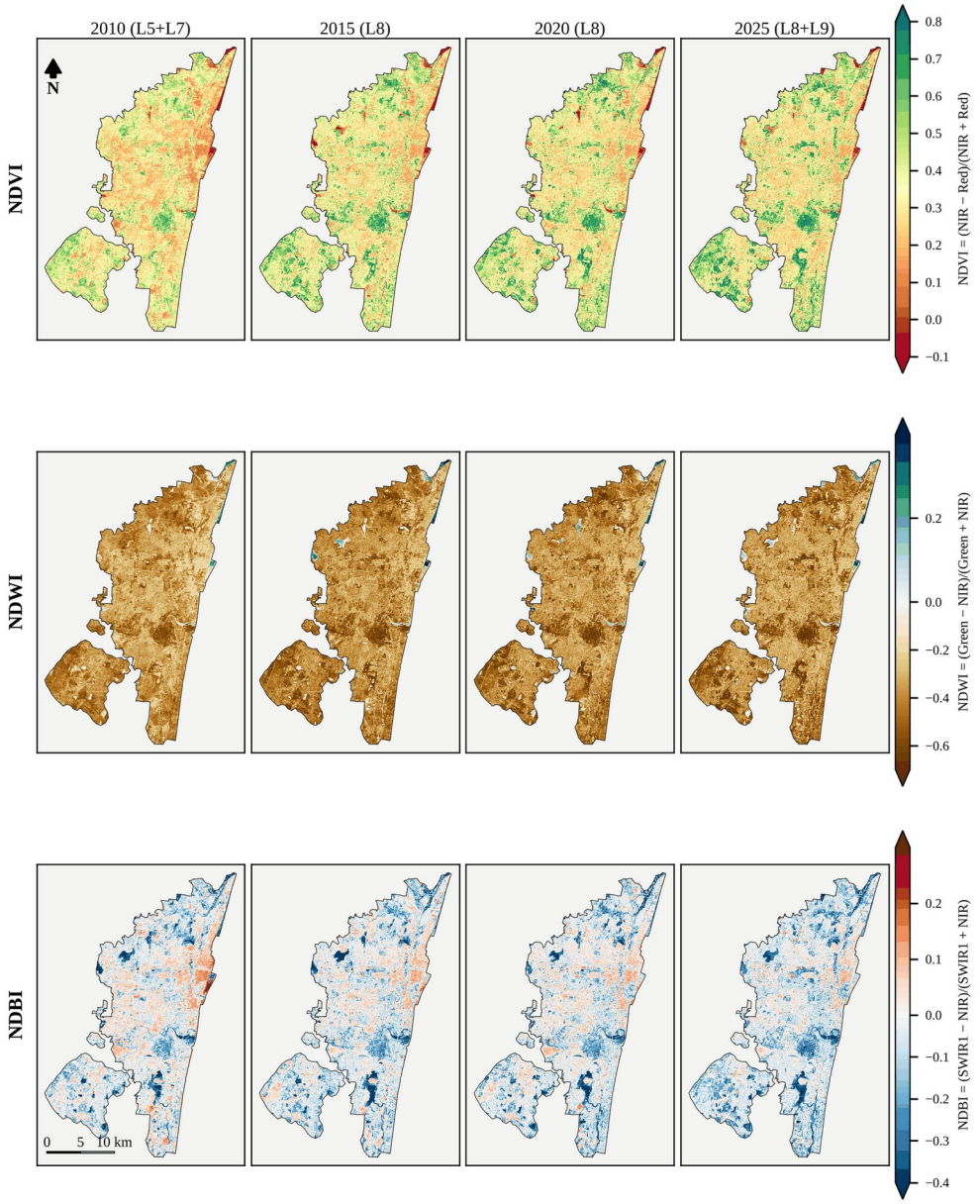


Fig. 3.3 NDVI, NDWI and NDBI for 2010–2025, each on one fixed scale.

Rendered from the project's numerical index GeoTIFFs (30 m, locked grid; 592,039 valid pixels per year). 2010 from the Landsat 5 + 7 predictor stack; 2015–2025 from the full-year composites.

3.4 PROCESSING WORKFLOW

Fig. 3.4 shows the processing chain applied to each year. Two rules held throughout: validation points were never used to fit a classifier or select training data, and every reported area was computed from an exported and re-loaded classified raster rather than from an on-the-fly classification.

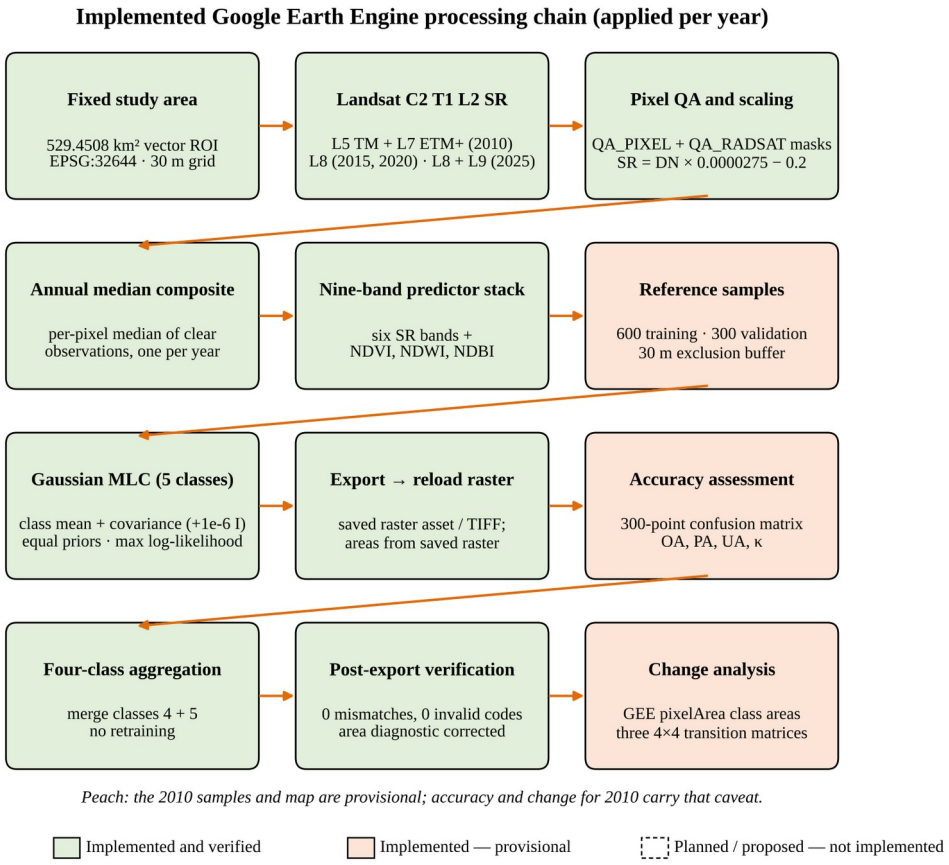


Fig. 3.4 Implemented Earth Engine processing chain.

Drawn for this report from the project scripts and console outputs.

3.5 LAND-COVER CLASSES

Five classes were mapped: **Built-up** (buildings, roads, paved and industrial surfaces), **Vegetation** (dense or tree-dominated vegetation), **Water** (rivers, lakes, tanks), **Agricultural/Open** (agriculturally identifiable land) and **Barren/Fallow** (unused vacant or open land, which by the project's definition may be green, grassy or scrubby). For final reporting, the last two are merged into **Agricultural & Vacant/Unused Open Land** (Section 3.9).

3.6 TRAINING AND VALIDATION DATA

For each year the final training set contains **600 pixels (120 per class)**, drawn from labelled reference polygons with a fixed random seed (42), and the validation set contains **300 points (60 per class)**. A 30 m exclusion buffer separates the two, and the quality checks for every year report zero overlap. How the reference data were produced differs by year and is part of the development story in Chapter 4.

3.7 GAUSSIAN MAXIMUM LIKELIHOOD CLASSIFIER

Earth Engine has no built-in MLC, so the classifier was implemented with Earth Engine array operations. For each class c, the mean vector μ_c and the 9 × 9 covariance matrix Σ_c are estimated from its 120 training pixels, and a small ridge term guarantees a stable inverse. Each pixel x receives the Gaussian log-likelihood score (equal priors, constant omitted) and is assigned to the class with the highest score:

$$\Sigma'_c = \Sigma_c + 10^{-6}\, I, \quad g_c(x) = -\tfrac{1}{2} \ln|\Sigma'_c| - \tfrac{1}{2} (x - \mu_c)^T \Sigma'^{-1}_c (x - \mu_c) \tag{3.5}$$

The first term penalises classes with large covariance; the second is the squared Mahalanobis distance from the class mean, i.e. how far the pixel lies from what the classifier learned for that class.

3.8 ACCURACY ASSESSMENT

With n_{ij} validation points of reference class i assigned to map class j and N = 300, overall, producer's and user's accuracy are computed as below; Cohen's kappa is reported for continuity [13], [16]. All values in this report were recomputed from the stored confusion matrices and agree with the Earth Engine outputs.

$$OA = \Sigma_i\, n_{ii} / N, \quad PA_i = n_{ii} / n_{i+}, \quad UA_j = n_{jj} / n_{+j} \tag{3.6}$$

3.9 AREAS, AGGREGATION AND TRANSITIONS

Class areas. Each saved classified raster assigns one class label to every cell of the locked 30 m grid (EPSG:32644). The area of a class *c* is obtained by masking the raster to that class, multiplying the mask by `ee.Image.pixelArea()`, which gives the area of each cell in square metres, and summing the result over the study-area geometry with `ee.Reducer.sum()`. In practice all classes were measured in one pass, by adding the class band to the pixel-area image and grouping the sum by class value. With $L(p)$ the label of pixel p, a_p its Earth Engine pixel area in m² and $[\cdot]$ equal to 1 when the condition holds and 0 otherwise:

$$A_c = 10^{-6} \times \Sigma_p\, a_p \cdot [L(p) = c] \quad (\text{km}^2) \tag{3.7}$$

A full 30 m × 30 m cell nominally represents 900 m², or 0.0009 km², so a pixel count multiplied by 0.0009 gives an approximate area. The implemented method does not rely on that shortcut: it sums the area that Earth Engine assigns to each cell inside the study geometry, so cells crossed by the study-area boundary are treated by Earth Engine's masking rules instead of being counted as whole pixels. Areas were measured on the exported and re-loaded rasters, because area reductions on the live classification exceeded

Earth Engine's interactive limits. The two conventions give slightly different totals (see the note at the end of this section).

Four-class aggregation. Four-class maps are produced by relabelling the saved five-class rasters (classes 1–5 → 1, 2, 3, 4, 4) **without retraining**, and the four-class confusion matrix is obtained by summing rows and columns 4 and 5 of the five-class matrix.

Post-classification comparison. Change between an earlier map A and a later map B is measured by post-classification comparison (PCC) [17]. Because every saved raster lies on the same grid, each cell of A has exactly one corresponding cell in B, and the comparison reads the two labels of each cell. This is a pixel-wise raster overlay: no polygon Union or Intersection geoprocessing is involved, and no geometry is created or clipped. Each pair of labels is encoded as one transition code:

$$T = 10 \times A + B \tag{3.8}$$

The tens digit is the earlier class and the units digit the later class. In the five-class maps, for example, code 11 is Built-up → Built-up, 41 Agricultural/Open → Built-up, 51 Barren/Fallow → Built-up, 14 Built-up → Agricultural/Open and 15 Built-up → Barren/Fallow; in the four-class maps, code 14 is Built-up → Agricultural & Vacant/Unused Open Land. The transition matrix is produced in the same way as the class areas: the code image is added as a band to `ee.Image.pixelArea()`, and the area is summed separately for each code with a grouped `ee.Reducer.sum()`. The 25 possible codes of a five-class comparison (16 for four classes) fill the cells of a matrix M, whose entry M_{ij} is the area mapped as class i in the earlier year and class j in the later year (Fig. 3.5). Row totals reproduce the earlier year's class areas and column totals the later year's, which provides a check on every matrix.

Gains, losses and net change. For any class j, the gross gain is the area that entered j from all other classes, the gross loss is the area that left j for all other classes, and the net change is their difference:

$$\text{Gain}_j = \Sigma_{i \neq j} M_{ij}, \quad \text{Loss}_j = \Sigma_{k \neq j} M_{jk}, \quad \text{Net}_j = \text{Gain}_j - \text{Loss}_j \tag{3.9}$$

In matrix terms, the gain is the class's column total minus its diagonal cell and the loss its row total minus the diagonal, so the net change equals the change in the class's mapped area between the two years. Because the matrix contains every from–to combination, these quantities can be computed for every class, not only Built-up, and because both maps cover the same footprint, the net changes of all classes sum to zero. Annual area totals alone give only the net change; gross gains and losses require the matrix. Chapter 6 applies these calculations to the original maps and Section 8.8 to the fusion-adjusted series (Table 8.11).

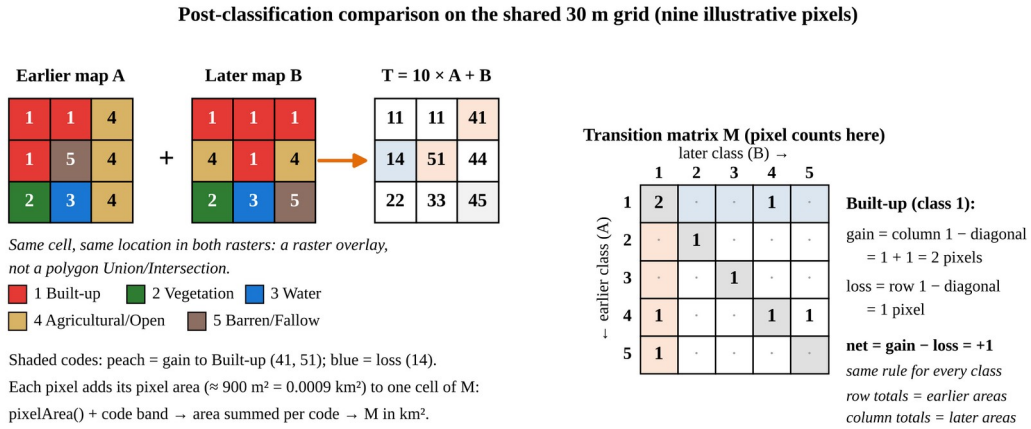


Fig. 3.5 Post-classification comparison of two maps on the shared 30 m grid: transition coding, matrix accumulation and class gains and losses.

Schematic drawn for this report with nine illustrative pixels, not project data. In the project the same operations were applied to every cell of the study area in Earth Engine, with each pixel contributing its pixelArea() rather than a count.

Note on area totals. The grouped Earth Engine areas total 529.77 km² in every year, slightly more than the 529.45 km² vector area, and counting whole exported pixels gives 532.84 km². These are different measurement conventions; boundary-cell treatment is a plausible but unreconciled explanation. All areas reported here are Earth Engine **mapped** areas, not error-adjusted estimates [15].

3.10 POST-CLASSIFICATION BUILT-UP FUSION

Chapter 8 describes how a late phase of the project produced a second, deliberately separate kind of map: a **fusion-adjusted candidate**. Fusion here is a deterministic, rule-based correction of an existing classified map. No classifier is trained or re-run and no predictor is added to the Gaussian model: the locked MLC raster L is copied, and a pixel is relabelled from Agricultural/Open (class 4) or Barren/Fallow (class 5) to Built-up (class 1) only when every evidence layer $E_1 \ldots E_k$ specified for that year agrees. Every other pixel, including every Built-up, Vegetation and Water pixel, keeps its MLC label:

$$L'(x) = 1 \text{ if } L(x) \in \{4, 5\} \text{ and } E_1(x) \wedge \ldots \wedge E_k(x); \quad \text{otherwise } L'(x) = L(x) \quad (3.10)$$

The evidence layers are external Built-up products, matched to the map year where an archive exists, and, in two cases, a corrected map of a later year (Table 3.2). All tests are evaluated on the fixed 30 m MLC grid. Products finer than 30 m are converted to the fraction of each 30 m cell that is positive, and a cell passes only above a stated fraction; GAIA is tested as 'impervious by' a cutoff year; GHSL is tested at its native 100 m cell, so a positive GHSL cell does not imply that every 30 m pixel inside it is built-up. The products define 'built-up' differently from each other and from the MLC class, so agreement between them is supporting evidence, not ground truth. No rule was calibrated to reach an area

target, but several rules were compared on the same validation points, so the choice among them is exploratory (Section 8.6).

Table 3.2 External evidence used in the fusion experiments.

Evidence	What it records	Native grid	Used for	Test on the 30 m grid
GAIA v10 [25]	Year a pixel became impervious, 1985–2018 (cannot revert)	30 m	2010, 2015, 2020, 2025	Impervious by 2010, 2015 or 2018 (change-year index ≥ 9, ≥ 4 or ≥ 1)
WSF 2015 [27]	Settlement extent, 2015	10 m	2015	Positive fraction of the 30 m cell ≥ 0.67
ESA WorldCover v100 [28]	Land cover 2020; class 50 Built-up	10 m	2020	Built-up fraction of the cell ≥ 0.67
Dynamic World V1 [29]	Per-image probability of 'built' from Sentinel-2	10 m	2025 (79 images)	10 m median built probability ≥ 0.65 with ≥ 6 valid observations; passing fraction of the cell ≥ 0.67
GHSL GHS-BUILT-S R2023A [26]	Built-up surface per cell (m^2)	100 m	2015 screen; 2010	2015: ≥ 6,000 m^2 per cell; 2010: > 0 (support) or = 0 (absence)
Corrected later-year map (this project)	Fusion-adjusted Built-up	30 m	2015 (uses 2020); 2010 (uses 2015)	Built-up in the later candidate — a temporal-consistency screen, not independent evidence

Earth Engine IDs: `Tsinghua/FROM-GLC/GAIA/v10`, `DLR/WSF/WSF2015/v1` (band `WSF`), `ESA/WorldCover/v100` (band `Map`), `GOOGLE/DYNAMICWORLD/V1` (band `built`), `JRC/GHSL/P2023A/GHS_BUILT_S/<year>` (band `built_surface`). No year-matched 10 m product exists for 2010.

Two features of these rules shape the rest of the report. First, only pixels that the MLC map labelled Agricultural/Open or Barren/Fallow can be promoted, so a corrected map differs from its original only by a transfer of area from those two classes to Built-up. Second, the retained 2015 rule uses the corrected 2020 candidate as one of its evidence layers, so the 2020 correction had to be produced first, and the corrected 2015 and 2020 maps are not independent of each other (Section 8.5, Fig. 8.4).

Each candidate is scored on the same validation points as its source MLC map, so the comparison is paired: for every point it is known whether fusion corrected it, worsened it or left it unchanged. For 2025 the archived 300-point sample could not be recovered, so a new sample of 60 pixels per class (seed 84) was drawn from the 33 existing validation master polygons, excluding pixels within 30 m of training polygons, and both the original and the fused 2025 maps were scored on it. Because the validation points had already been used during development, all fusion accuracies are exploratory rather than independent estimates. Areas of the retained candidates were computed, as for the original maps, from rasters exported on the fixed grid and reloaded.

Three kinds of result. The report therefore presents results from three processing stages, summarised in Table 3.3. Values from different stages are never combined. For example,

mapped Built-up rises by +24.83 km² between 2015 and 2020 in the original maps (Table 6.2) but by +4.50 km² in the fusion-adjusted series (Table 8.10); each value is correct for its own maps and is always reported with its method.

Table 3.3 Three kinds of classified output reported in this study.

Output	How it was produced	Classes	Retraining	Where reported	2015→2020 Built-up net change
Original five-class Gaussian MLC	Supervised classification of each year's nine-band stack (Sections 3.6–3.7)	5	Trained once per year; locked	Chapters 4–5; Table B.1	+24.83 km²
Four-class aggregation	Relabelling of the saved five-class rasters, classes 4 and 5 → 4 (Section 3.9)	4	None; reporting simplification	Chapters 5–6; Tables 5.1, 6.1, 6.2	+24.83 km² (Built-up is unaffected by the merge)
Five-class fusion-adjusted candidates	Rule-based promotion of class 4 or 5 pixels to Built-up where year-specific evidence agrees (this section; Chapter 8)	5	None; separate experimental rasters	Chapter 8; Tables B.5, B.6	+4.50 km²

The 2010 map of the fusion-adjusted series is the original, provisional MLC map; the 2010 two-way fusion alternative was subsequently reconstructed and exported for documentation only (Section 8.7; Fig. 8.10).

CHAPTER 4
IMPLEMENTATION JOURNEY AND EXPERIMENTAL DEVELOPMENT

With the method of Chapter 3 in place, the maps were produced backwards in time, from the best-observed year to the most difficult. Fig. 4.1 summarises the path actually followed, including the routes that were abandoned; this chapter explains what prompted each step and what it led to.

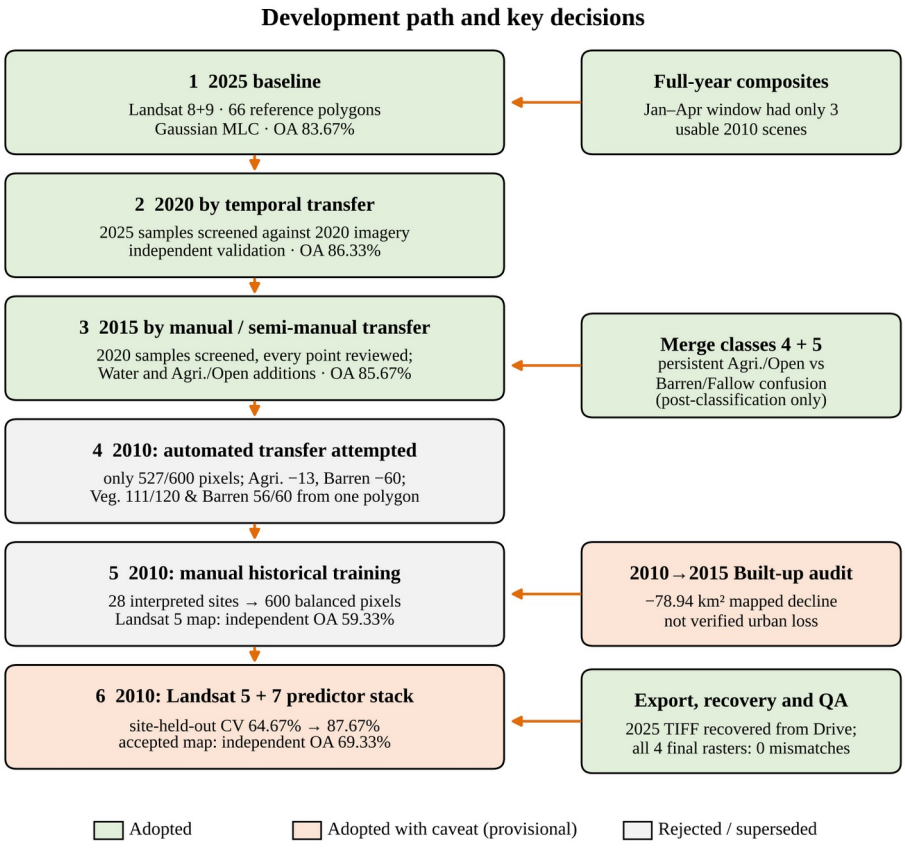


Fig. 4.1 Development path and key decisions.

Drawn for this report from the project scripts, console outputs and checkpoint records. OA = independent five-class overall accuracy; CV = site-held-out cross-validation on training data only.

4.1 DEVELOPING THE INITIAL CLASSIFICATION WORKFLOW

An initial Python pipeline on a cloud data catalogue was considered and replaced by Google Earth Engine, which allowed whole-year compositing without downloading imagery. The early design also changed in two other ways already described: the study boundary was fixed at 529.45 km², and the seasonal window was replaced by full-year composites when 2010 turned out to have only three usable dry-season scenes. Both changes were made before any map was accepted, so every year was produced under the same design.

4.2 ESTABLISHING THE 2025 BASELINE

2025 was classified first because it has the most observations (40 Landsat 8 and 9 scenes; median 20 clear observations per pixel) and because current land cover can be interpreted most confidently. Reference data were digitised as 66 training polygons, yielding 2,016 candidate training pixels, from which the balanced 600-pixel training set was drawn; 300 validation points were drawn separately (seed 84) after the 30 m exclusion. The resulting map reached an independent overall accuracy of **83.67%**. Its main error, half of the Barren/Fallow validation points mapped as Agricultural/Open, already signalled the class problem that later led to the merge.

4.3 EXTENDING CLASSIFICATION TO 2020 AND 2015

Re-digitising every year from scratch would have been slow, so reference data were carried backwards by **temporal transfer**. For 2020, 2025 samples were screened against the 2020 composite and kept where the land cover was judged unchanged; an independent 2020 validation set was assembled separately. The 2020 map reached **86.33%**. For 2015, the 2020 samples were screened in the same way, every point was then reviewed manually, and Water and Agricultural/Open polygons were added where the transferred data were insufficient. The 2015 map reached **85.67%**.

4.4 THE PRACTICAL LIMITATIONS OF MANUAL TRAINING

Although the three recent classifications achieved acceptable validation performance, each had required substantial manual interpretation: digitising polygons, screening transferred samples and, for 2015, reviewing every training point by hand. Two older years remained, and they were harder still. 2010 combined three difficulties: the fewest observations (seven Landsat 5 scenes, a median of two clear observations per pixel), a sensor break (TM rather than OLI, so 2015 spectral signatures could not simply be reused), and no reference data of its own. Its land cover also had to be interpreted from older historical imagery. Repeating the full manual procedure twice more was not realistic within the project's time, and this workload — not a research question formulated in advance — prompted an attempt to reuse the checked 2015 reference data automatically.

4.5 ATTEMPTING AUTOMATED HISTORICAL TRAINING

The automated approach started from the 600 locked 2015 training points. Each year's bands were normalised robustly within that year, and each 2015 point was screened in 2010 imagery: it was labelled stable only if it had enough clear 2010 observations, its nearest 2010 class signature was its original class with a sufficient margin, and most neighbouring pixels agreed. The thresholds were inherited from the earlier transfer framework and were

not tuned to validation data. Only 289 points were stable (277 uncertain, 34 changed). The stable points were then used as seeds for an automated expansion in 2010 feature space: 2,601 candidate pixels were generated, 1,923 passed screening, and 727 remained after removing duplicate Landsat pixels. A balanced selector could fill only **527 of the 600** required places (Table 4.1).

Table 4.1 Outcome of the automated 2010 training selection.

Class	Selected / 120	Source polygons	Problem
Built-up	120	10	—
Vegetation	120	2	111 of 120 pixels from one polygon
Water	120	4	Few source polygons
Agricultural/Open	107	8	13 short
Barren/Fallow	60	2	60 short; 56 of 60 pixels from one polygon
Total	**527 / 600**		Not adopted as final training

The automated screening therefore produced too few balanced examples, but the shortfall was the lesser problem. Even with the quota filled, a classifier trained on these data would have learned the spectral signature of two particular places rather than of the Vegetation and Barren/Fallow classes. Further automated rescue attempts followed; their console outputs were not retained, so their individual results are not reported, but none produced a sufficient and spatially diverse training set without weakening the quality rules. The automated set was rejected, and the project returned to manual historical interpretation for 2010.

Two classifier results from this period, recovered from conversation records rather than original console logs, point the same way. An early 2010 classifier (Stage 05F1) reached 100% on its own training pixels but only **40.67%** on the 300 independent validation points ($\kappa = 0.258$; 122 correct), with Vegetation and Barren/Fallow recall of 13.33% and 15.00% and 151 of the 300 points assigned to Agricultural/Open. It is not claimed that 05F1 used exactly the 527 selected samples. A later, supplementary test (Stage 07J), which transferred the 2015 model directly to 2010 without adequate harmonisation, reached **44.67%** ($\kappa = 0.308$; 134 correct); it was not the reason automation was abandoned. Because the two protocols differ, the figures are not a controlled ranking, and neither is the accepted 2010 map.

4.6 REBUILDING THE 2010 REFERENCE DATASET AND PREDICTOR STACK

New training polygons were digitised from historical imagery at **28 sites** (five or six per class), all inside the study area and clear of the validation zones. From 4,287 candidate pixels, a site-balanced set of exactly 600 pixels was drawn, so that no single site could dominate a class. A separate 2010 validation set of 300 points from 20 polygons was reserved.

The first manual model, on the Landsat 5 stack, fitted its training data well (86% resubstitution accuracy) but reached only **59.33%** on the independent validation points. Diagnostics located the problem. Training pixels re-sampled identically, so it was not a data-handling error. The errors were concentrated among Vegetation, Agricultural/Open and Barren/Fallow, and the separability of their fitted distributions explains why (Table 4.2): the pairs were moderately separable with all nine bands (JM 1.21–1.64), but Vegetation and Barren/Fallow were almost inseparable on the indices alone (JM 0.19), consistent with a Barren/Fallow class that admits green, scrubby land. Removing the indices did not help either: a six-band model reached 60.67% site-held-out cross-validation, against 64.67% for nine bands.

Table 4.2 Jeffries–Matusita distances between the fitted 2010 class distributions (Landsat 5 stack).

Class pair	All 9 bands	6 reflectance bands	3 indices
Vegetation vs Agricultural/Open	1.640	1.236	1.070
Vegetation vs Barren/Fallow	1.210	0.925	0.187
Agricultural/Open vs Barren/Fallow	1.636	1.140	0.836

JM ranges from 0 (complete overlap) to 2 (complete separation). It describes the fitted training distributions, not map accuracy.

Because the training pixels themselves were sound, the next step was to improve the image information available for 2010: the stack was rebuilt with Landsat 7 ETM+ observations added to Landsat 5. Sensor limitations were one plausible contributor to the weak result, alongside scene availability and the representativeness of the 28 sites; the experiments do not isolate their separate effects. The same 600 training pixels were re-evaluated by five-fold **site-held-out cross-validation**, in which whole sites are held out together (Table 4.3). Accuracy rose from 64.67% to 87.67%, with the largest gains in the three confused classes. This comparison used training data only.

Table 4.3 Site-held-out cross-validation of the 2010 training data (training data only).

Predictor stack	OA (%)	κ	Built-up	Veg.	Water	Agri./ Open	Barren
Landsat 5 only	64.67	0.558	85.83	51.67	90.00	48.33	47.50
Landsat 5 + 7	87.67	0.846	98.33	75.83	99.17	76.67	88.33

Per-class values are producer's accuracies (%).

The Landsat 5 + 7 map, however, reached an independent overall accuracy of only **69.33%**. Cross-validation measures how well the model generalises between the 28 training sites; the independent validation measures it at 20 different polygons, and the gap indicates that the model generalised much less well to new locations — consistent with, though not proof of, training sites that did not represent the full variety of 2010 land cover. The map was accepted, but only as a **provisional** classification (Section 5.5). Its first comparison with 2015 then revealed an unexpectedly large decline in mapped Built-up area, investigated in Chapter 7.

4.7 EVALUATING RANDOM FOREST AS AN ALTERNATIVE

The Random Forest comparison was carried out after the provisional 2010 map existed, during the period in which the 2010–2015 anomaly was being investigated (project stage numbering places it between the external-dataset cross-check and the likelihood audit of Chapter 7; exact dates were not preserved). It is reported here as a supplementary, retrospective classifier comparison. Its motivation was the gap between 87.67% cross-validation and 69.33% independent accuracy in 2010: one possible explanation was the classifier itself. The Gaussian MLC assumes that each class forms a single, roughly elliptical cluster in the nine-dimensional predictor space. Agricultural/Open and especially Barren/Fallow, which by definition include bare plots, fallow fields and green scrub, are unlikely to be unimodal, and the JM distances in Table 4.2 showed exactly these classes overlapping. A classifier that makes no distributional assumption might separate them better. Random Forest (RF) was therefore tested, first for 2010 and 2015 and then for 2020 and 2025.

RF builds an ensemble of decision trees, each grown on a bootstrap sample of the training pixels, with each split chosen from a random subset of the predictors; the predicted class is the majority vote of the trees [22]. It can represent multimodal classes and non-linear boundaries and is insensitive to the scaling of the predictors, which is why it is widely used for land-cover mapping [23]. The price is that it learns whatever the training pixels contain, including site-specific signatures, and it has no explicit class model of the kind that Chapter 7 later decomposes.

To isolate the effect of the classifier, everything else was held fixed. Both classifiers used the same nine predictors, the same 600 training pixels and the same 300 independent validation points, so their confusion matrices can be compared cell by cell. RF used 150 trees, three variables per split, a minimum leaf population of three, a bag fraction of 0.7 and a fixed random seed, and it was also assessed by five-fold site-held-out cross-validation on the training data. Two qualifications apply. The 2010 RF script was not archived; the recovered output reports the same settings and the same 300 validation points, but the exact 2010 predictor stack it read cannot be confirmed independently. For 2025, the identifiers of the original balanced point sets could not be confirmed, so new 600/300 samples were rebuilt from the saved 2025 master polygons with the same seeds and 30 m exclusion, and **both** classifiers were re-fitted and evaluated on them. The 2025 comparison is therefore properly paired, but it is not a re-test of the official 2025 map: its MLC accuracy of 85.67% differs from the official 83.67% because the test points differ.

Table 4.4 Gaussian MLC and Random Forest on identical validation points.

Year	MLC OA (%)	RF OA (%)	RF − MLC (pp)	MLC κ	RF κ	RF site CV (%)	MLC 4-class‡	RF 4-class‡
2010*	69.33	67.33	−2.00	0.617	0.592	85.67	81.00	78.00
2015	85.67	80.33	−5.33	0.821	0.754	82.50	88.67	85.00
2020	86.33	82.67	−3.67	0.829	0.783	90.50	95.67	90.33
2025†	85.67	79.67	−6.00	0.821	0.746	82.50	96.67	92.67

*Provisional 2010 map. †2025 paired experiment on samples reconstructed from the master polygons (official 2025 MLC accuracy: 83.67%). ‡Derived by merging classes 4 and 5 in the same confusion matrices; no four-class model was trained. Site CV uses the 600 training pixels only and is not an independent accuracy.

RF was less accurate than MLC in all four experiments, by 2.00 to 6.00 percentage points (Table 4.4, Fig. 4.2a), and the derived four-class values keep the same ordering. Fig. 4.2b shows where the differences come from. Vegetation and Water were mapped almost identically by both classifiers (within 3.3 points in every year), so the classifiers differ only on Built-up and the two open-land classes, the classes whose definitions overlap.

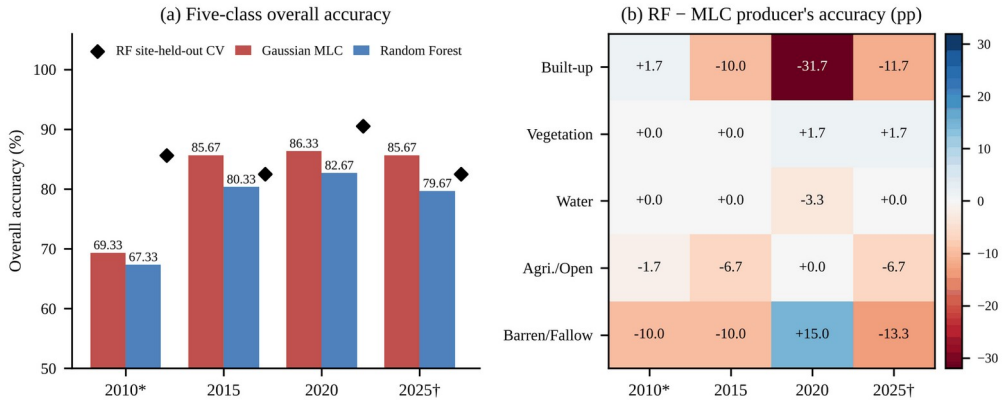


Fig. 4.2 Overall and class-level results of the Random Forest comparison.

Plotted for this report from the recovered five-class confusion matrices. (a) Independent five-class overall accuracy; diamonds: RF site-held-out cross-validation on training data. (b) Difference in producer's accuracy, RF minus MLC (percentage points). *2010 provisional; †reconstructed paired 2025 samples.

2010. RF recovered one more Built-up reference point (producer's accuracy 93.33% against 91.67%) but labelled 20 open-land reference points as Built-up, against 12 for MLC, so Built-up user's accuracy fell from 82.09% to 72.73%. Barren/Fallow producer's accuracy fell from 33.33% to 23.33% (14 of 60 correct), and Vegetation remained at 60.00% with the same confusion with open land (Fig. 4.3). RF therefore did not resolve the confusion that makes 2010 provisional; it shifted errors towards Built-up, the class at the centre of the 2010–2015 anomaly.

2015 and 2020. In 2015 RF was weaker on Built-up (71.67% against 81.67%; 15 Built-up points mapped as Agricultural/Open) and on both open-land classes, while 2015 Water stayed at 66.67% with both classifiers. That shared Water error points to the reference data and annual compositing (seasonal tanks dry for much of the year), not to the classifier. In 2020 RF did what the open-land problem seemed to call for: Barren/Fallow producer's accuracy rose from 48.33% to 63.33%. It did so at the cost of Built-up, whose producer's accuracy collapsed from 86.67% to 55.00% as 20 of 60 Built-up reference points were mapped as Barren/Fallow (three for MLC).

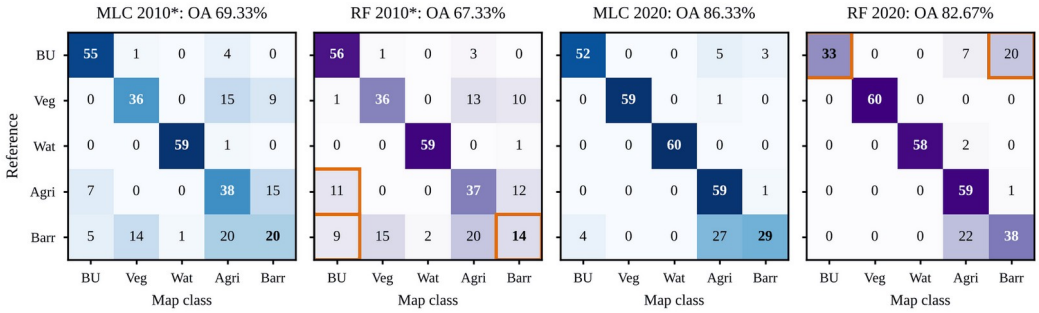


Fig. 4.3 Paired five-class confusion matrices, Gaussian MLC and Random Forest, 2010 and 2020.

Rows: reference (60 points per class); columns: map class (BU Built-up, Veg Vegetation, Wat Water, Agri Agricultural/Open, Barr Barren/Fallow). Outlined: cells discussed in the text. All RF matrices are in Appendix B.

Reconstructed 2025. On the rebuilt samples RF lost 11.7 points of Built-up and 13.3 points of Barren/Fallow producer's accuracy (36.67% against 50.00%), with 31 Barren/Fallow points mapped as Agricultural/Open by RF and 29 by MLC. The open-land confusion was essentially the same for both classifiers.

Cross-validation versus independent validation. RF site-held-out cross-validation reached 85.67% for 2010, 82.50% for 2015, 90.50% for 2020 and 82.50% for the reconstructed 2025 samples. These are not independent accuracies. In 2010 the RF gap between cross-validation and independent validation (85.67% against 67.33%) mirrors the MLC gap (87.67% against 69.33%). Because both classifiers show a similar gap, it is unlikely to be caused by the Gaussian model alone; limitations of the inputs and reference data — the representativeness of the 28 training sites, sensor differences, annual compositing and mixed pixels — remain plausible contributors, but the experiments do not isolate their individual effects.

Limits of the comparison. The differences were not tested for statistical significance. A paired test such as McNemar's [24] needs the per-point predictions of both classifiers, which were not retained, so differences of 2–6 points on 300 points should be read as a consistent direction rather than proven superiority. No RF classified maps, area totals or transition rasters were produced for the record, so the comparison says nothing directly about how RF would have mapped the 2010–2015 change. The RF results are recovered transcriptions of the original Earth Engine outputs, recomputed and checked for internal consistency; their provenance is recorded in Appendix A.

Why Gaussian MLC remained the official classifier. It achieved higher independent accuracy in every documented experiment; RF's losses fell on Built-up, the class that matters most for an urban change series; one classifier for all four years keeps the series

methodologically consistent; and MLC's explicit likelihoods made the read-only anomaly audit of Chapter 7 possible. The comparison does not show that RF is generally inferior. It shows that, with these predictors and this reference data, changing the algorithm did not solve the problems that mattered.

4.8 REFINING THE REPORTING CLASSIFICATION AND SCOPE

By this point the same confusion between Agricultural/Open and Barren/Fallow had appeared in 2025, 2020 and, more severely, 2010. Rather than re-train every year with redefined classes, the two classes were merged after classification, and the saved five-class rasters were relabelled. No new classifier was trained; the accuracy evidence for this decision is presented in Section 5.3. The merge was motivated by the open-land confusion, not by the 2010 anomaly, and it does not change any Built-up label. At the same time, the effort that 2010 had required made it clear that 2005 — which would need the same historical interpretation — could not be completed to the same standard in the time available. Its Landsat 5 inventory (19 scenes) and stack had been prepared, but 2005 was removed from the completed scope, and the final series covers 2010, 2015, 2020 and 2025.

4.9 COMPLETING AND VERIFYING THE FINAL MAPS

When the four-class rasters were prepared, the 2025 classification was not among the project's Earth Engine assets: the original 2025 script had exported it to Google Drive as a GeoTIFF instead. The original file was recovered from Drive and re-uploaded without rebuilding the classifier. Its class areas matched the areas archived when the 2025 map was first produced to better than 10^{-6} km² in all five classes, and the 2025 accuracy reported here is that of the archived 300-point validation record. All four final four-class rasters were then re-loaded and compared with a fresh merge of their five-class sources, with **zero** mismatching pixels and zero invalid class codes in every year. An area diagnostic that briefly showed 528.79 km² was traced to the mask being applied twice in the measurement, not to the rasters, and was discarded.

These four rasters are the locked Gaussian MLC baseline of this report. The Built-up correction described in Chapter 8 produced separate candidate rasters and did not alter any of these assets, their training samples or their validation points.

CHAPTER 5
CLASSIFICATION AND VALIDATION RESULTS

5.1 FINAL LAND-COVER MAPS

Fig. 5.1 shows the four final maps on one grid, boundary and palette, so differences between panels are differences between classifications. The three later maps show a dense Built-up core with open land and vegetation concentrated in the south-west and along the fringes. The 2010 panel differs visibly, with much more Built-up and much less open land; Chapter 7 shows why that difference should not be read as change.

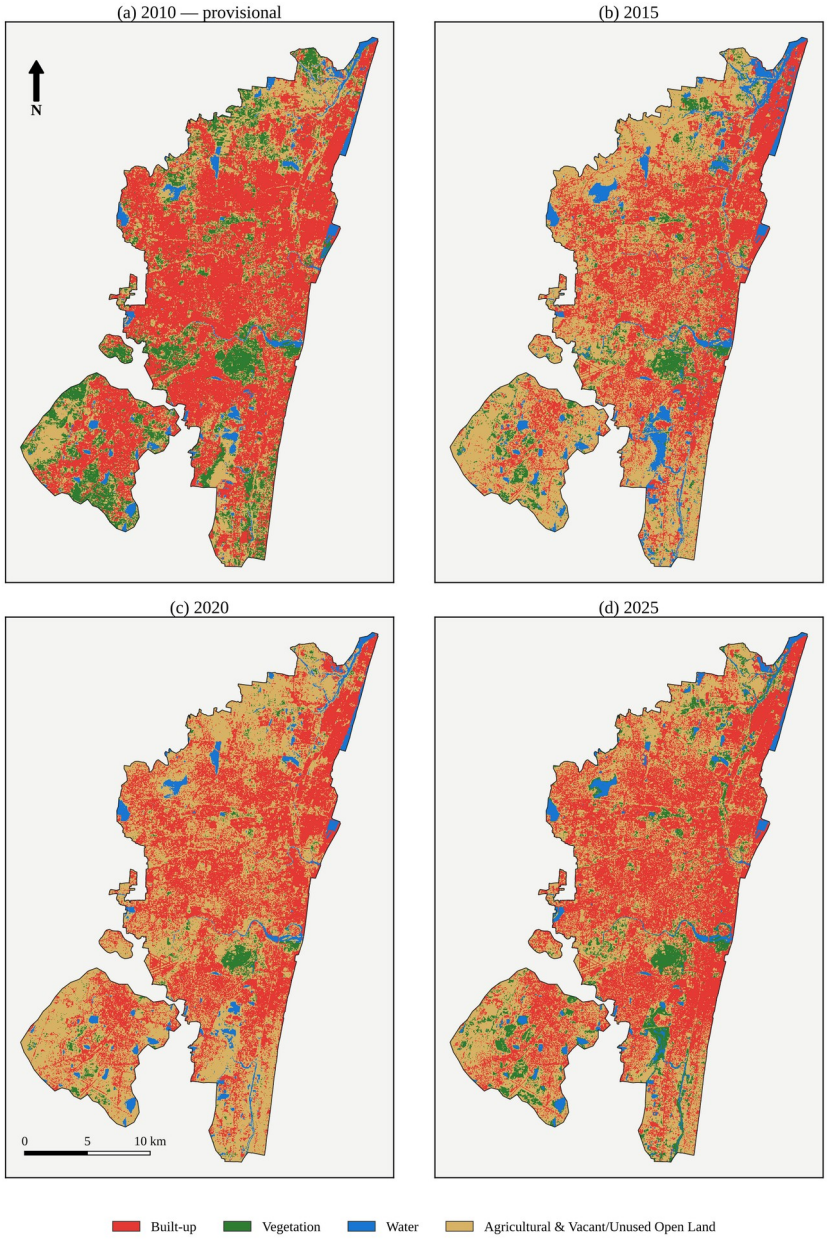


Fig. 5.1 Four-class LULC maps of the Chennai Urban Study Area, 2010–2025.

Rendered from the exported final GeoTIFFs; 30 m cells, nearest-neighbour display. The 2010 map is provisional.

5.2 OVERALL ACCURACY

Table 5.1 gives the accuracy of every map before and after the merge. The three accepted maps reach 83.7–86.3% overall accuracy with five classes; 2010 reaches 69.3%.

Table 5.1 Overall accuracy and kappa of the five-class and four-class maps.

Year	5-class correct	5-class OA (%)	5-class κ	4-class correct	4-class OA (%)	4-class κ	Change (pp)
2010*	208/300	69.33	0.617	243/300	81.00	0.736	+11.67
2015	257/300	85.67	0.821	266/300	88.67	0.838	+3.00
2020	259/300	86.33	0.829	287/300	95.67	0.940	+9.33
2025	251/300	83.67	0.796	290/300	96.67	0.953	+13.00

*Provisional. 300 independent validation points per year. The four-class gain results from merging two classes, not from a better classifier.

5.3 CLASS-LEVEL ERRORS AND THE MERGE

Table 5.2 Producer's (PA) and user's (UA) accuracy of the five-class maps (%).

Class	2010 PA	UA	2015 PA	UA	2020 PA	UA	2025 PA	UA
Built-up	91.67	82.09	81.67	98.00	86.67	92.86	91.67	100.00
Vegetation	60.00	70.59	100.00	98.36	98.33	100.00	91.67	100.00
Water	98.33	98.33	66.67	97.56	100.00	100.00	100.00	100.00
Agricultural/Open	63.33	48.72	90.00	72.00	98.33	64.13	86.67	57.78
Barren/Fallow	33.33	45.45	90.00	73.97	48.33	87.88	48.33	72.50

Table 5.2 shows where the errors lie. Water and Vegetation are mapped reliably from 2015 onwards (producer's accuracies 91.7–100%), except 2015 Water (66.67%), where 20 Water reference points were mapped as open land, plausibly seasonal tanks that are dry in an annual median. Built-up has high user's accuracy in every year (82–100%). The persistent error is between the two open-land classes: Barren/Fallow producer's accuracy is 48.33% in both 2020 and 2025 because about half of its reference points are mapped as Agricultural/Open, and Agricultural/Open user's accuracy is correspondingly low (57.8–72.0%). In 2010 the confusion spreads further, with Vegetation, Agricultural/Open and Barren/Fallow mutually confused and only 20 of 60 Barren/Fallow points correct.

The full matrices (Appendix B) make the pattern concrete. In 2025 the only substantial off-diagonal cell is Barren/Fallow mapped as Agricultural/Open (31 of 60). In 2010 the same cell holds 20 points, but Vegetation and Barren/Fallow are also confused in both directions, which no merge of classes 4 and 5 can remove. These errors share one cause. Barren/Fallow, as defined in this project, admits green, grassy and scrubby unused land, which in an annual median is spectrally close to fallow farmland and, in 2010, to vegetation.

Presenting the two classes as reliably distinct would overstate what the maps can support. Final reporting therefore merges them into **Agricultural & Vacant/Unused Open Land** by post-classification aggregation, and the five-class results remain part of the record.

The Random Forest comparison of Section 4.7 supports this reading. On the same validation points RF reproduced the Vegetation and Water results and the open-land confusion; where it differed from MLC, it moved errors between Built-up and the open-land classes rather than removing them. The class-level errors in Table 5.2 are therefore better explained by overlapping class definitions, reference data and annual compositing than by the parametric form of the classifier.

5.4 WHAT THE FOUR-CLASS ACCURACY MEANS

After the merge, overall accuracy is 81.00%, 88.67%, 95.67% and 96.67% for 2010–2025 (Table 5.1). The gain is largest where most errors were between classes 4 and 5 (2025: +13.0 percentage points) and smallest in 2015 (+3.0), whose main error is Water mapped as open land. The higher figures are correct descriptions of the four-class maps, but they are an accounting consequence of no longer counting within-pair disagreement as error. Built-up, Vegetation and Water decisions are identical before and after the merge, the merged class combines land uses with different meanings, and it carries 120 validation points against 60 for each other class. The merge is a change of reporting scheme, not an improvement of the classifier.

5.5 RELIABILITY OF THE 2010 MAP

The 2010 map is provisional for three reasons, stated once here and assumed in the rest of the report. First, its independent accuracy (69.33% five-class; 81.00% four-class) is clearly below the other years. Second, the gap between its training-only cross-validation (87.67%) and its independent accuracy indicates that the model generalised poorly beyond its 28 training sites; site representativeness is a likely contributor, alongside sensor and compositing effects. Third, the validation results of the earlier Landsat 5–only model were visible before the Landsat 5 + 7 stack was adopted. The documented basis for that choice was the training-only cross-validation, and no labels or settings were tuned to validation points, but the validation set was no longer entirely unseen, so 69.33% may be mildly optimistic. The map is labelled provisional in its asset name, in this report and in the application.

5.6 WHAT OVERALL ACCURACY DOES NOT SHOW

The accuracies above describe agreement at 300 sampled points per year. They do not show whether a class is missing over large, spatially coherent areas. In 2015, for example, the map is 85.67% accurate overall and its Built-up user's accuracy is 98.00% — almost every point mapped as Built-up is Built-up — yet 11 of the 60 Built-up reference points (18.3%) are mapped as open land. With only 60 Built-up points per year, an omission concentrated in particular neighbourhoods can coexist with good overall figures. Chapter 8 shows that this is what happened in the later maps and reports the accuracy of the fusion-adjusted candidates (Section 8.6); the official accuracies of the original maps remain those given here.

CHAPTER 6
MULTI-TEMPORAL LAND-COVER CHANGE

6.1 MAPPED CLASS AREAS

Table 6.1 and Fig. 6.1 give the mapped area of each class. Across the three accepted maps (2015–2025), Built-up area rises from 200.88 to 256.37 km² (+55.49 km²; from 37.9% to 48.4% of the mapped area), Water falls from 39.73 to 20.39 km², and Vegetation fluctuates (33.65, 14.11 and 44.27 km²), a pattern more consistent with differences in annual greenness and class confusion than with steady loss or gain. The 2010 column is shown for completeness only.

Table 6.1 Mapped class areas, km² (% of mapped area).

Class	2010*	2015	2020	2025
Built-up	279.82 (52.8%)	200.88 (37.9%)	225.72 (42.6%)	256.37 (48.4%)
Vegetation	81.22 (15.3%)	33.65 (6.4%)	14.11 (2.7%)	44.27 (8.4%)
Water	21.78 (4.1%)	39.73 (7.5%)	22.04 (4.2%)	20.39 (3.8%)
Agricultural & Vacant/Unused Open Land	146.95 (27.7%)	255.50 (48.2%)	267.91 (50.6%)	208.74 (39.4%)
Total	**529.77**	**529.77**	**529.77**	**529.77**

*Provisional. Earth Engine pixel-area sums over the saved four-class rasters.

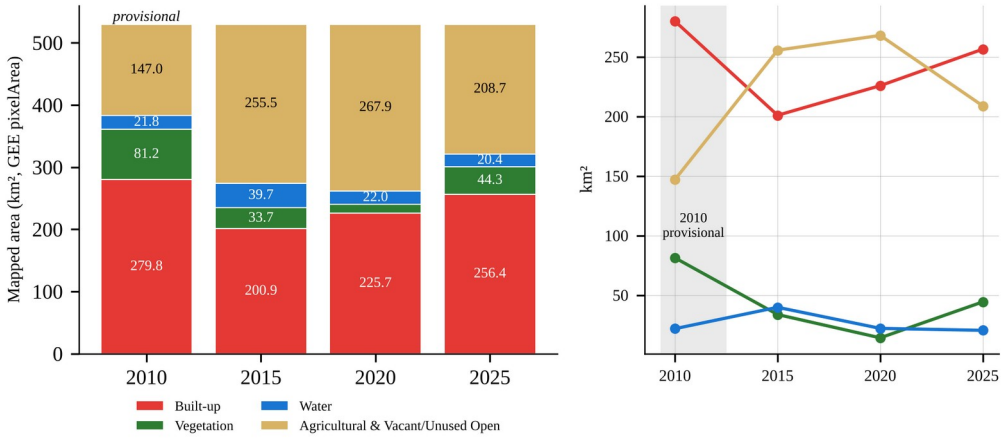


Fig. 6.1 Mapped class areas by year.

Left: stacked areas (km²). Right: trends; the shaded band marks the provisional 2010 map.

6.2 TRANSITIONS BETWEEN YEARS

Fig. 6.2 gives the three adjacent-year transition matrices, computed from the saved four-class rasters by the per-pixel transition coding and grouped pixel-area sums of Section 3.9 (Eq. (3.8), Fig. 3.5). Between the accepted maps, the largest flow into Built-up comes from open land (49.67 km² in 2015→2020 and 59.69 km² in 2020→2025). Each period also

shows a substantial reverse flow from Built-up to open land (28.03 and 29.07 km²). Table 6.2 separates these gross flows from the net change.

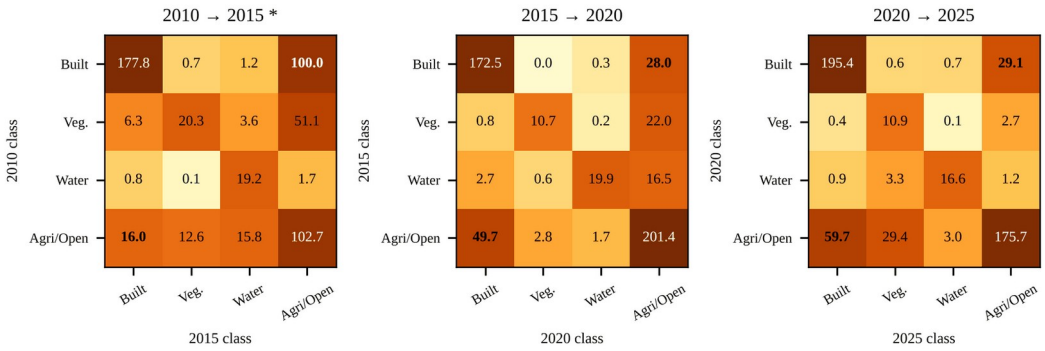


Fig. 6.2 Four-class transition matrices between adjacent years (km²).

Rows: earlier year; columns: later year. *2010 provisional. Earth Engine pixel-area sums grouped by transition code.

Table 6.2 Gross and net Built-up flows between adjacent years (km²).

Period	Stable Built-up	Into Built-up	of which from open land	Out of Built-up	of which to open land (14)	Net change
2010 → 2015*	177.83	23.06	15.97	101.99	100.05	−78.94
2015 → 2020	172.55	53.17	49.67	28.33	28.03	+24.83
2020 → 2025	195.37	61.00	59.69	30.35	29.07	+30.66

*Involves the provisional 2010 map.

6.3 WHERE THE CHANGE OCCURS

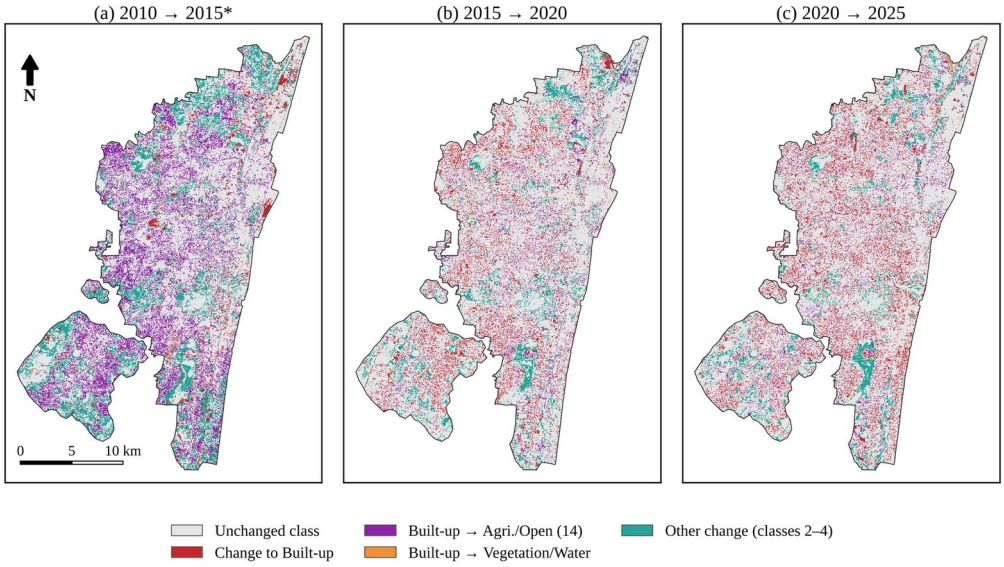


Fig. 6.3 Mapped class transitions in each period.

Per-pixel comparison of the aligned four-class GeoTIFFs; shown for location only (official transition areas are the Earth Engine values above). *Involves the provisional 2010 map.

Fig. 6.3 maps where the transitions occur. In 2015 → 2020 and 2020 → 2025, conversions to Built-up (red) are scattered across the whole study area rather than concentrated at the fringe, and they are interleaved with reverse changes. Such scattered patterns can combine real change, mixed-pixel effects and disagreement between two maps that each have roughly 3–16% error [17]. The 2010 → 2015 panel is different in kind: Built-up → open land (purple) covers large contiguous parts of the established city — the pattern that prompted the investigation in Chapter 7.

6.4 NET VERSUS GROSS CHANGE

Each period also contains reverse flows of 28–29 km² from Built-up to open land. Gross transition areas such as these may include real change, mixed-pixel effects at the Built-up/open-land boundary (low-density development, construction sites, vacant plots) and disagreement between the two maps. Net change (+24.83 km² in 2015 → 2020 and +30.66 km² in 2020 → 2025) summarises the balance of mapped gains and losses, but it is also affected by classification error. Because both quantities inherit errors from both maps, either can overestimate or underestimate physical change [15], [17]; both are mapped quantities and are read cautiously here.

Table 6.2 applies the gain and loss definitions of Eq. (3.9) to Built-up only. The same calculation can be made for every class from the matrices of Fig. 6.2; for the fusion-adjusted series it is reported for all five classes in Section 8.8 (Table 8.11). The two sets of values describe different maps (Table 3.3) and are not combined: the original 2015 → 2020 Built-up net change of +24.83 km², for example, corresponds to +4.50 km² in the fusion-adjusted series.

All values in this chapter describe the original MLC maps. Chapter 8 shows that these maps omit substantial Built-up land in 2015, 2020 and 2025, and reports the fusion-adjusted series that results from correcting them (Section 8.8). The original values are retained as the locked baseline against which those corrections are measured.

CHAPTER 7
INVESTIGATING THE 2010–2015 BUILT-UP ANOMALY

The provisional 2010 map completed the four-year series, but its first comparison with 2015 produced a result that could not simply be reported. Rather than accept it, or adjust the classifications until the change looked plausible, the project examined it through a sequence of investigations, each prompted by the limits of the one before. This chapter follows that sequence: the transitions themselves, a revised 2015 training set, external development products at city and pixel level, dated historical imagery, a reconstruction of both classifiers, their likelihood diagnostics, and the year-specific validation errors.

7.1 THE UNEXPECTED BUILT-UP DECLINE

The provisional 2010 map has 279.82 km² of Built-up against 200.88 km² in 2015, a mapped decline of **78.94 km²** in five years for a city that grew over the period; 2010 Built-up is larger even than in 2025 (256.37 km²). The problem was not only a difference in citywide totals. The five-class transition analysis (Table 7.1) showed that the two maps assigned very different classes to the same locations: of the 2010 Built-up pixels, 177.83 km² stayed Built-up, 72.21 km² became Agricultural/Open and 27.84 km² became Barren/Fallow. These two open-land transitions, original codes 14 and 15, together form the **100.05 km²** four-class transition 14 (Built-up → Agricultural & Vacant/Unused Open Land) used in Chapter 6, while only 15.97 km² moved from open land to Built-up. Taken at face value, these mapped assignments would mean large-scale urban abandonment.

Table 7.1 Five-class transitions of 2010 Built-up pixels, 2010 → 2015.

Original five-class code	2010 class → 2015 class	Area (km²)	Four-class code
11	Built-up → Built-up	177.827394	11
14	Built-up → Agricultural/Open	72.207435	14
15	Built-up → Barren/Fallow	27.842303	14
	Codes 14 + 15 combined	**100.049737**	14

Earth Engine pixel-area sums of the saved rasters. Original codes 14 and 15 are distinct five-class transitions; after aggregation they form the single four-class code 14. These are mapped category assignments, not verified conversions.

7.2 A FIRST HYPOTHESIS: TOO FEW URBAN EXAMPLES IN 2015

The first response to the decline was the simplest one. Visual comparison of the 2015 map with its own Landsat composite showed dense built-up areas labelled as open land, so the 2015 training set might simply contain too few examples of that kind of urban surface. A revised 2015 training set, **V2A** (stage 07U2, script `07U2_2015_V2_training_and_MLC.js`), tested this. Eleven candidate historical polygons

were digitised; two lay outside the study boundary and one had to be clipped to it, and nine were accepted after screening against the boundary and the validation exclusion areas, keeping 30 m from every validation location (asset `chennai_training_polygons_2015_v2_clean9`). They supplied **73** new training pixels — 32 Built-up, 21 Agricultural/Open and 20 Barren/Fallow — and the same number of original pixels was withdrawn from each of those classes, so the revised set kept 527 original pixels and 120 pixels per class. (This 527 is unrelated to the 527 pixels of the automated 2010 selection in Section 4.5.)

Table 7.2 Revised 2015 training experiment V2A against the locked 2015 model (same nine predictors, same 300 validation points).

Model	Training pixels	OA	Built-up correct	Built-up → Agri./Open	Built-up → Barren	Decision
Locked 2015 MLC	600 original (120 per class)	257/300 (85.67%)	49/60 (81.67%)	8	3	Retained
V2A (07U2)	527 original + 73 new (120 per class)	258/300 (86.00%)	49/60 (81.67%)	7	4	Rejected

The planned experimental export `chennai_mlc_lulc_2015_V2A_exp_v1` is not confirmed as completed. V2A values are recovered from project records; the full V2A matrix was not retained.

Overall accuracy rose by one point, but Built-up recall was **unchanged at 49 of 60**. The 11 Built-up omissions remained; one of them merely moved from Agricultural/Open to Barren/Fallow. Because the extra examples did not improve the class they were meant to improve, V2A was rejected and the locked 2015 model was kept. The result suggested that the problem was not only a shortage of urban examples, and prompted the external and pixel-level checks that follow.

7.3 EXTERNAL CROSS-CHECKS WITH GAIA AND GHSL

The next question was whether other, independently produced development products showed a similar decline. Two global datasets were examined over the same study area: **GAIA** (Global Artificial Impervious Area), an annual 30 m Landsat-derived map of land that has become predominantly impervious, 1985–2018 [25], and **GHSL** built-up surface (GHS-BUILT-S), a 100 m grid of built-up surface per cell in five-year epochs [26]. The archived results of this cross-check (project stage 07W) are in Table 7.3. Between 2010 and 2015 GAIA impervious area increased by 27.44 km² and GHSL built-up surface by 11.17 km², while the MLC Built-up class decreased by 78.94 km² (Fig. 7.1a). Neither external product showed a contraction.

Table 7.3 Archived external cross-check areas within the Chennai Urban Study Area (km²).

Product	2005	2010	2015	Change 2010→2015
GAIA impervious area	281.682122	318.932724	346.367985	+27.44
GHSL built-up surface	121.666332	133.761846	144.930702	+11.17
Gaussian MLC Built-up (this project)	not classified	279.821871	200.882401	−78.94

GAIA and GHSL values are archived stage 07W results recovered from the project conversation of 13 October 2026; the original 07W script and console output, including the exact product versions and processing settings, were not retained, and the values were not re-computed for this report. Dataset descriptions follow the official documentation [25], [26].

These comparisons raised doubt about a literal reading of the decline, but they are not ground truth. Both are model-derived products with their own errors. GAIA records only the year in which a pixel became impervious, so its time series is monotonic by design and cannot represent demolition or reversion [25]; GHSL measures built-up surface per 100 m cell and its epochs may involve temporal interpolation [26]. Neither defines an urban class in the same way as the MLC Built-up class, so their absolute areas cannot be compared with ours or with each other (Fig. 7.1 therefore compares directions of change, not areas).

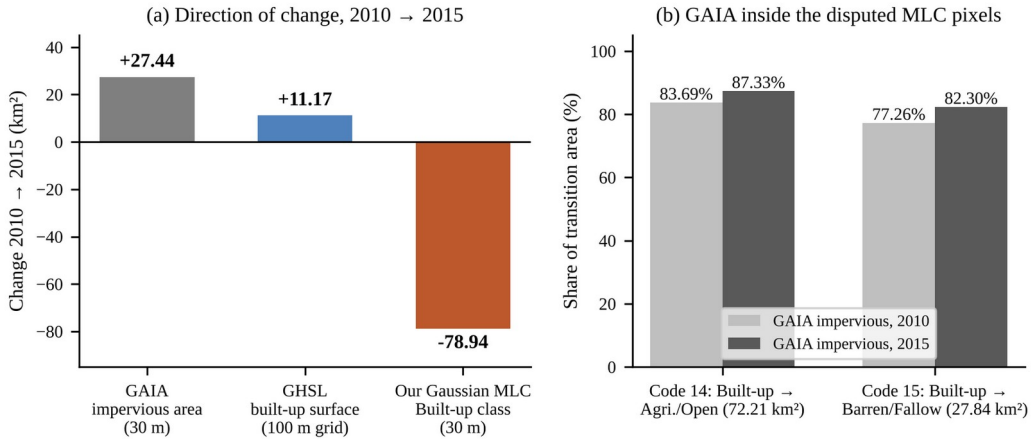


Fig. 7.1 External cross-checks of the 2010–2015 mapped Built-up decline.

(a) Change in each product's own measure, 2010 → 2015 (km²); the products measure different quantities, so only the direction of change is comparable. (b) Share of the area of original MLC transitions 14 and 15 that GAIA marks as impervious in 2010 and in 2015. Plotted for this report from the archived stage 07W values in Table 7.3 and Table 7.4.

7.4 SPATIAL DISAGREEMENT INSIDE THE DISPUTED PIXELS

Citywide totals could hide spatial offsetting, so the cross-check was then narrowed to the pixels that mattered: GAIA impervious coverage was measured inside the areas of original MLC transitions 14 and 15 (Table 7.4, Fig. 7.1b). In the 72.21 km² that the 2015 map moved from Built-up to Agricultural/Open, GAIA marked 83.69% as impervious in 2010 and 87.33% in 2015; in the 27.84 km² moved to Barren/Fallow, 77.26% and 82.30%. In other words, many locations that the 2015 land-cover map labelled as open land were still

identified as developed by an external product, and the external share increased rather than fell.

Table 7.4 GAIA impervious coverage inside the disputed MLC transitions (archived stage 07W).

Original MLC transition	Area (km²)	GAIA impervious, 2010	GAIA impervious, 2015
14 — Built-up → Agricultural/Open	72.207435	83.69% (0.836946)	87.33% (0.873329)
15 — Built-up → Barren/Fallow	27.842303	77.26% (0.772607)	82.30% (0.822987)

Fractions of each transition's area. They measure agreement between two classification products, not misclassification rates or ground-truth accuracy.

These fractions are more directly relevant to the anomaly than citywide totals, but they still describe disagreement between two products with different definitions and temporal assumptions; they are not measured error rates, and they do not establish how much of the disputed area was physically built-up. They did, however, justify looking at the disputed locations directly.

7.5 HISTORICAL IMAGERY AT AUDITED LOCATIONS

Five audit locations were examined on dated historical Google Earth imagery (Table 7.5, Fig. 7.2). At the three Built-up → open-land sites (AUDIT_04, AUDIT_08, AUDIT_32), established residential building is visible where the 2015 map shows open land; AUDIT_13, a control, is Built-up in both maps and in the imagery; and AUDIT_16, a university campus, is a genuinely mixed case of buildings and trees at 30 m. This is stronger local evidence than any citywide comparison, because it shows the land cover itself. It confirms classification disagreement at the audited locations; it does not quantify misclassification across the whole 100.05 km².

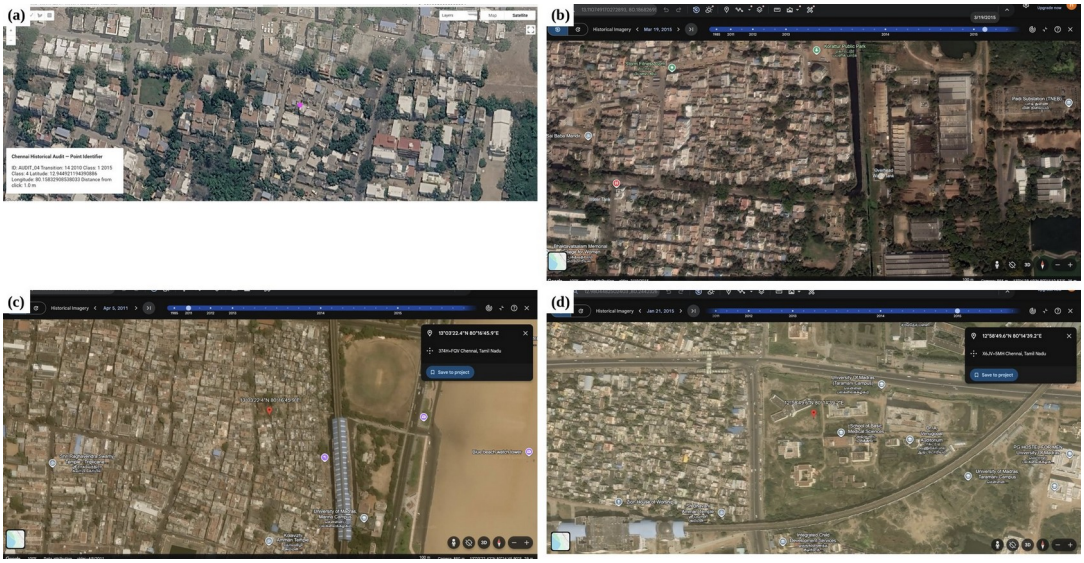


Fig. 7.2 Audit evidence at disputed locations and controls.

(a) Earth Engine point identifier for AUDIT_04 (transition 14) on the current basemap — locates the point but is not dated historical evidence. (b) Google Earth historical imagery, 19 Mar 2015, AUDIT_32: dense residential building where the 2015 map shows open land. (c) 5 Apr 2011, AUDIT_13: stable Built-up control. (d) 21 Jan 2015, AUDIT_16: university campus, a mixed case. Unaltered user screenshots, cropped for layout; imagery © respective providers. No dated screenshot of AUDIT_08 was retained.

Table 7.5 Historical audit locations.

Site	Location (lon, lat)	Mapped 2010 → 2015	2010 margin	2015 margin	Historical imagery
AUDIT_04	80.1583, 12.9449	Built-up → open (14)	14.66	−0.83	Residential, built-up
AUDIT_08	80.2275, 13.1095	Built-up → open (14)	2.03	−40.83	Residential, built-up
AUDIT_32	80.1868, 13.1107	Built-up → open (14)	19.86	−1.35	Residential, built-up
AUDIT_13	80.2794, 13.0562	Built-up stable (11)	32.90	38.01	Stable Built-up control
AUDIT_16	80.2442, 12.9804	Built-up → open (15)	4.59	−1.08	University campus, mixed

Five-class transition codes. Margin: Built-up log-likelihood score minus best open-land score (Section 7.7).

7.6 RECONSTRUCTING THE CLASSIFIERS

Before interpreting the classifiers' decisions, it had to be established that the disputed transitions reflected the stored classifications and their actual decision rules, not a coding error or an accidental raster substitution. A read-only audit (stage 07AB) — a diagnostic, not another retraining experiment — rebuilt the 2010 and 2015 Gaussian models from the archived training data — nine-band predictors, class means, regularised covariance matrices and equal priors — and recomputed the class likelihoods. Re-sampled predictors showed zero differences, and the reconstructed classes matched the saved rasters at all 300 checked locations in each year (**0 mismatches**). This supports the integrity of the calculation at the tested samples; it is not a full-raster equivalence proof and says nothing about historical accuracy.

7.7 WHY THE 2015 CLASSIFIER PREFERRED OPEN LAND

A stratified sample of 340 pixels from five original transition groups was then examined (Table 7.6). For each pixel the Built-up margin — the Built-up log-likelihood minus the best open-land log-likelihood — was computed in both years, and the 2015 preference was split into the two terms of Eq. (3.5): a multivariate spectral-distance (Mahalanobis) part and a covariance (log-determinant) part.

Table 7.6 Likelihood and index diagnostics for 340 sampled pixels by original transition.

Five-class transition	n	2010 Built-up margin	2015 Built-up margin	Mean ΔNDVI	Mean ΔNDBI
11 Built-up → Built-up	60	+37.86	+32.41	+0.032	−0.006
14 Built-up → Agri./Open	120	+10.41	−17.06	+0.088	−0.045
15 Built-up → Barren/Fallow	80	+21.21	−19.83	+0.055	−0.014
41 Agri./Open → Built-up	40	−39.53	+69.76	−0.061	+0.058
51 Barren/Fallow → Built-up	40	−26.73	+93.89	−0.050	−0.004
Total	**340**				

Positive margin: Built-up preferred. Δ = 2015 − 2010 annual-median value. Source: stage 07AB audit output.

For code 14 pixels, Built-up was preferred in 2010 by a modest margin (+10.41) and lost in 2015 by 17.06; of that 2015 open-land advantage, 16.61 came from spectral distance and only about 0.45 from the covariance term. For code 15 the advantage of 19.83 split into 18.79 and about 1.04. Put simply, in 2015 these pixels looked spectrally closer to the open-land training examples than to the Built-up examples, and they were also greener in the 2015 annual median (mean ΔNDVI +0.088 for code 14). Stable Built-up pixels kept large margins, and the reverse transitions behaved in the opposite way. This explains the **classifier's decision**; it does not show that the land was physically converted.

7.8 TRAINING REPRESENTATIVENESS AND VALIDATION ERRORS

The anomaly connects directly to the difficulties of Chapter 4: few clear Landsat 5 observations, a failed automated transfer, only 28 manually interpreted sites, and a 2010 model whose cross-validation (87.67%) far exceeded its independent accuracy (69.33%). The validation matrices show the relevant errors. In 2010, Built-up user's accuracy was 82.09%, with 12 open-land reference points mapped as Built-up; in 2015, Built-up producer's accuracy was 81.67%, with 11 Built-up reference points mapped as open land. The two models therefore erred in **opposite directions** around the Built-up/open-land boundary — the 2010 map tending to over-assign Built-up and the 2015 map to under-assign it — which is the combination that inflates an apparent decline. Plausible contributors include the representativeness of the 2010 training sites, the change of sensor, seasonal mixing in annual medians, vegetation mixed with buildings in 30 m pixels, and the different year-specific reference examples. These are observed patterns and plausible mechanisms; the investigations did not isolate the effect of any single cause.

The supplementary Random Forest comparison (Section 4.7) adds one further observation: on the same 2010 validation points RF was less accurate than MLC (67.33% against 69.33%) and mapped more open-land reference points as Built-up (20 against 12).

Changing the classifier therefore did not remove the confusion. RF maps and change rasters were not retained, so RF provides no estimate of the 2010–2015 change.

7.9 WHAT COULD AND COULD NOT BE CONCLUDED

The investigations agree in direction: external products, imagery at audited sites and the validation errors all point to classification disagreement rather than urban loss, and the likelihood audit explains how the 2015 classifier reached its decisions. None of them measures how much of the 100.05 km² was physically built-up. The classifications were therefore not altered at this stage. Relabelling suspicious areas, or adjusting models until the change looked like expected growth, would have tuned the results to an assumed answer and introduced confirmation bias. The 2010 map was kept as provisional, and the later four-class merge was justified by the open-land confusion alone; it leaves every Built-up label unchanged and does not resolve the anomaly.

> **Conclusion of the investigation.** The 2010–2015 mapped Built-up decline of 78.94 km² is **not verified physical urban loss**. Multiple investigations indicate substantial classification disagreement, including at locations with historically visible residential structures, but the exact spatial extent of the disagreement and the individual contributions of its possible causes were not established.

The investigation left one question open: whether the disagreement lay mainly in a 2010 map that over-assigned Built-up or in later maps that under-assigned it. Chapter 8 follows that question across all four years.

CHAPTER 8
CORRECTING BUILT-UP OMISSION ACROSS THE SERIES

Chapter 7 established that the 2010–2015 decline was not verified urban loss, but left open where the disagreement mainly lay: in a 2010 map that over-assigned Built-up, in later maps that under-assigned it, or both. This chapter follows that question across all four years. It describes, in the order in which they were tried, a series of attempts to recover missing Built-up land — spatial-context predictors, segmentation, Random Forest with automatically generated training labels and, finally, post-classification fusion — and what each attempt taught. Throughout, one rule applied: the aim was a Built-up record supported by imagery and independent evidence, **not** a rising area curve; no total was imposed, no 2010 raster was chosen for its trend, and no pixel was relabelled merely to meet a target. Stage numbers (08A–08G) refer to the project's working record; the Earth Engine scripts that survive are listed in Appendix A.

8.1 FROM A 2010 ANOMALY TO A SERIES-WIDE OMISSION PROBLEM

After the four maps were complete, the same neighbourhoods were compared across the 2010, 2015, 2020 and 2025 maps and their annual composites (stages 08A–08C). Many dense built-up areas that are clear in the composites appeared as Agricultural/Open or Barren/Fallow in 2015, 2020 and 2025, while the 2010 map more often recognised them as Built-up. In the original series, mapped Built-up area is 279.82 km² in 2010 but only 200.88, 225.72 and 256.37 km² in the three later years: the provisional 2010 map has more Built-up than even 2025.

Two further checks narrowed the problem. A Gaussian diagnostic (08D) reproduced the saved 2015 class at ten audited sites with zero differences, so the labels were the model's genuine decisions, not an export or band-order fault. A 30 m footprint inspection of the same ten sites on historical imagery (08E) judged four to be likely 2015 Built-up omissions, one a correct Built-up label, two genuinely non-Built-up and three mixed or uncertain. At the four likely omissions the 2015 model had preferred open land by small likelihood margins (about −1.1 to −6.8), against −121.15 at a genuine open-land control; these margins are recovered from the project record and were not re-computed. Ten sites cannot measure citywide error, but they showed that the later maps were missing real Built-up land, not only that 2010 had too much.

This is also why the validation results of Chapter 5 had not revealed the problem. A balanced sample of 60 Built-up points per year can show good overall accuracy and very high Built-up user's accuracy while missing an omission that is concentrated in particular

parts of the city (Section 5.6). Conversely, a larger Built-up area or a rising trend is not evidence of accuracy, because commission errors inflate area too. Both omission and commission therefore had to be tracked in every experiment that followed.

To gauge the possible scale, a read-only atlas (08G-1) intersected the original class 4 and 5 pixels of each year with GAIA impervious area (Table 8.1). In 2015, 153.73 km² mapped as open land was impervious in GAIA by 2015; in 2020 and 2025 (GAIA by 2018, its last epoch) the figures were 148.56 and 118.87 km², and 77.63 km² was impervious by 2015 yet mapped as open land in all three later years. GAIA impervious area is not the same category as the MLC Built-up class, and the 2015 and later rows use different GAIA cutoffs, so these are areas of **candidate** omission, not confirmed error.

Table 8.1 GAIA disagreement with the original MLC open-land classes (stage 08G-1, read-only).

Disagreement	Area (km²)
2015: GAIA impervious by 2015 ∩ original class 4 or 5	153.73
2020: GAIA impervious by 2018 ∩ original class 4 or 5	148.56
2025: GAIA impervious by 2018 ∩ original class 4 or 5	118.87
GAIA impervious by 2015 but class 4 or 5 in all of 2015, 2020 and 2025	77.63
Mapped 2015 Built-up → 2020 class 4 or 5	28.03
Mapped 2020 Built-up → 2025 class 4 or 5	29.07

The first script referenced a non-existent 2025 asset name and failed; the values are from the corrected run using `chennai_mlc_lulc_2025_from_original_tif`.

8.2 ADDING SPATIAL CONTEXT TO THE PREDICTORS

V2A (Section 7.2) had shown that more urban training examples did not help, and the likelihood audit (Section 7.7) that the 2015 model preferred open land on spectral grounds. A natural next hypothesis was that single 30 m spectra cannot separate roofs, concrete and dry impervious surfaces from bare soil and fallow land, but that their surroundings can: urban fabric is heterogeneous and consistently 'built' over a neighbourhood, while an open field is more uniform. Stage 08F tested this on 2015 with everything else frozen — the same 600 training and 300 validation points, the same 19 Landsat 8 scenes, archived red and NDBI values reproduced exactly — by adding two context predictors to the nine-band Gaussian MLC: the mean NDBI and the standard deviation of the red band over a window.

Fig. 8.1 shows how the two context predictors were computed. For a 3 × 3 window, the nine 30 m cells centred on a pixel, a 90 m × 90 m square, are read from the 2015 composite; the mean of their NDBI values and the standard deviation of their red-band reflectance become two new values of the centre pixel. The 5 × 5 window does the same over 25 cells, a 150 m × 150 m square. Computed in this way for every pixel, the two values were

appended to the nine-band stack, so each pixel was described by 9 + 2 = 11 predictors, and the Gaussian MLC estimated 11-element mean vectors and 11 × 11 covariance matrices in Eq. (3.5). No additional training samples were generated from the neighbouring pixels: the original 600 training and 300 validation points stayed where they were, and only the values attached to each of them grew from nine to eleven. The window is therefore a description of each point's surroundings, not additional data.

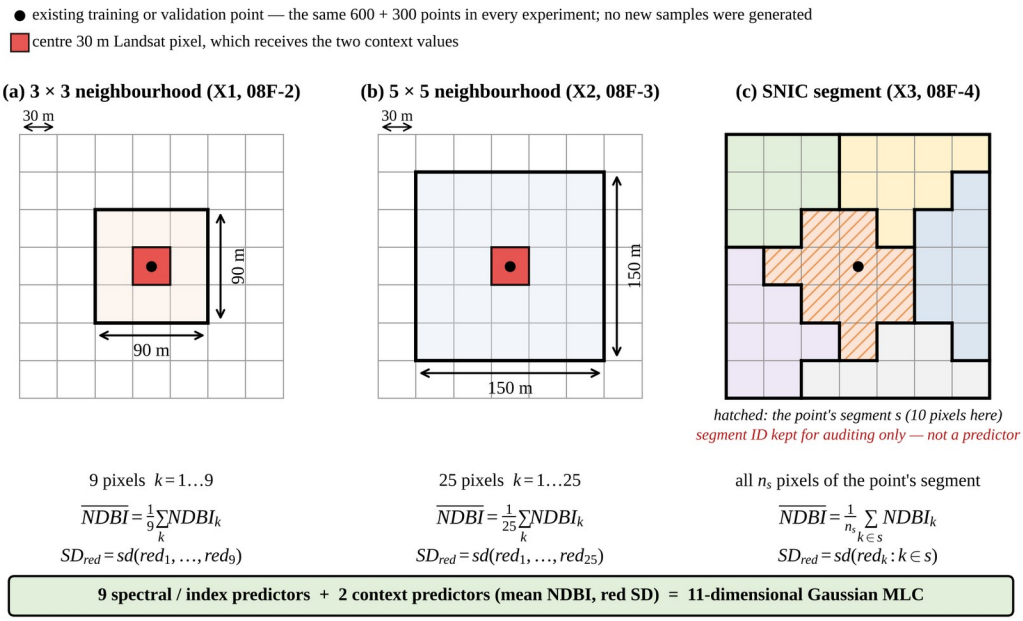


Fig. 8.1 How the context predictors of experiments X1–X3 were computed: (a) 3 × 3 window, (b) 5 × 5 window and (c) SNIC segment.

Schematic drawn for this report from the stage 08F record; cell values and segment shapes are illustrative. In all three experiments the two context values were attached to the same 600 training and 300 validation points, giving an 11-dimensional Gaussian MLC; the SNIC segment identifier was kept for auditing only.

With a 3 × 3 window (90 m; experiment X1, 08F-2) overall accuracy fell to 255/300 (85.00%) and Built-up stayed at 49/60. With a 5 × 5 window (150 m; X2, 08F-3) overall accuracy returned to 257/300 but Built-up was still 49/60, and two more open-land points were now mapped as Built-up, so Built-up user's accuracy fell from 98.00% to 94.23% (Table 8.2). Fixed square windows therefore added no Built-up detections and, at the larger size, added commission errors. Both were rejected. This is an informative negative result; it does not exclude other spatial methods.

Table 8.2 Context and segmentation experiments on the 2015 validation set (11-band Gaussian MLC unless stated).

Experiment	Added predictors	OA	κ	Built-up correct	Built-up FP	Decision
Locked 2015 MLC (9 bands)	—	257/300 (85.67%)	0.821	49/60	1	Retained
X1 3 × 3 (08F-2)	Window mean NDBI, red SD	255/300 (85.00%)	0.813	49/60	1	Rejected
X2 5 × 5 (08F-3)	Window mean NDBI, red SD	257/300 (85.67%)	0.821	49/60	3	Rejected
X3 SNIC, dynamic (08F-4)	Segment mean NDBI, red SD	258/300 (86.00%)	0.825	51/60	1	Exploratory
X3 SNIC, point snapshot (08F-4A/B)	Same, recomputed at 900 points	257/300 (85.67%)	—	50/60	1	Exploratory
X3 SNIC, frozen raster (08F-4D)	Same, from exported 12-band raster	258/300 (86.00%)	—	51/60	1	Exploratory

FP: non-Built-up validation points mapped as Built-up. All rows reuse the same 300 points, so differences of one or two points are within the noise of repeated selection. Matrices in Appendix B.

8.3 SEGMENT-BASED CONTEXT AND ITS REPRODUCIBILITY

Fixed windows mix a pixel with whatever happens to surround it. Segmentation instead groups pixels into spectrally coherent patches, so context can be measured within the patch a pixel belongs to. Experiment X3 (08F-4) used Earth Engine's SNIC segmentation [30] on the 2015 nine-band stack (seed spacing 5 pixels, compactness 0.1, 8-connectivity, neighbourhood 64, fixed grid; no class labels used) and added the within-segment mean NDBI and red-band standard deviation to the Gaussian MLC (Fig. 8.1c). Every pixel received the mean NDBI and red-band standard deviation of the segment to which it belongs, so all pixels of a segment share the same two context values while keeping their own nine spectral values, and the model again had 11 dimensions. Segment identifiers were kept for auditing only, never as predictors. The first run reached 258/300 (86.00%), with Built-up 51/60 and still only one false positive — a small gain on the target class. However, 33 validation points lay in 11 segments that also contained training pixels; that is not label leakage, because segments are built without labels, but those points are not spatially independent of training.

A paired audit of this gain repeatedly exceeded Earth Engine's interactive memory, so the segment predictors were first exported at the 900 training and validation points (asset `chennai_08f4a_X3_point_snapshot_v1`) and then as a complete 12-band raster — nine bands, two segment predictors and an audit-only segment identifier (`chennai_08f4c_X3_SNIC_frozen_11bands_segments_v1`). The three versions did not agree exactly. The point snapshot gave 257/300 with Built-up 50/60; the frozen raster reproduced the first run's totals (258/300; Built-up 51/60; one false positive) but not its per-

class matrix. Comparing frozen and snapshot values at the 900 points, the nine spectral bands were identical, but segment mean NDBI differed by up to 0.0628 and red SD by up to 0.0114, 12 segment identifiers differed, and training points fell in 143 rather than 141 segments. An early audit also reported zero changes because numeric 0/1 flags had been filtered as Boolean true; the corrected audit found five gains and five regressions. The cause of the drift was not isolated; evaluation scale and tiling are plausible, since SNIC output depends on them.

Against the locked map, the frozen X3 changed 13 validation labels: six became correct and five became wrong. Among the 60 Built-up points there were three corrections and one regression (49 → 51), all outside segments shared with training. Over the 269 points in unshared segments, both maps were correct at 229, with five gains and five regressions; the single net gain in overall accuracy came from the 31 shared-segment points, all Agricultural/Open (28 → 29 correct). A net improvement of two Built-up points, obtained after many trials on the same labels and at a high computational cost, was too small to justify replacing the 2015 map. X3 was recorded as an exploratory result, the frozen raster was kept as its only authoritative predictor source, and incremental tuning of context features on the 2015 validation labels was stopped.

8.4 RANDOM FOREST WITH GAIA-ASSISTED TRAINING

The earlier Random Forest benchmark (Section 4.7) had been less accurate than MLC with the original training data. If the real limitation was the representativeness of those data, a classifier given many more examples of the confused situations might do better, and GAIA offered a way of finding such examples automatically. Stage 08G-3 tested this for 2015 in seven steps (Fig. 8.2):

(1) **Overlay.** The locked 2015 Gaussian MLC map was compared cell by cell with GAIA's year of first imperviousness on the same 30 m grid.

(2) **Grouping.** The comparison defined four groups of eligible pixels (Table 8.3): pixels mapped as Agricultural/Open or Barren/Fallow that GAIA records as impervious by 2010, and pixels mapped as the same two classes that GAIA does not record as impervious by 2015. A pixel thus counted as Built-up evidence only if GAIA had seen it impervious well before the map year, and as open-land evidence only if GAIA had not seen it impervious even by the map year.

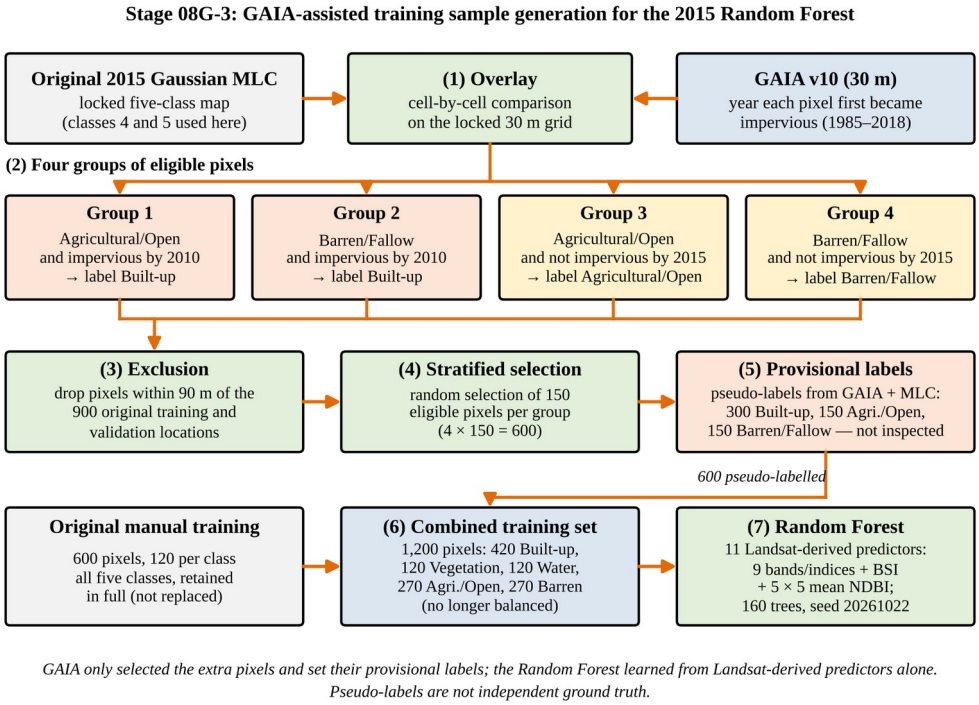
(3) **Exclusion.** Pixels within 90 m of any of the 900 original training and validation locations were excluded, so that no new sample lay in the immediate neighbourhood of an existing reference point.

(4) **Stratified selection.** 150 eligible pixels were selected at random from each group, so that the four groups contributed equally regardless of their sizes (600 pixels in total).

(5) **Provisional labels.** Pixels from groups 1 and 2 were labelled Built-up; pixels from groups 3 and 4 kept their MLC class. These are **pseudo-labels**: they were assigned by the agreement of two maps, not by inspecting imagery.

(6) **Combination.** The 600 pseudo-labelled pixels were added to all 600 original, manually labelled training pixels (Table 8.4); no original sample was removed or replaced.

(7) **Training.** The predictor values of all 1,200 pixels were extracted from the 2015 Landsat composite, and a new Random Forest was trained on them.



GAIA only selected the extra pixels and set their provisional labels; the Random Forest learned from Landsat-derived predictors alone. Pseudo-labels are not independent ground truth.

Fig. 8.2 Automatic generation of the GAIA-assisted training set for the 2015 Random Forest (stage 08G-3).

Drawn for this report from the stage 08G-3 record. Numbers in brackets correspond to the steps in the text.

Table 8.3 The four groups of GAIA-assisted pseudo-labelled pixels (stage 08G-3).

Group	Original MLC class	GAIA condition	Assigned training label	Samples
1	Agricultural/Open	Impervious by 2010	Built-up	150
2	Barren/Fallow	Impervious by 2010	Built-up	150
3	Agricultural/Open	Not impervious by 2015	Agricultural/Open	150
4	Barren/Fallow	Not impervious by 2015	Barren/Fallow	150
Total				**600**

Eligible pixels lie at least 90 m from all 900 original training and validation locations. Labels are provisional pseudo-labels, not verified ground truth.

Table 8.4 Composition of the combined 2015 Random Forest training set.

Class	Original manual	GAIA-assisted additional	Final
Built-up	120	300	420
Vegetation	120	0	120
Water	120	0	120
Agricultural/Open	120	150	270
Barren/Fallow	120	150	270
Total	**600**	**600**	**1,200**

All five original classes remain represented; the original 600 samples were kept in full. The combined set is not balanced.

Two distinctions matter for interpreting the result. GAIA was used only to choose the additional pixels and to set their provisional labels; the Random Forest never received GAIA as an input and learned from 11 Landsat-derived predictors: the nine bands and indices, a bare-soil index (BSI) and the 5 × 5 mean NDBI of Section 8.2. And because the original training data were kept in full, all five classes remained represented: the forest was not trained on the four automatically generated groups alone. It used 160 trees, minimum leaf population 3, bag fraction 0.7 and seed 20261022. The design had two costs, visible in Table 8.4: the combined set was no longer balanced (420 Built-up pixels against 120 each of Vegetation and Water), and half of its samples carried labels inferred from two other maps rather than verified ground truth.

On the original 2015 validation points RF reached 259/300 (86.33%) and raised Built-up recall from 49/60 to **55/60**, but Built-up false positives rose from one to **six** (user's accuracy 98.00% → 90.16%), and the paired comparison showed 14 points corrected and 12 worsened. Citywide, mapped Built-up rose from 200.88 to 318.36 km² (+117.47 km²): 126.39 km² became Built-up and 8.91 km² ceased to be. Of the new Built-up area, 94.11 km² was GAIA impervious by 2010, 6.85 km² converted in GAIA between 2011 and 2015, and 25.42 km² was not GAIA impervious by 2015. Its 79.9% overlap with historical GAIA is not independent support, because GAIA had generated the training labels. The 2015 RF map was exported as a documented alternative (`chennai_2015_08G3_gaia_assisted_RF_experimental_v1`) but not selected.

The same design failed for 2020 (08G-4; 21 scenes, positive labels from GAIA by 2015, negatives with no GAIA conversion through 2018). Overall accuracy fell from 259/300 (86.33%) to **237/300 (79.00%)**; Built-up recall fell from 52/60 to 43/60 while false positives rose from four to **31** (user's accuracy 92.86% → 58.11%), mostly Barren/Fallow points mapped as Built-up; 10 points were corrected and 32 worsened. Moving the positive-label cutoff to GAIA by 2018 (08G-4A) produced an identical validation matrix (equal

matrices do not prove identical maps). Both 2020 RF candidates were rejected and none was exported.

Two lessons followed. Retraining could raise Built-up recall, but only by also teaching the model to call bare and fallow land Built-up, and a design that helped in one year did not transfer to the next. Retraining the classifier again and again was therefore unlikely to give a defensible series; what was needed was a way to add Built-up only where several independent sources agreed, while leaving the rest of each map untouched.

8.5 FROM RETRAINING TO EVIDENCE-BASED FUSION

That requirement defines the fusion approach of Section 3.10 (Fig. 8.3). It differs from all the preceding experiments in kind: the Gaussian MLC maps are not retrained, no predictor is added and no class other than Built-up can be gained. The evidence for each year was chosen by what the archives can support. GAIA covers 1985–2018 at 30 m, so it is available for every year but ends in 2018 and cannot record demolition. A second, independent and year-matched product was then required: ESA WorldCover exists for 2020, Dynamic World (Sentinel-2) supplies frequent observations for 2025, and WSF 2015 matches 2015. No such 10 m product exists for 2010, which is why the 2010 test had to rely on GHSL's 100 m surface and on later-year information.

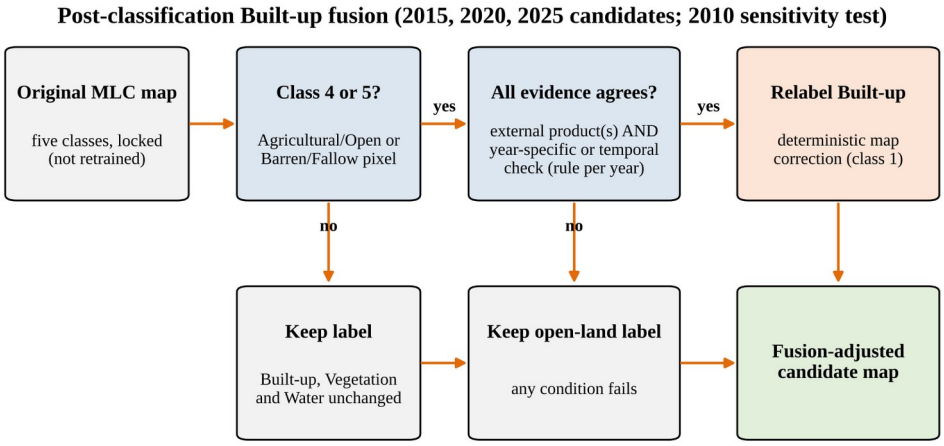


Fig. 8.3 How post-classification Built-up fusion works.

Drawn for this report from the stage 08G scripts. Evidence and thresholds for each year are listed in Table 3.2.

2020 (08G-4B, 08G-4C). The first fusion rule promoted original class 4 or 5 pixels where GAIA was impervious by 2015 and WorldCover 2020 Built-up covered at least 0.67 of the 30 m cell. It added 69.81 km² (44.68 from class 4, 25.13 from class 5), raised overall accuracy to 265/300 (88.33%) and Built-up to 58/60, with false positives unchanged at four; six points were corrected and none worsened. With the WorldCover test held fixed,

moving the GAIA cutoff to 2018 added only 1.21 km² more, corrected one further point (266/300, 88.67%; Built-up 59/60) and again worsened none. This GAIA-2018 + WorldCover rule became the 2020 candidate (296.74 km² Built-up).

2015 (08G-5, 08G-6, 08G-9). For 2015 the analogous rule used GAIA by 2015 and WSF 2015 (after a first run failed because the WSF band is named `WSF`, not `settlement`). It was far too permissive: it added 143.73 km², about 93.5% of all GAIA disagreement in 2015, so WSF was filtering very little. Built-up recall reached 59/60, but false positives rose from one to 12, overall accuracy fell to 256/300 (85.33%), and ten corrections were offset by eleven new errors. It was rejected. A stricter, pre-specified screen that also required GHSL 2015 built-up surface of at least 6,000 m² per 100 m cell went to the other extreme: it removed 141.52 of the 143.73 km², kept 2.21 km², and recovered only two Built-up points (259/300). It was not selected. The rule finally retained added a **temporal-consistency** condition instead: a 2015 class 4 or 5 pixel is promoted only if it passes the broad GAIA + WSF test **and** is Built-up in the corrected 2020 map. This added 91.36 km² (64.95 from class 4, 26.41 from class 5), filtered out 52.37 km² that 2020 did not support, raised overall accuracy to 266/300 (88.67%) and Built-up to 58/60 with false positives unchanged at one, and corrected nine points while worsening none. Original 2015 Built-up was always kept, including 13.51 km² that is not Built-up in the 2020 candidate.

The retained 2015 rule makes the order of work essential (Fig. 8.4). The 2020 map had to be corrected first, from its own evidence: an original 2020 class 4 or 5 pixel became Built-up where GAIA was impervious by 2018 and ESA WorldCover 2020 Built-up covered at least 0.67 of the 30 m cell. Only then could 2015 be corrected, because the corrected 2020 map is one of the 2015 evidence layers: a 2015 class 4 or 5 pixel became Built-up only where GAIA was impervious by 2015, WSF 2015 covered at least 0.67 of the cell **and** the same cell was Built-up in the corrected 2020 candidate. In both stages only Agricultural/Open and Barren/Fallow pixels could be promoted, every other pixel kept its MLC label, and neither the Gaussian MLC nor the Random Forest was retrained. The dependency is also the method's main weakness: part of the 2015 evidence is 2020 evidence, so the corrected 2015 and 2020 maps are not independent, and Built-up land missing from the 2020 candidate cannot be recovered in 2015 by this rule.

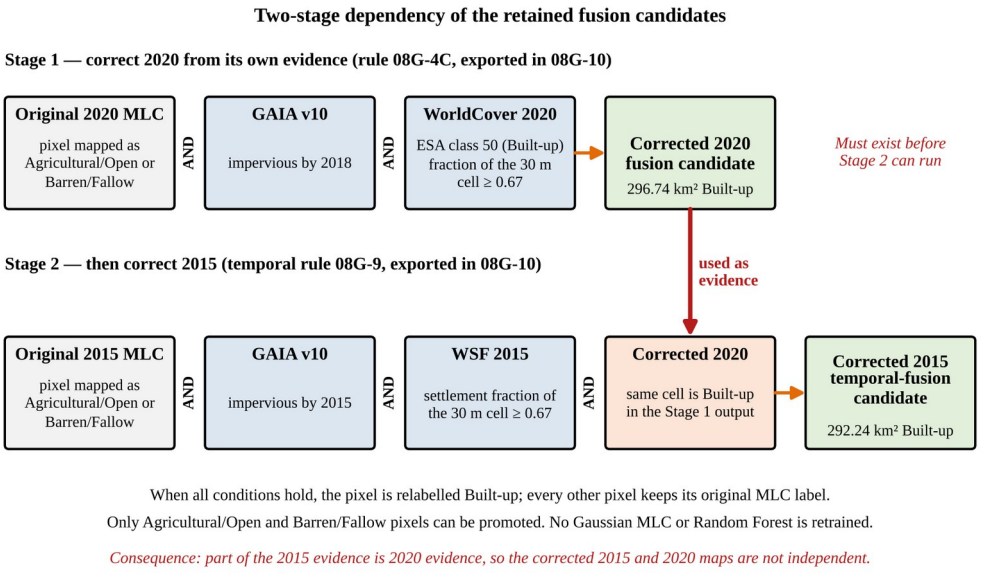


Fig. 8.4 Two-stage dependency of the retained 2020 and 2015 fusion candidates.

Drawn for this report from the selected 2020 rule (08G-4C, implemented in 08G-10) and the 2015 temporal rule (08G-9). Thresholds as in Table 3.2; Built-up areas from Table 8.9.

2025 (08G-7, 08G-8). For 2025 the second source was Dynamic World: a 10 m pixel counted as built if its median built probability over the 79 images of 2025 was at least 0.65 with at least six valid observations, and a 30 m cell passed if at least 0.67 of it did; GAIA impervious by 2018 was also required. The rule added 86.23 km² (29.11 from class 4, 57.12 from class 5), giving 342.60 km². A further branch that would have added Built-up without GAIA support, where Dynamic World was very confident (median ≥ 0.80 over ≥ 0.89 of the cell), added nothing. The 2025 candidate therefore recovers GAIA-supported areas that Dynamic World still sees as built in 2025; it does not separately capture expansion after 2018.

Table 8.5 Fusion variants tested, with paired results on each year's validation points.

Year · stage	Rule (promote class 4/5 → Built-up if …)	Added (km²)	Built-up (km²)	OA	BU correct	BU FP	Fixed / worse	Decision
2020 · 4B	GAIA by 2015 ∧ WorldCover ≥ 0.67	69.81	295.53	88.33%	58/60	4	6 / 0	Superseded
2020 · 4C	GAIA by 2018 ∧ WorldCover ≥ 0.67	71.02	296.74	88.67%	59/60	4	7 / 0	Retained candidate
2015 · 5	GAIA by 2015 ∧ WSF ≥ 0.67 (broad)	143.73	344.61	85.33%	59/60	12	10 / 11	Rejected
2015 · 6	broad ∧ GHSL ≥ 6,000 m² (strict)	2.21	203.09	86.33%	51/60	1	2 / 0	Rejected
2015 · 9	broad ∧ Built-up in 2020 candidate (temporal)	91.36	292.24	88.67%	58/60	1	9 / 0	Retained candidate
2025 · 7/8	GAIA by 2018 ∧ Dynamic World ≥ 0.65 over ≥ 0.67	86.23	342.60	84.67%‡	58/60	3	4 / 0	Retained candidate
2025 · 7	no GAIA ∧ Dynamic World ≥ 0.80 over ≥ 0.89	0.00	—	—	—	—	—	No additions

Original MLC values on the same points: 2015 257/300, BU 49/60, FP 1; 2020 259/300, BU 52/60, FP 4; 2025‡ 250/300, BU 54/60, FP 3 on the new polygon-derived sample (Section 3.10). BU = Built-up; FP = other classes mapped as Built-up.

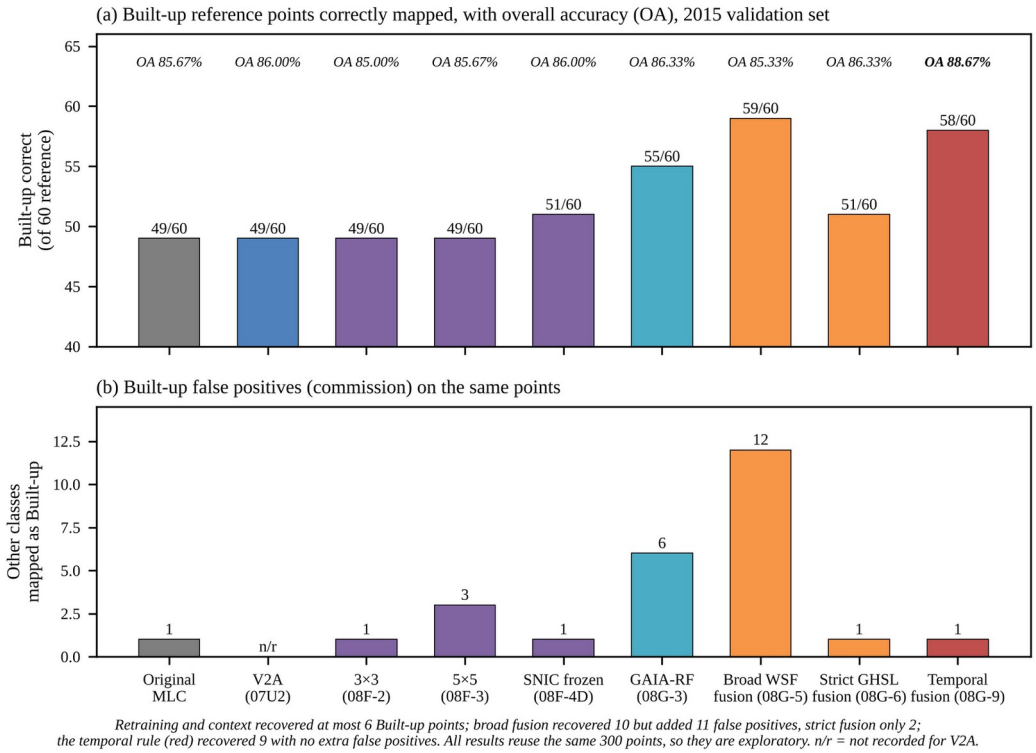


Fig. 8.5 The sequence of 2015 experiments on the same validation points.

Plotted for this report from the stage 07U2, 08F and 08G outputs. Built-up false positives for V2A were not recorded.

Fig. 8.5 places all the 2015 experiments side by side. Revised training and context features left Built-up recall at 49–51 of 60; GAIA-assisted RF reached 55 at the cost of five extra false positives; broad fusion reached 59 at the cost of eleven; strict fusion barely moved; and the temporal rule recovered nine Built-up points with no new false positive. That pattern, not the overall accuracy alone, is the reason the temporal rule was retained.

8.6 VALIDATION OF THE RETAINED CANDIDATES

The three retained rules were exported as separate candidate assets in stage 08G-10, leaving every original raster untouched (Appendix A), and their areas were re-measured from the reloaded rasters (08G-11, 08G-13). Table 8.6 and Fig. 8.6 summarise their paired validation; the full matrices are in Appendix B.

Table 8.6 Paired validation of the retained fusion candidates against the original MLC maps.

Year	Validation points	Original OA	Fused OA	BU correct (orig. → fused)	BU FP (orig. → fused)	Fixed / worsened
2015	Original 300 (60 per class)	257/300 (85.67%)	266/300 (88.67%)	49 → 58	1 → 1	9 / 0
2020	Original 300 (60 per class)	259/300 (86.33%)	266/300 (88.67%)	52 → 59	4 → 4	7 / 0
2025‡	New 300 from 33 master polygons (seed 84)	250/300 (83.33%)	254/300 (84.67%)	54 → 58	3 → 3	4 / 0

‡The archived 2025 sample was not recovered; the official 2025 MLC accuracy remains 83.67% on that sample. The new points cluster within 33 polygons. All points were used during method development, so these accuracies are exploratory, not independent.

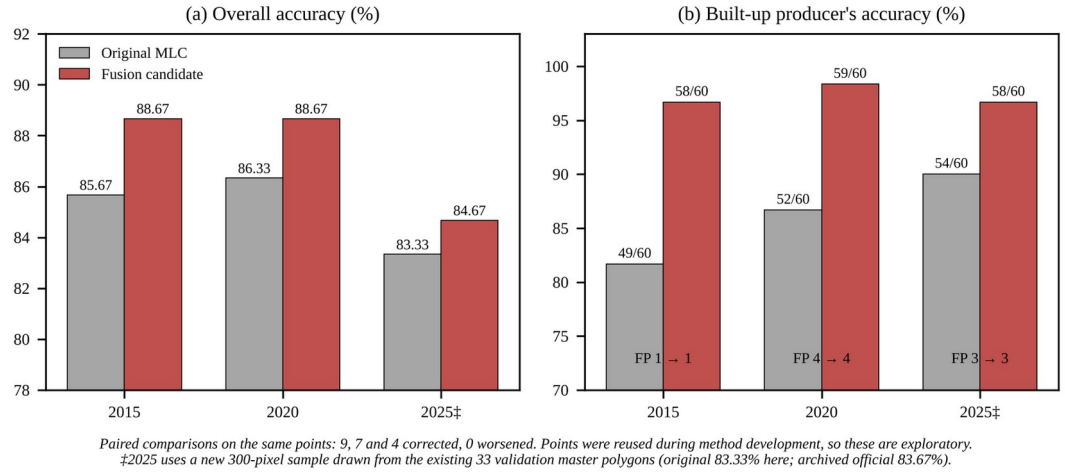


Fig. 8.6 Overall accuracy and Built-up producer's accuracy, original and fusion-adjusted maps.

Plotted for this report from Table 8.6.

In all three years fusion recovered Built-up points without adding false positives on these samples, and no point that the original map had right was made wrong. The improvement is concentrated where it was intended: other classes are unchanged by construction, so the Water and open-land errors of the original maps remain (in 2020, for example, 27 Barren/Fallow points are still mapped as Agricultural/Open). Three qualifications apply. The points had been used to compare rules, so the figures are optimistic. The 2025 sample is new, but it comes from the same 33 polygons as the earlier sample and is spatially clustered. And because the 2015 rule uses the corrected 2020 map, the 2015 and 2020 candidates are not independent of each other.

8.7 THE 2010 SENSITIVITY TEST

Before 2010 could be treated in the same way, the correct 2010 raster had to be confirmed. An early version of the analysis script had loaded `chennai_mlc_lulc_2010`, which gave only 127.52 km² of Built-up. Stage 08G-12 therefore scored all four same-grid 2010 rasters on the 300 original 2010 validation points (Table 8.7). Only

`chennai_mlc_lulc_2010_L5L7_manual_exp_v1` matched the documented lineage and accuracy of the provisional map, and it was confirmed on that basis, not because of its area. (The figures for `chennai_mlc_lulc_2010` — 40.67%, 22 Built-up points correct, 12 false positives — coincide with those of the early 05F1 classifier in Section 4.5.)

Table 8.7 Same-grid 2010 rasters scored on the 300 original 2010 validation points (stage 08G-12).

Asset	Built-up (km²)	OA	BU correct	BU FP	Status
chennai_mlc_lulc_2010	127.52	40.67%	22/60	12	Archive
chennai_mlc_lulc_2010_manual_v1	270.92	59.33%	54/60	18	Landsat 5 manual
chennai_mlc_lulc_2010_v2c	108.97	44.00%	16/60	4	Archive
chennai_mlc_lulc_2010_L5L7_manual_exp_v1	**279.82**	**69.33%**	**55/60**	**12**	Provisional map

2010 has no year-matched 10 m Built-up product, so the 2010 test (08G-15) combined GAIA impervious by 2010, GHSL 2010 built-up surface at 100 m, and the corrected 2015 map as later-year evidence (itself dependent on 2020). Two variants were run in memory only. The **add-only** rule promoted 2010 class 4 or 5 pixels where GAIA was impervious by 2010, the GHSL 2010 cell had built-up surface and the 2015 candidate was Built-up. The **two-way** rule did the same and also demoted 2010 Built-up pixels to the 2015 candidate's class 4 or 5 where GAIA was not impervious by 2010, the GHSL cell had no built-up surface and the 2015 candidate was open land. The demotion assigns a later-year label, not a verified 2010 one; its purpose was to remove likely commission errors.

Table 8.8 The 2010 sensitivity test on the 300 original 2010 validation points (stage 08G-15).

2010 version	OA	BU correct	BU FP	BU recall	BU precision	Built-up (km²)	Status
Original L5/L7 MLC	208/300 (69.33%)	55/60	12	91.67%	82.09%	279.82	Retained (exported)
Add-only fusion	209/300 (69.67%)	57/60	13	95.00%	81.43%	308.47	Rejected
Two-way fusion	214/300 (71.33%)	57/60	7	95.00%	89.06%	304.72	Retained (not exported)

Two-way: +28.65 km² promoted (24.77 from class 4, 3.88 from class 5) and 3.75 km² demoted (2.07 to class 4, 1.69 to class 5); net +24.89 km². Paired against the original: add-only 2 corrected / 1 worsened; two-way 7 corrected / 1 worsened.

The two-way version is the better of the two on these points: it recovers two more Built-up points and removes five false positives. It also reverses the direction of the first interval. With the original 2010 map, mapped Built-up rises from 279.82 km² (2010) to 292.24 km² (2015); with the two-way map it falls from 304.72 to 292.24 km². Choosing the original

because it gives a monotonic series, or the two-way map because it scores two points higher on reused validation data, would both be choices made for the wrong reason. Both are therefore retained: the exported original as the 2010 map of the reported series, and the subsequently reconstructed and exported two-way result as a documented sensitivity case (Fig. 8.10). The add-only version was not preferred, since it gained one point overall while adding a false positive.

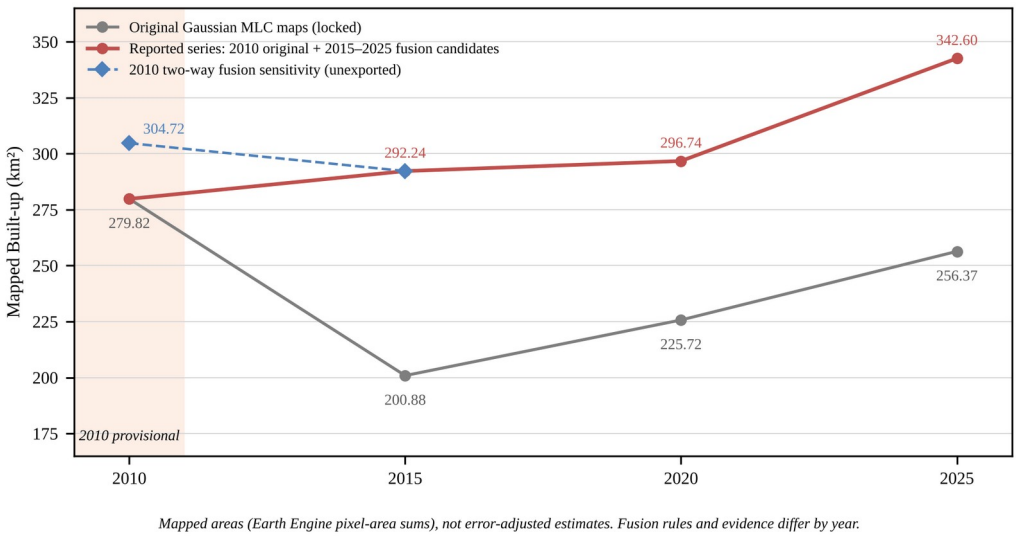
8.8 FUSION-ADJUSTED BUILT-UP AREAS AND TRANSITIONS

Table 8.9 gives the five-class areas of the reported series — the original 2010 map and the three fusion-adjusted candidates — computed on the same grid and study area (stage 08G-13), and Fig. 8.7 compares its Built-up curve with that of the original maps. Mapped Built-up rises in every interval: by 12.42 km² in 2010 → 2015, 4.50 km² in 2015 → 2020 and 45.86 km² in 2020 → 2025, a total of 62.78 km² (+22.44%). With the two-way 2010 map the first interval becomes −12.47 km² and the 2010 → 2025 total +37.89 km² (Table 8.10). The direction of 2010 → 2015 change is therefore **not robust** to the choice of 2010 method; the overall 2010 → 2025 increase is positive under both.

Table 8.9 Five-class mapped areas of the reported series (km²; stage 08G-13).

Class	2010* original	2015 fusion	2020 fusion	2025 fusion
Built-up	279.82	292.24	296.74	342.60
Vegetation	81.22	33.65	14.11	44.27
Water	21.78	39.73	22.04	20.39
Agricultural/Open	114.46	103.15	132.43	58.20
Barren/Fallow	32.49	60.99	64.46	64.31
Valid area	**529.77**	**529.77**	**529.77**	**529.77**

*Provisional. Earth Engine pixel-area sums over exported rasters. The 2010 two-way alternative (304.72 km² Built-up) was subsequently reconstructed and exported for documentation (Fig. 8.10); it is not substituted into the class-area table or transition analysis reported here.



Mapped areas (Earth Engine pixel-area sums), not error-adjusted estimates. Fusion rules and evidence differ by year.

Fig. 8.7 Mapped Built-up area: original MLC maps and the fusion-adjusted reported series.

Plotted for this report from Table 8.9, the original-map areas of Chapter 6 and Table 8.8.

Table 8.10 Built-up change in the reported series (km²).

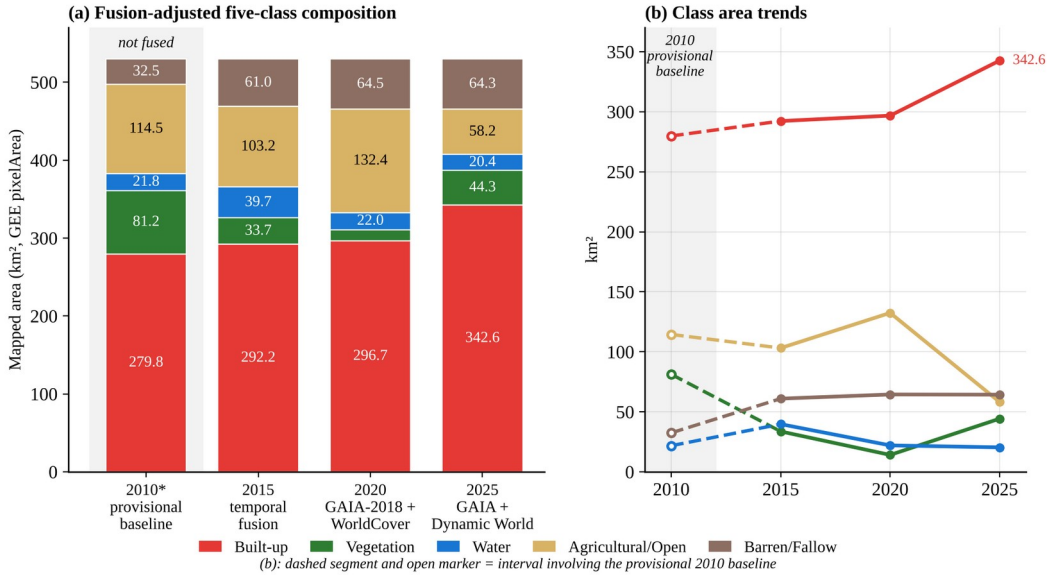
Interval	Gross gain	Gross loss	Net (2010 original)	Net (2010 two-way)
2010 → 2015	51.50	39.08	+12.42	−12.47
2015 → 2020	18.00	13.51	+4.50	+4.50
2020 → 2025	62.39	16.53	+45.86	+45.86
2010 → 2025			**+62.78**	**+37.89**

Gross flows are from the 08G-13 transition matrices (Appendix B) and exist only for the original-2010 version. Mapped quantities, not error-adjusted estimates.

These are mapped differences between maps built by different rules, and the small first two increases are well within the uncertainty of the maps. Part of the 2015→2020 agreement is built in, because 2015 promotions required Built-up in 2020. The other four classes are examined below. The full 25-cell transition matrices are given in Appendix B as **post-classification mapped transitions**.

The full fusion-adjusted composition. Built-up was the target of the correction, but the retained candidates are complete five-class maps, so the composition of all classes after fusion was also examined. Fig. 8.8 is the post-fusion counterpart of the original-results figure (Fig. 6.1). It plots the five-class areas of Table 8.9 — the provisional original 2010 baseline and the three retained candidates — as a stacked composition and as class trajectories, and summarises them in a compact table. Because fusion moves pixels only from Agricultural/Open and Barren/Fallow to Built-up, Vegetation and Water are identical to the original maps in every year, and the whole correction appears as a transfer from the two open-land classes to Built-up (compare Table B.2). The figure therefore shows the full

LULC trajectory that results from the Built-up omission correction, not only its Built-up curve.



(c) Summary of the after-fusion series (km²)

Class	2010*	2015	2020	2025	Net 2010–2025
Built-up	**279.82**	**292.24**	**296.74**	**342.60**	**+62.78**
Vegetation	81.22	33.65	14.11	44.27	−36.95
Water	21.78	39.73	22.04	20.39	−1.39
Agricultural/Open	114.46	103.15	132.43	58.20	−56.26
Barren/Fallow	32.49	60.99	64.46	64.31	+31.82

**2010 = original L5/L7 Gaussian MLC (provisional baseline, not fusion-adjusted). 2015–2025 = retained fusion-adjusted candidates. Only Built-up was corrected: Vegetation and Water equal the original maps; the correction appears as a transfer from classes 4 and 5 to Built-up. Built-up net by interval: +12.42, +4.50, +45.86 km². 2010–2015 becomes −12.47 km² with the two-way 2010 alternative (Section 8.7).*

Fig. 8.8 Fusion-adjusted LULC composition and area trends across Chennai, 2010–2025.

After-fusion (corrected) series, separate from the original classification trend of Fig. 6.1. 2010: original L5/L7 Gaussian MLC (chennai_mlc_lulc_2010_L5L7_manual_exp_v1), provisional baseline, not fused. 2015: temporal-fusion candidate. 2020: GAIA-2018 + WorldCover candidate. 2025: GAIA + Dynamic World candidate. Five-class Earth Engine pixel-area sums from Table 8.9 (stage 08G-13); mapped areas, not error-adjusted estimates.

Table 8.11 extends the Built-up gain and loss analysis of Table 8.10 to all five classes and all three intervals. It uses Eq. (3.9) and the five-class transition matrices of the same series computed in stage 08G-13 (Table B.6); it is not derived from the four-class matrices of Chapter 6, which describe the original maps, nor from annual totals, which give only net change. Each class's net change equals the difference between its mapped areas in Table 8.9, and the net changes of the five classes sum to zero within rounding (−0.01, +0.02 and −0.01 km² in the three intervals), as they must on a fixed footprint.

Table 8.11 Gross gain, gross loss and net change of all five classes in the fusion-adjusted series (km²; stage 08G-13).

Interval	Class	Gross gain (km²)	Gross loss (km²)	Net change (km²)
2010 → 2015*	Built-up	51.50	39.08	+12.42
	Vegetation	13.38	60.95	−47.57
	Water	20.55	2.60	+17.95

Interval	Class	Gross gain (km²)	Gross loss (km²)	Net change (km²)
	Agricultural/Open	59.36	70.67	−11.31
	Barren/Fallow	49.96	21.46	+28.50
2015 → 2020	Built-up	18.00	13.50	+4.50
	Vegetation	3.46	23.00	−19.54
	Water	2.12	19.81	−17.69
	Agricultural/Open	60.73	31.45	+29.28
	Barren/Fallow	31.90	28.43	+3.47
2020 → 2025	Built-up	62.39	16.53	+45.86
	Vegetation	33.34	3.18	+30.16
	Water	3.79	5.44	−1.65
	Agricultural/Open	21.86	96.09	−74.23
	Barren/Fallow	38.20	38.35	−0.15

*Involves the provisional 2010 baseline. Gain = column total − diagonal; loss = row total − diagonal, from the two-decimal matrices of Table B.6 (agreeing with sums of the off-diagonal cells within 0.01 km²; Built-up values agree with Table 8.10 within rounding). Mapped class differences, not independently verified physical conversions. No transition analysis is reported for the subsequently exported 2010 two-way sensitivity alternative.

Read together, Fig. 8.8 and Table 8.11 show the following mapped trajectories. **Built-up** grows in every interval (+12.42, +4.50 and +45.86 km²), from 52.8% of the mapped area in 2010 to 64.7% in 2025, but its gross flows are large relative to the net: in 2010→2015, 51.50 km² entered and 39.08 km² left Built-up, a turnover that largely reflects disagreement between a provisional 2010 map and a corrected 2015 map, and this first interval reverses under the two-way 2010 alternative (Section 8.7). **Agricultural/Open** carries most of the correction and most of the volatility: it gains 60.73 km² and loses 31.45 km² in 2015→2020, then loses 96.09 km² in 2020→2025, 40.63 km² of it to Built-up and 25.49 km² to Vegetation, ending at 58.20 km², about half of its 2010 area. **Barren/Fallow** nearly doubles between 2010 and 2015 (+28.50 km²) and then stays at 61–64 km², although 28–38 km² enters or leaves it in each later interval, much of it exchanged with Agricultural/Open. **Vegetation** falls from 81.22 km² in 2010 to 14.11 km² in 2020 and recovers to 44.27 km² in 2025, its 2020→2025 gain coming mainly from Agricultural/Open (25.49 km²). **Water** nearly doubles in 2015 (39.73 km²) and returns to 20–22 km² in 2020 and 2025, the 2015→2020 loss going mainly to Agricultural/Open (13.84 km²).

Apart from Built-up, these classes were not corrected, so their swings carry the original classification errors, the Agricultural/Open–Barren/Fallow confusion (Section 5.3) and the seasonal effects of annual composites (Section 10.2). They are better read as differences between classifications than as ecological change, and none of them is a verified conversion on the ground.

A discontinued alternative. At one stage a parallel 'Analysis B' summed GHSL built-up surface over the study area for each epoch, as an independent trend. It was removed from the final work: it involved no classification of our own, it restated a modelled product (whose 2025 epoch is itself a projection), and the project's purpose was to evaluate its own land-cover maps. Its values are not reported as a result. GHSL nevertheless remains part of the record as evidence in two fusion experiments: the rejected strict 2015 screen and the 2010 sensitivity test.

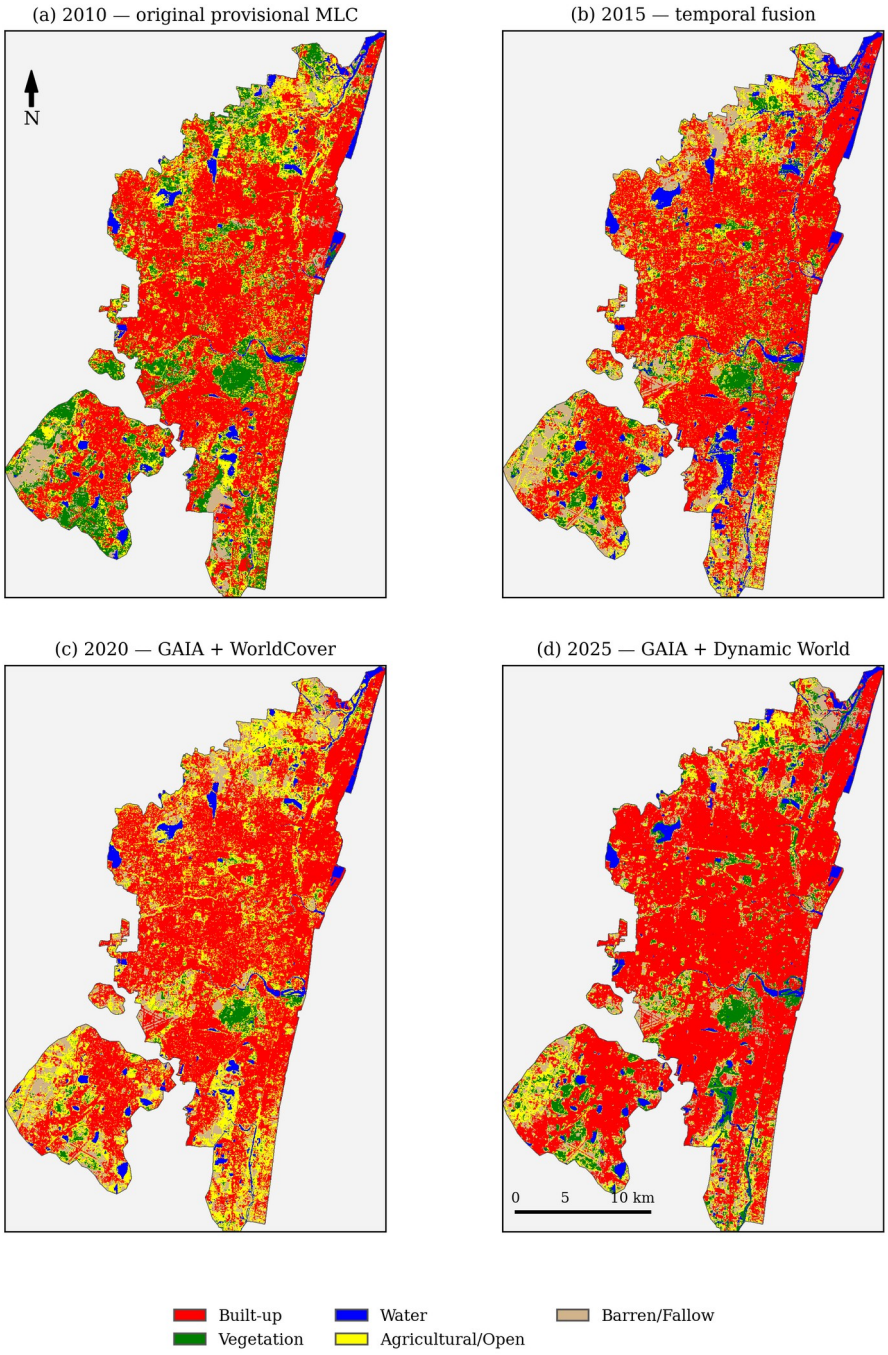


Fig. 8.9 Reported-series LULC maps: original provisional 2010 MLC and retained 2015–2025 fusion classifications.

Source: supplied exported class-code GeoTIFFs; 30 m, EPSG:32644, nearest-neighbour display. White/background pixels outside the mapped study area are excluded.

Interpretation. The common boundary, north-up orientation and five-class legend permit direct comparison of the reported-series maps. The later panels are post-classification corrections of the existing Gaussian MLC outputs, not newly trained classifiers. Their spatial patterns document the retained classifications; they do not independently establish physical land-cover change.

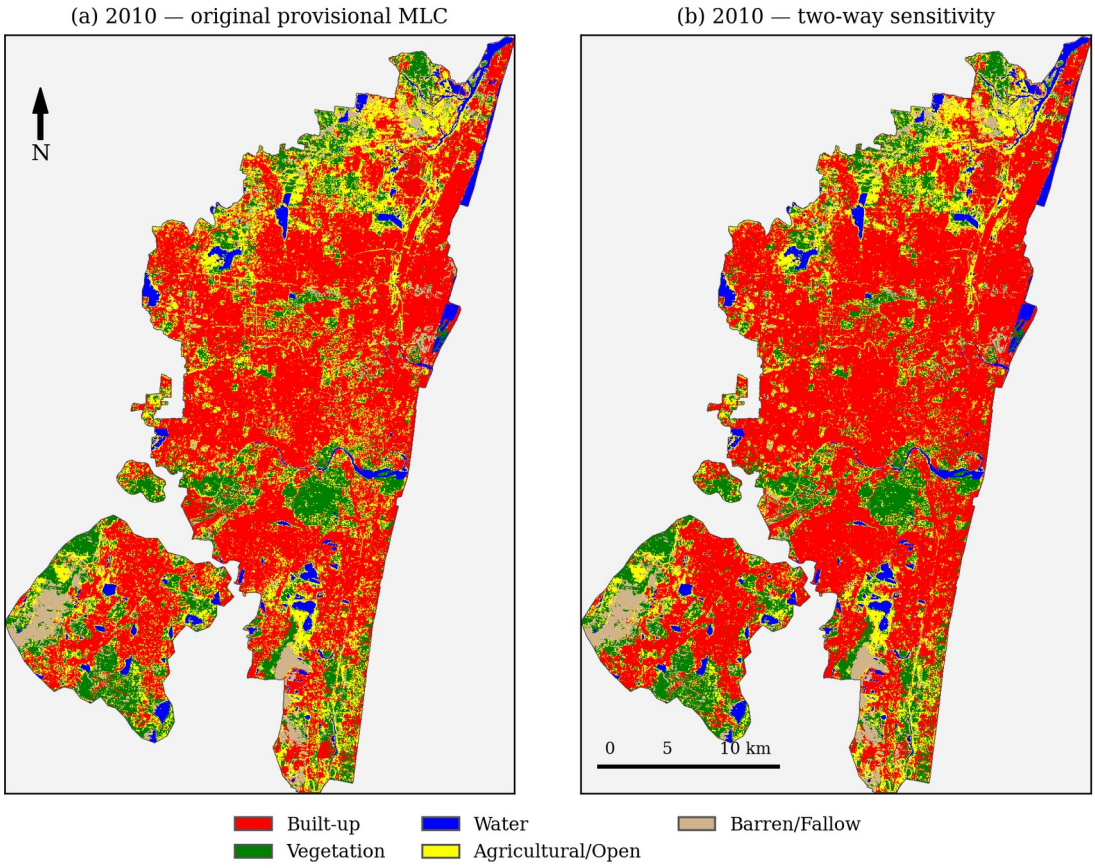


Fig. 8.10 Original 2010 provisional MLC and reconstructed two-way fusion sensitivity alternative.

Source: supplied exported class-code GeoTIFFs; 30 m, EPSG:32644, nearest-neighbour display. White/background pixels outside the mapped study area are excluded.

Interpretation. The two-way map is an exploratory sensitivity alternative, reconstructed using Stage 08G-15 and subsequently exported for documentation. The archived accuracy changes from 69.33% to 71.33%, and mapped Built-up from 279.82 to 304.72 km². The original map remains the official 2010 baseline; the alternative is not substituted into the reported series or transition matrices.

8.9 WHAT THE CORRECTION DOES AND DOES NOT SHOW

The fusion candidates are better records of Built-up land than the original maps at the validation points, and they remove the implausible collapse of Built-up after 2010 without any rule having been tuned to produce a rising curve. They do not prove how much Chennai built. The evidence differs from year to year (WSF, WorldCover, Dynamic World, GHSL), as do the resolutions and definitions behind it; the 2015 candidate depends on the 2020 candidate, and the 2010 alternative on 2015; the original 2010 classifier is weak (69.33%); validation points were reused across many experiments; the 2025 sample is new but clustered; and areas and transitions are mapped quantities without error adjustment or

confidence intervals [15]. The candidates are therefore reported as **fusion-adjusted map estimates** — experimental corrections — alongside, not instead of, the locked MLC maps.

Conclusion of the correction. Systematic Built-up omission in the 2015, 2020 and 2025 maps was confirmed at audited sites and by external evidence. Retraining with more examples, context features, segmentation or GAIA-labelled Random Forest did not correct it reliably. Rule-based fusion with year-specific evidence recovered 4–9 Built-up validation points per year without new false positives, giving mapped Built-up of 279.82 (2010), 292.24, 296.74 and 342.60 km². The first interval reverses under the equally documented 2010 two-way alternative (304.72 km²), so only the overall 2010–2025 increase is robust to the 2010 choice, and its magnitude remains uncertain.

CHAPTER 9
SYSTEM ARCHITECTURE AND INTERACTIVE GIS APPLICATION

9.1 PURPOSE AND ARCHITECTURE

Static figures cannot show a 30 m map of 529 km² at full resolution or let a reader compare years at one location, and repeated interactive requests during development ran into Earth Engine usage limits. After the final rasters were exported, a local application was therefore built to serve them directly. Fig. 9.1 shows the two layers. The analysis layer is the Earth Engine pipeline of Chapters 3 and 4, which produces the four-class rasters, the four annual true-colour composites and the study-area boundary. The application layer reads these files locally: a Python FastAPI backend serves map tiles and statistics [18], [19], and a React and Leaflet frontend displays them [20], [21]. Categorical tiles are resampled by nearest neighbour so that no fractional class values are created, and all published areas are the Earth Engine values rather than local pixel counts.

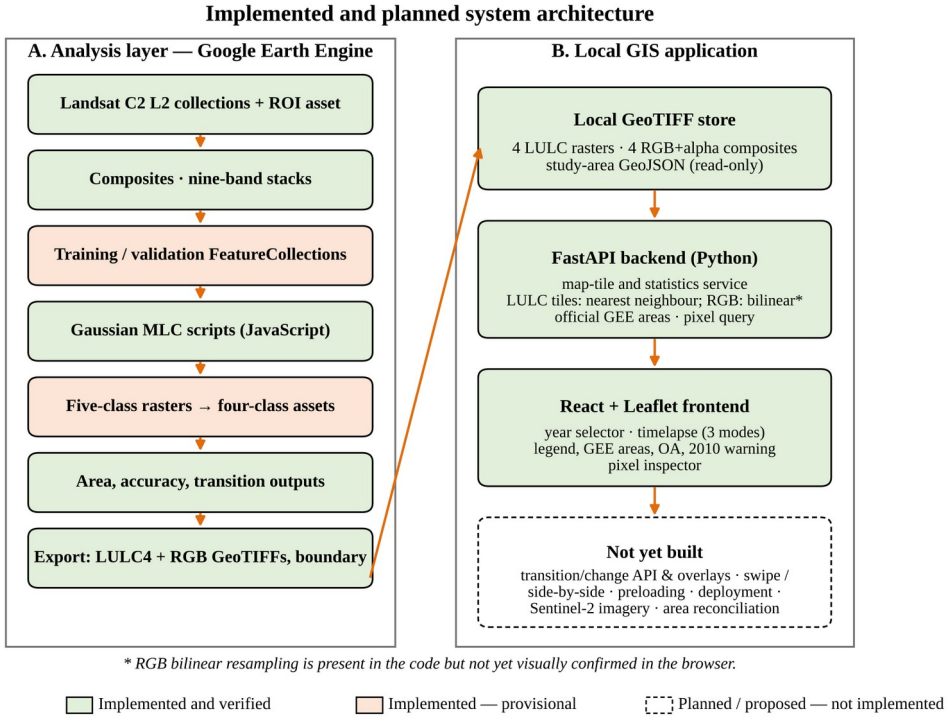


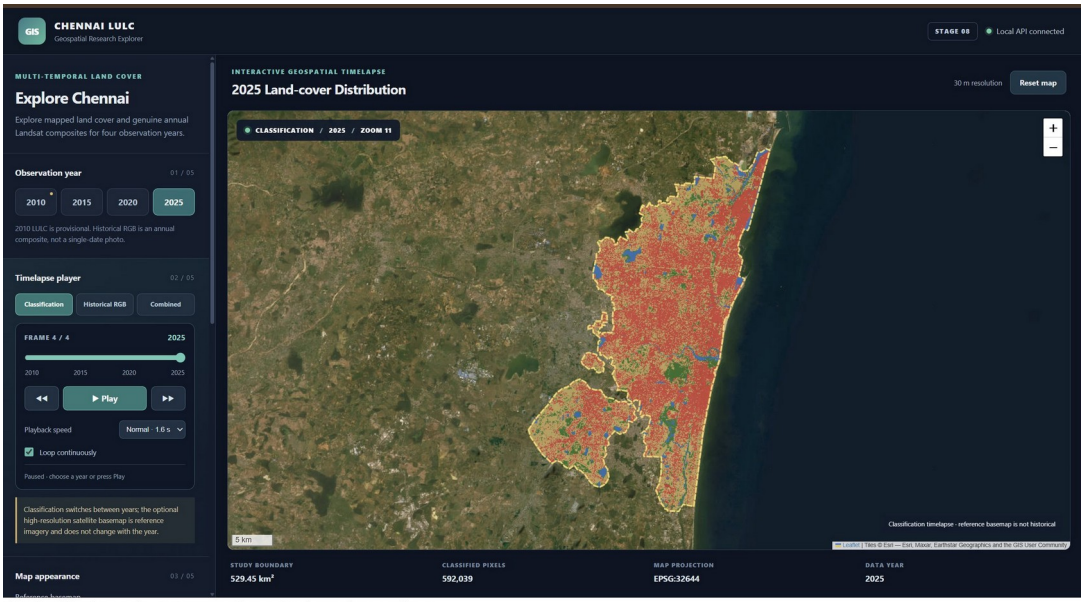
Fig. 9.1 Implemented and planned system architecture.

Drawn for this report from the application repository and handoff notes. Dashed: not implemented.

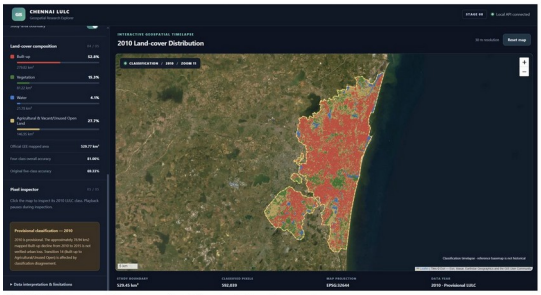
The application currently serves the locked four-class MLC maps described in Chapters 4 –6. The fusion-adjusted candidates of Chapter 8 have not yet been connected to it; adding them would require separately named layers with their method tags, and the subsequently exported 2010 two-way result would remain a separately identified sensitivity layer.

9.2 INTERFACE AND TIMELAPSE MODES

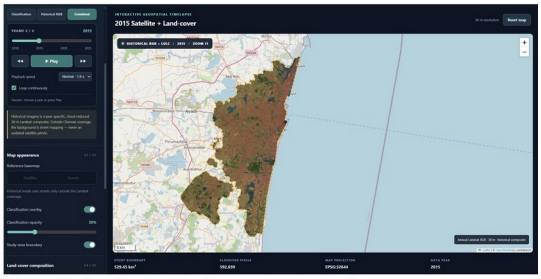
The interface (Fig. 9.2) combines a control panel with a large map. The panel holds a year selector with 2010 marked provisional, a four-frame timelapse player, map controls, the land-cover composition with official areas and accuracies, a pixel inspector and a limitations section. Three timelapse modes are available. **Classification** shows the LULC map over a modern reference basemap, which is labelled as not historical. **Historical RGB** shows the year's own annual Landsat composite, with street mapping outside the study area so that no undated imagery is presented as historical. **Combined** overlays the classification on that composite at adjustable opacity.



(a) Classification mode, 2025: year selector, timelapse player and map



(b) 2010 composition panel with provisional warning



(c) Combined mode, 2015 (classification at 30% opacity)

Fig. 9.2 The local GIS application.

Screenshots of the running application supplied by the author. Reference basemap © Esri and contributors; street mapping © OpenStreetMap contributors.

9.3 COMPLETED AND PLANNED FEATURES

Table 9.1 Application features by status.

Status	Features
Completed and verified	Local ingestion and checking of all rasters; tile and statistics service with passing integration tests; Classification, Historical RGB and Combined modes; timelapse playback; official areas and accuracies; 2010 provisional warning; pixel inspector
Implemented, not yet confirmed	Smooth (bilinear) resampling of the true-colour tiles; layout on small screens
Planned	Fusion-adjusted candidate layers with method tags; change overlays and interactive transition matrices; swipe or side-by-side comparison; reconciliation of local and Earth Engine areas; public deployment

CHAPTER 10

GEOGRAPHICAL GENERALISABILITY OF THE CHENNAI GAUSSIAN MLC

The Chennai maps answered how one workflow behaved across years within a fixed study area. A later experiment asked a different question: could the fitted 2020 Gaussian MLC classify an unseen region without learning any local training labels? Direct transfer to a Thanjavur city pilot performed poorly. Two preprocessing-only experiments then tested whether differences in observation timing and reflectance distributions could explain part of that failure. A final automated screening of three coastal cities examined cross-model agreement, with a different and explicitly weaker reference basis.

10.1 WHY AN EXTERNAL BENCHMARK WAS NEEDED

The related study by Nandi et al. [1] concerned Berhampore and included forecasting, whereas this project developed a historical Chennai mapping series. The project did not first reproduce the paper's exact study region, reference data and experimental protocol. The Thanjavur extension therefore provides an external geographical stress test of our Chennai-trained classifier; it does not establish an exact replication of the base paper or a controlled numerical comparison with its reported performance.

The purpose was to measure the limits of transfer, including an unsuccessful result. The five original classes were retained throughout: Built-up, non-crop Vegetation, Water, Agricultural/Open and Barren/Fallow. This experiment is separate from the four-class application layers and the Built-up fusion candidates in Chapter 8. No target result changed a locked Chennai map.

> **Interpretation.** This is an external stress test of the Chennai model, not a reproduction of the Berhampore paper. Its contribution is to establish where transfer fails and what limited recovery is possible.

10.2 WHAT WAS HELD FIXED

The source was the original 2020 Chennai MLC, fitted to 600 training pixels, 120 per class. Each class retained its nine-dimensional mean and sample covariance, with a 10^{-6} diagonal regularisation and equal priors (Section 3.7). The nine predictors remained blue, green, red, NIR, SWIR1, SWIR2, NDVI, NDWI and NDBI, in that order. No Thanjavur or coastal-city label re-estimated these Gaussian parameters.

All transfer runs used 2020 Landsat 8 Collection 2 Tier 1 Level 2 surface reflectance, QA_PIXEL bits 0–5 and QA_RADSAT masking, optical scaling DN × 0.0000275 − 0.2, and a 30 m EPSG:32644 analysis. The source validation result was 259/300 = 86.33% on

the independent, balanced Chennai sample. That reference-class-balanced design differs from the map-stratified Thanjavur design below, so their percentages are not a like-for-like estimate of geographical accuracy loss.

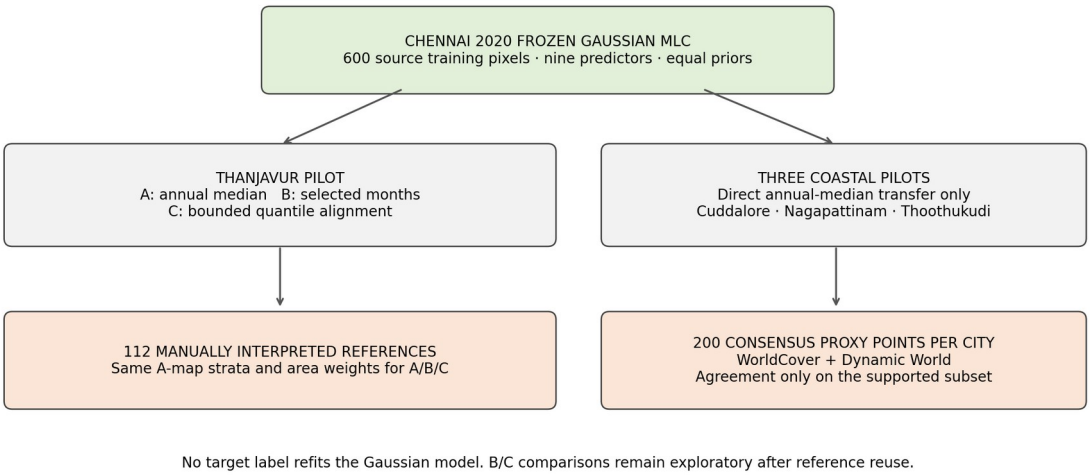


Fig. 10.1 Frozen-model transfer and the two distinct reference pathways.

Drawn from the benchmark methods. Only Thanjavur receives A/B/C comparison; the three-city branch applies direct frozen transfer and measures agreement with automated products.

> **Interpretation.** The learned decision rule is held constant. Changes in B and C therefore measure input-preprocessing sensitivity rather than the benefit of retraining on local labels.

10.3 THE THANJAVUR PILOT AND THE FIRST TRANSFER RESULT

Thanjavur (Tanjore) city was selected as the unseen target, rather than the whole district. An authoritative municipal polygon was unavailable, so a screenshot outline was traced into a provisional pilot boundary of 27.1007 km². Its valid classified raster area was 26.9913 km², approximately 99.60% of that vector area. The same boundary and valid area were retained for experiments A, B and C. It should not be described as the official corporation boundary.

Experiment A applied the unchanged Chennai model directly to the full-year target median. Thanjavur had 64 intersecting Landsat scenes, against 21 at Chennai, yet the resulting map assigned only 1.8732 km² (6.94%) to Built-up and 0.0106 km² (0.039%) to Vegetation. Agricultural/Open and Barren/Fallow occupied 50.66% and 41.03% of the classified area. These are model assignments, not verified land-cover proportions. Their spatial pattern prompted reference checking rather than acceptance of the map.

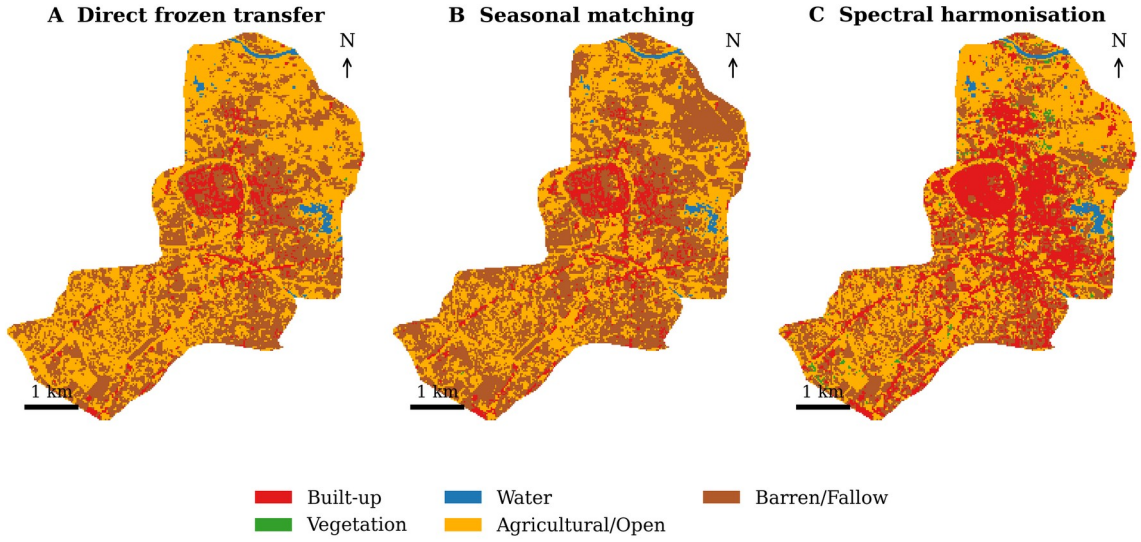


Fig. 10.2 Thanjavur 2020 classification maps under direct transfer (A), seasonal matching (B) and spectral harmonisation (C).

Source: supplied 30 m class-code GeoTIFFs, EPSG:32644. Identical extent, grid scale and class colours are used across A/B/C; white denotes no data. The provisional pilot is not an official municipal boundary. Maps show classifier assignments, not reference land cover.

The new export archive supplies all six class-code rasters, matching RGB quicklooks, ROI geometries, the final manual-reference points and an area manifest. Figure 10.2 renders the actual categorical rasters without smoothing. These exports supersede the earlier screenshot-only evidence and permit a direct spatial comparison of the three input protocols.

> **Interpretation.** The baseline assigns 91.69% of valid area to Agricultural/Open or Barren/Fallow. This is a warning requiring reference interpretation, not evidence that those are the true proportions of the city.

The manually interpreted labels were external to source-model fitting. Independence from fitting does not, however, make them an untouched confirmatory test set for B and C: their errors informed the later hypotheses. The final exported 112-point GeoJSON is authoritative for the completed reference set; the earlier progress-55 file and 252-point assisted queue document intermediate stages and must not be substituted for it.

10.4 MANUAL REFERENCES AND THE AREA ADJUSTED ESTIMATE

The initial reference plan selected about 252 points by the baseline map's predicted classes. After 55 had been interpreted, a deterministic reduced sample was adopted: 25 points in each of the Built-up, Water, Agricultural/Open and Barren/Fallow map strata, plus all 12 selected Vegetation points (seed 20261026). Twenty-one previously interpreted points were retained and the remaining 91 were reviewed against 2020 Sentinel-2 seasonal imagery. The final 112 point IDs were unique, all had valid labels and none had a missing

baseline prediction. Earlier WorldCover and Dynamic World suggestions were screening aids and were not accepted as the final manual truth labels.

The reference labels contained 55 Built-up, 8 Vegetation, 20 Water, 25 Agricultural/Open and only 4 Barren/Fallow points. Those totals describe interpreted land cover. They are different from the 25/12/25/25/25 sampling strata, which describe what map A predicted before interpretation. Keeping those two quantities separate is essential to the accuracy calculation.

Table 10.1 Original map strata and the correct predictions retained for every comparison.

Baseline A stratum	n	Area weight (%)	A correct	B correct	C correct
Built-up	25	6.9400	24	23	24
Vegetation	12	0.0391	2	4	6
Water	25	1.3285	13	16	13
Agricultural/Open	25	50.6582	7	5	6
Barren/Fallow	25	41.0341	1	2	10

The estimate follows the baseline map sampling design [15]: $O\hat{A} = \Sigma_h W_h(c_h/n_h)$, where W_h is map A's area share, n_h its sampled stratum size and c_h the correct predictions within that original stratum. B and C use those same weights and point groups, even after their maps change. Replacing them with the candidate map's new class areas would change the reference design after selection.

For A, the unweighted sample agreement is 47/112 = 41.96%, but the area-adjusted estimate is 23.19%. The difference arises because the two largest mapped strata, Agricultural/Open and Barren/Fallow, have only 7/25 and 1/25 correct predictions. Their combined 91.69% area weight dominates the estimate. A rough baseline standard error is 4.93 percentage points, with an approximate 95% interval of 13.51–32.86%; reference ambiguity and spatial dependence are not included.

> **Interpretation.** The area-adjusted result is 23.19%, substantially below the raw 41.96% sample match. Fixed baseline strata must remain the basis of every paired comparison.

10.5 DIAGNOSING THE FAILED TRANSFER BEFORE CHANGING THE INPUTS

The first diagnostic compared the 600 source training points with the 112 manually labelled target points, alongside a separate pool of 2,561 unlabeled target pixels. A pooled mean alone could conceal differences in land-cover composition, so the source–target comparisons also considered each reference class separately. Target Built-up pixels were greener and less strongly built-up in index space than the Chennai training examples (Table

10.2). A plausible explanation is more vegetation within 30 m urban pixels, but the analysis did not isolate a single cause.

Table 10.2 Mean predictor differences at Built-up reference locations.

Predictor	Chennai training	Thanjavur reference
NDVI	0.189623	0.373359
NDBI	+0.070199	−0.017213
NDWI	−0.261485	−0.430359
NIR reflectance	0.204583	0.264178

Monthly extraction then tested whether annual medians mixed different observation seasons. The original CSVs contain 600 Chennai and 112 Thanjavur point records, each with monthly NDVI, NDWI, NDBI and clear-observation counts. A zero count means no usable observation and −9999 is a missing-value sentinel. Chennai had 4,513 clear point-months out of 7,200 (62.7%), against 1,177/1,344 (87.6%) at Thanjavur. Source coverage was zero in June and September, although target coverage was 100% and 92%.

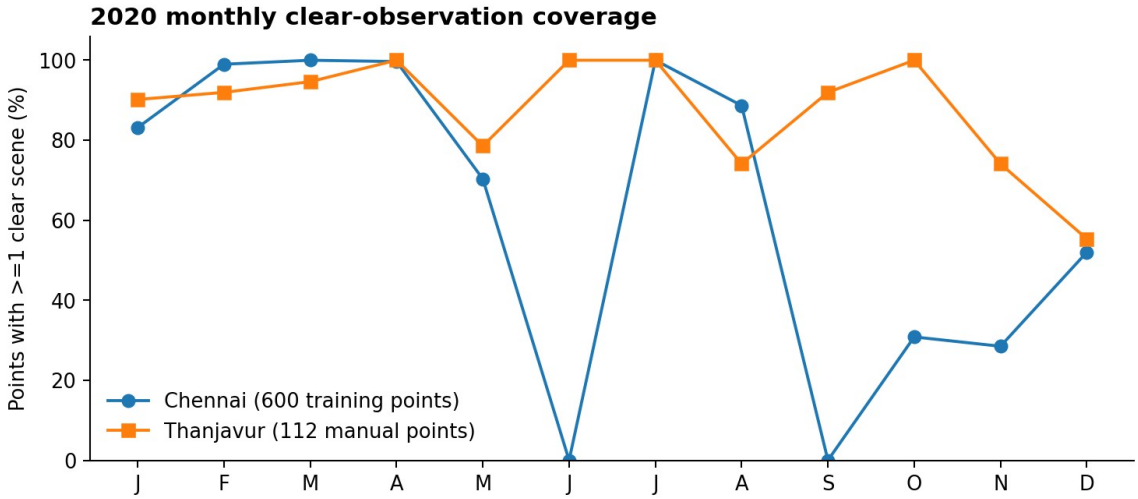


Fig. 10.3 Monthly availability of clear observations at the sampled locations.

Availability means at least one clear observation at a sampled point in that month. The samples differ in design and class composition, so the graph is a diagnostic of these locations, not a census of ROI-wide clear pixels.

Seasonal availability was only part of the explanation. Built-up NDVI remained higher in Thanjavur during well-observed shared months, including February (0.336 versus 0.188) and July (0.390 versus 0.198). Agricultural/Open target means varied from 0.672 in January to 0.437 in May and 0.701 in November, describing sampled-site means rather than one field's crop cycle. Eighteen out-of-range Chennai Water NDVI values from low-denominator cases were excluded from temporal interpretation.

Interpretation. Unequal seasonal coverage is real, but the Built-up spectral shift persists in shared months. Observation timing alone is insufficient to explain the transfer failure.

The diagnostics separate availability from spectral response. Clear-count summaries ask whether a monthly value can be estimated; class-conditional index summaries ask whether comparable land-cover labels occupy similar predictor distributions. Neither comparison alone demonstrates a causal mechanism. Source training points and target reference points also have different sampling designs, so pooled averages may change because of class composition rather than a uniform radiometric shift.

10.6 EXPERIMENT B SOURCE MONTH RESTRICTION

The observation diagnostic suggested a simple test: restrict the target composite to months adequately represented at the source. Months with at least 50% source-point coverage were fixed as January–May, July–August and December. This retained 46 of 64 target scenes. It is a coarse source-month restriction, not exact matching of every pixel's seasonal observations. The nine predictors and frozen class parameters otherwise stayed unchanged.

B classified all 112 reference locations and preserved the 26.9913 km² valid area, but its area-adjusted accuracy fell to 20.66%, a change of −2.52 percentage points from A. Its unweighted count rose from 47 to 50 correct points, which would have suggested improvement if the sampling weights had been ignored. In the original Agricultural/Open stratum, however, correct predictions fell from 7/25 to 5/25. That stratum carries about half of the target area weight, outweighing improvements in smaller strata (Table 10.1).

The result rejected this particular month restriction as an improvement. It did not rule out more complex seasonal effects, and the months were not subsequently adjusted to chase a better score on the same reference points. Implementation required filtering the Landsat collection before preprocessing and retaining system:time_start; a previous editor version lost the timestamp and could not apply calendarRange. The recorded successful method is B V3.

Interpretation. The specified seasonal restriction fails: weighted accuracy falls by 2.52 percentage points even though three more sampled points become correct. The design weights explain the apparent contradiction.

10.7 EXPERIMENT C BOUNDED UNLABELED QUANTILE ALIGNMENT

Because the urban spectral shift remained within shared months, the next test changed reflectance preprocessing without changing the model. For each of the six bands, full-year source and target ROI quartiles were estimated from unlabeled imagery at 90 m. A bounded affine transform aligned the target interquartile range and median towards the source.

Classification still ran at 30 m, with NDVI, NDWI and NDBI recalculated after the six reflectance corrections.

$$IQR = P75 - P25$$

$$gain = clip(IQR_{source} / max(IQR_{target}, 0.0001), 0.85, 1.15)$$

$$offset = clip(P50_{source} - gain \times P50_{target}, -0.03, +0.03)$$

$$aligned\ target\ reflectance = gain \times target\ reflectance + offset$$

Table 10.3 Applied band corrections from the original exported parameter CSV.

Band	Gain	Offset
Blue	1.150000	+0.010061
Green	1.150000	−0.002426
Red	1.150000	−0.001617
NIR	1.047033	−0.030000
SWIR1	1.037080	−0.019383
SWIR2	1.034344	+0.004531

The Blue, Green and Red gains reached the 1.15 upper bound and the NIR offset reached −0.03. The correction used no local true labels for quantile estimation or model fitting. Nevertheless, the same 112 interpreted references had already informed the diagnostic reasoning, so the resulting comparison is exploratory. Whole-ROI quantile matching can also confound sensor/scene differences with different class proportions; it is not verified physical radiometric calibration.

> **Interpretation.** The transformation is bounded and label-free in its parameter estimation, but it is not physical radiometric calibration. Its exploratory selection followed examination of the manual reference results.

Recomputing the indices after adjustment is essential: applying a correction to reflectance while retaining the old indices would produce an internally inconsistent nine-dimensional vector. The gain limits constrain amplification of target variation and the offset limits constrain brightness shifts. Because these bounds are reached in four band parameters, the applied correction is materially different from unconstrained quantile matching.

10.8 WHAT THE A B C COMPARISON SHOWED

Table 10.4 Thanjavur results on the same 112 points and original A area weights.

Experiment	Unweighted correct	Sample (%)	Area adjusted (%)	Change vs A (pp)
A direct frozen	47/112	41.96	23.19	—
B month restriction	50/112	44.64	20.66	−2.52
C quantile alignment	59/112	52.68	35.94	+12.76

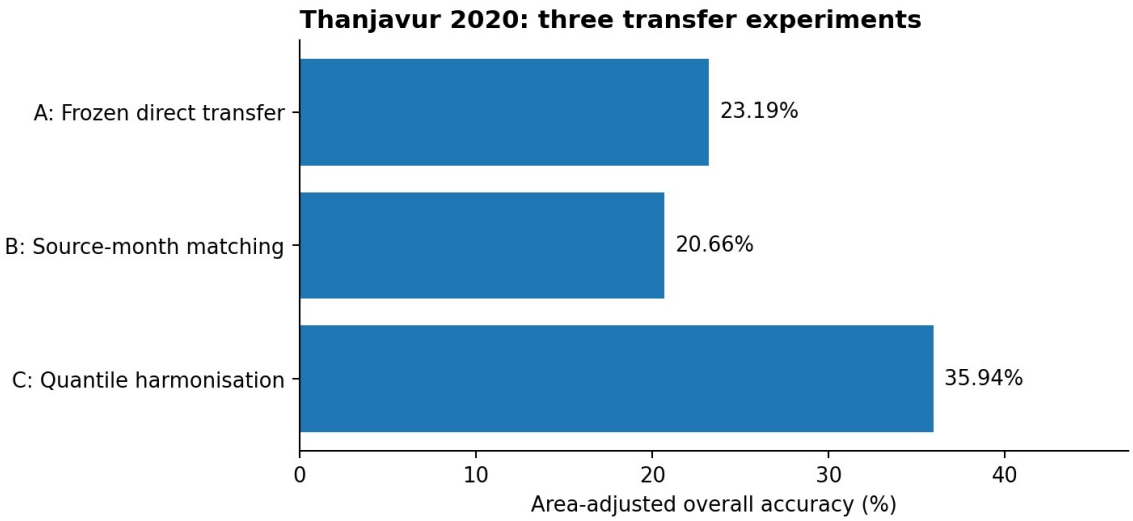


Fig. 10.4 Area-adjusted Thanjavur accuracy under three frozen-model input protocols.

A uses direct transfer. B and C change preprocessing only. All estimates retain the baseline A strata and area weights; B/C are exploratory after reference reuse.

C raised the area-adjusted estimate from 23.19% to 35.94%, an increase of 12.76 percentage points. The approximate paired standard error is 4.50 points and the normal-based 95% interval for C minus A is +3.93 to +21.58 points. This interval reflects the paired stratum calculation only. It does not account for spatial autocorrelation, interpretation errors, the sample-reduction decision or exploratory analysis, and cannot establish an unconditional improvement in a new population.

The weighted gain is concentrated in the original map's Barren/Fallow sampling stratum, where correct predictions increased from 1/25 to 10/25, contributing +14.77 percentage points. The original Agricultural/Open stratum declined from 7/25 to 6/25, contributing −2.03 points. These are baseline prediction strata, not true Barren/Fallow and Agricultural/Open reference classes.

The defensible outcome is partial recovery under one bounded input transform, while overall target accuracy remains low. The independent Chennai source figure of 86.33% is retained as source context; it is not placed on the same plot as a directly comparable target-area estimate.

THANJAVUR CLASS ERRORS AND MAPPED AREAS

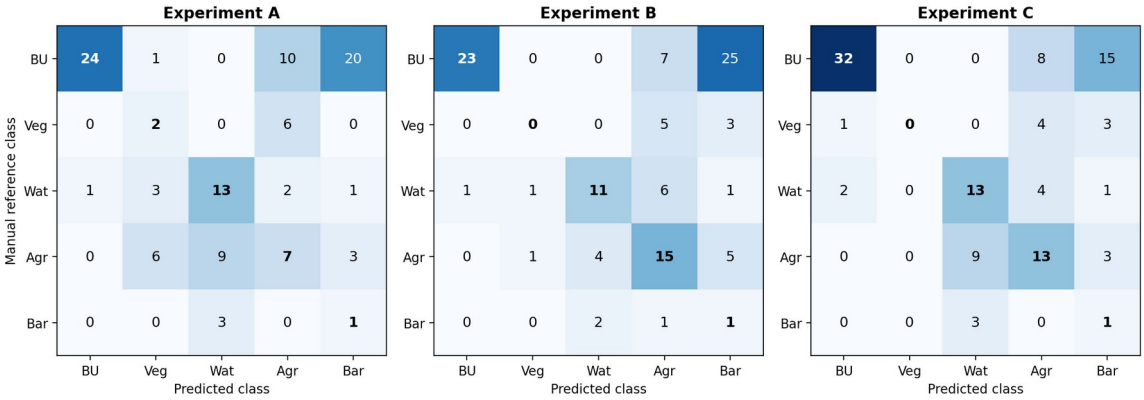


Fig. 10.5 Manual-reference confusion matrices for A, B and C.

Rows = interpreted class; columns = predicted class. BU = Built-up, Veg = Vegetation, Wat = Water, Agr = Agricultural/Open, Bar = Barren/Fallow. These are unweighted counts, not area-adjusted class error proportions.

Table 10.5 Correct predictions by actual manually interpreted reference class.

True reference class	Reference n	A correct	B correct	C correct
Built-up	55	24	23	32
Vegetation	8	2	0	0
Water	20	13	11	13
Agricultural/Open	25	7	15	13
Barren/Fallow	4	1	1	1

Reading by true reference class gives a different view from the sampling-stratum calculation. C recovers more Built-up points (24/55 to 32/55) and Agricultural/Open points (7/25 to 13/25), but classifies none of the eight Vegetation references as Vegetation. Water remains 13/20 correct and Barren/Fallow 1/4. These small, unequal true-class samples do not provide stable population producer accuracies.

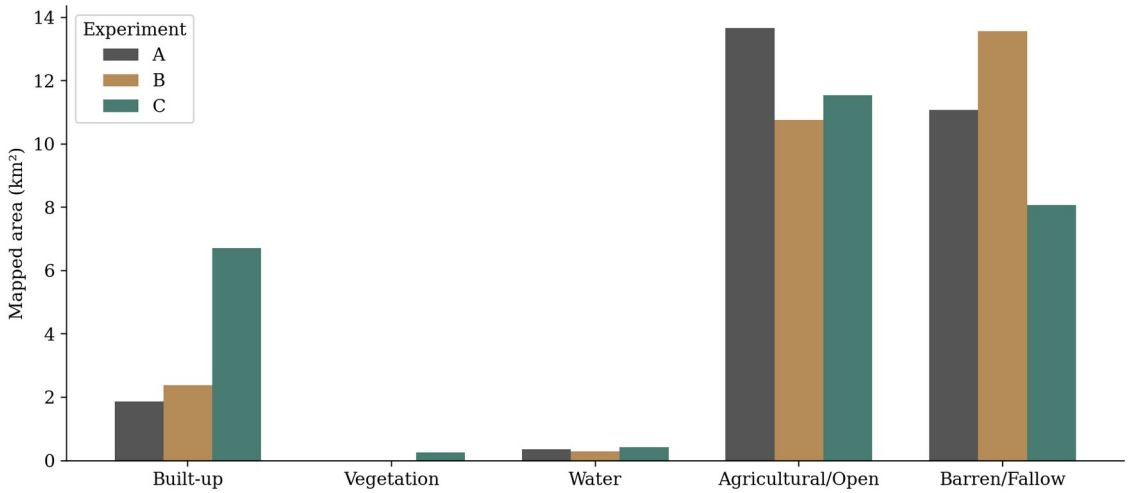


Fig. 10.6 Mapped class areas for all three Thanjavur experiments on the common valid footprint.

Model-mapped areas on the same 26.9913 km² valid footprint. Larger mapped Built-up area is not itself evidence of better spatial accuracy.

Mapped Built-up rises from 1.8732 to 6.7055 km², while Agricultural/Open falls from 13.6733 to 11.5379 km² and Barren/Fallow from 11.0757 to 8.0784 km². C still maps 0.2537 km² Vegetation and 0.4157 km² Water. Thus Vegetation does not vanish from the raster, although all eight manual Vegetation references are missed. Area totals and point-level class errors answer different questions.

> **Interpretation.** C provides partial recovery, not reliable generalisation: weighted accuracy reaches only 35.94%, and all eight true Vegetation references remain misclassified.

For a paired audit, define d_i as the C-correct indicator minus the A-correct indicator at the same reference point. Within each original map stratum, estimate the mean difference and sample variance of d. The weighted difference is $\Sigma W_h \bar{d}_h$, with approximate variance $\Sigma W_h^2 s^2(d_h)/n_h$ when finite-population corrections are omitted. Pairing retains the dependence between predictions on the same locations; treating the two accuracies as independent discards that information. The quoted normal interval addresses sampling variation under this approximation only.

10.9 EXTENDING SCREENING TO THREE COASTAL CITIES

The final extension asked whether a spectrally closer region would behave differently, while avoiding another manual interpretation exercise. Three city-centred 3 km radius circles were screened: Cuddalore (79.770000° E, 11.740000° N), Nagapattinam (79.843130° E, 10.763769° N) and Thoothukudi (78.134835° E, 8.764166° N). Each Earth Engine polygon had a vector area of about 27.9365 km². These circular pilots are not municipal boundaries, and coastal circles include water.

The screening score was the six-band mean of |target median − Chennai median| / max(|Chennai median|, 0.05), using unlabeled whole-ROI medians at 90 m. Lower values indicate a closer pooled spectral median among these candidates only. Cuddalore ranked first (0.076513; 22 Landsat scenes), Nagapattinam second (0.143042; 43 scenes) and Thoothukudi third (0.213905; 22 scenes). The score is a custom distance, not a probability of similarity or a prediction of classification accuracy. Every city then received the same direct frozen classifier, without C alignment or local training.

Because no additional human reference interpretation was undertaken, comparison used agreement between two existing 10 m prediction products: ESA WorldCover 2020 v100 [28] and the 2020 Dynamic World mode and mean class probabilities [29]. A pixel entered the proxy subset only when both products agreed under the class mapping in Table 10.6 and Dynamic World had at least three observations.

Table 10.6 Automated consensus definitions and minimum Dynamic World mean probabilities.

Project proxy class	WorldCover code	DW label	Probability
Built-up	50 built	6 built	≥0.30
Vegetation	10 tree	1 trees	≥0.40
Water	80 water	0 water	≥0.50
Agricultural/Open	40 cropland	4 crops	≥0.25
Barren/Fallow	60 bare	7 bare	≥0.25

For each city, 40 proxy points per class were selected (200 total, seed 20261027) and the MLC predictions sampled at 30 m. Overall agreement weights each class match rate by its share of the proxy-supported area. Non-water agreement excludes Water and renormalises the remaining supported-area weights. Coverage is the proxy-supported area divided by all valid classified area; outside it, performance is unknown.

The proxy classes are narrower than the project's full semantic definitions: trees do not represent every form of non-crop Vegetation, cropland does not cover all Agricultural/Open land, and bare surfaces do not identify all Barren/Fallow conditions. Agreement between two modelled maps can preserve shared errors and cannot turn these locations into independent ground truth.

Interpretation. The coastal experiment evaluates direct transfer against a restricted automated consensus. It does not add independently interpreted validation or apply the Thanjavur C correction.

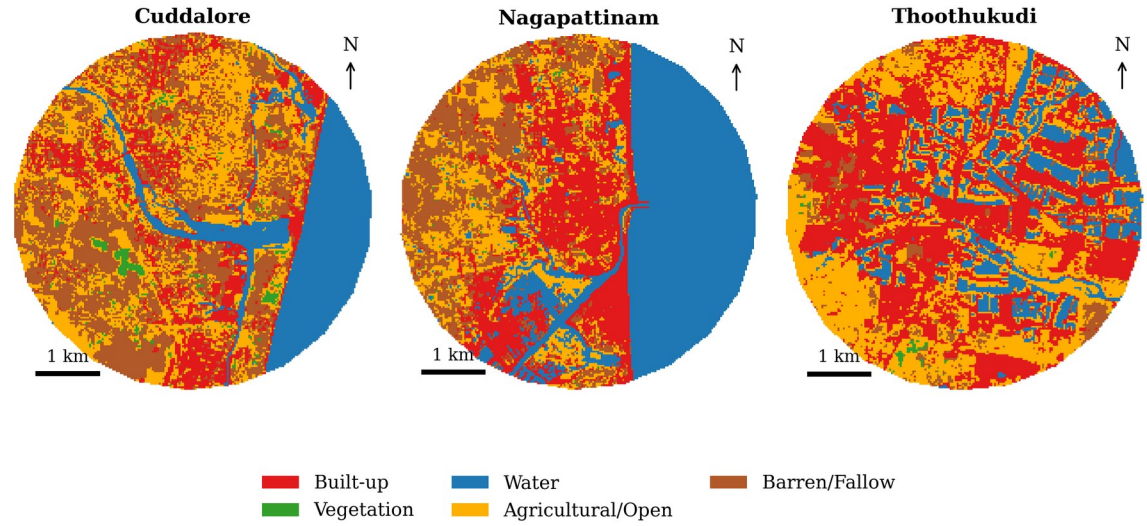


Fig. 10.7 Direct frozen-model classifications for the three coastal pilots, 2020.

Source: actual 30 m exported class-code rasters, EPSG:32644, displayed with the same five-class legend and scale. Each circular pilot uses a 3 km radius and is not an

administrative boundary. Offshore water is retained because it belongs to the defined pilot; it must therefore be considered when interpreting aggregate proxy agreement.

10.10 THREE CITY RESULTS AND THE WATER EFFECT

Table 10.7 Agreement on the automated consensus subset and its spatial coverage.

City	Supported agreement (%)	Non-water (%)	Coverage (%)	Water share of support (%)
Cuddalore	57.93	40.01	56.83	32.57
Nagapattinam	82.91	57.70	56.14	59.59
Thoothukudi	46.09	43.61	28.91	6.80

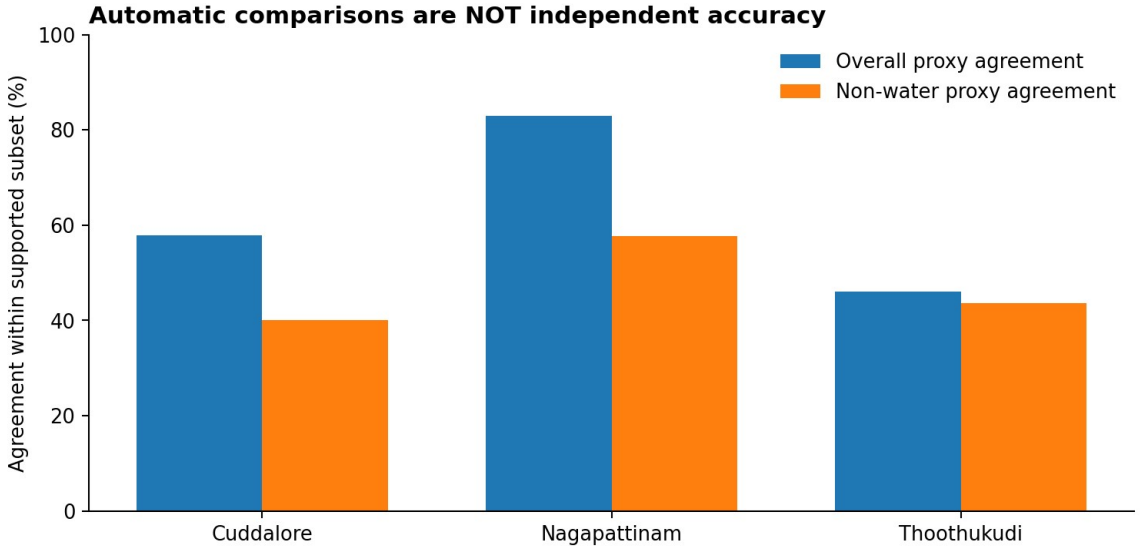


Fig. 10.8 Three-city cross-model agreement with and without Water.

Area-weighted agreement over the supported proxy subset only. These percentages are not independent whole-area accuracy estimates and are not equivalent to Thanjavur's manual-reference OA.

Nagapattinam's 82.91% headline is strongly influenced by Water, which occupies 59.59% of its supported proxy area and has 40/40 matching points. Excluding Water lowers agreement to 57.70%. Its circle maps 10.5754 km² as Water, illustrating the effect of coast-facing geometry on the comparison. Cuddalore, the closest pooled spectral match, does not have the highest aggregate proxy agreement. A whole-region median cannot guarantee within-class spectral transfer.

The valid classified areas are 27.8265 km² in Cuddalore, 27.8239 km² in Nagapattinam and 27.8197 km² in Thoothukudi. The proxy subset covers only 56.83%, 56.14% and 28.91%, respectively. Especially in Thoothukudi, the agreement result leaves most classified land outside its reference support.

WHAT THE CLASSWISE PROXY CHECK REVEALED

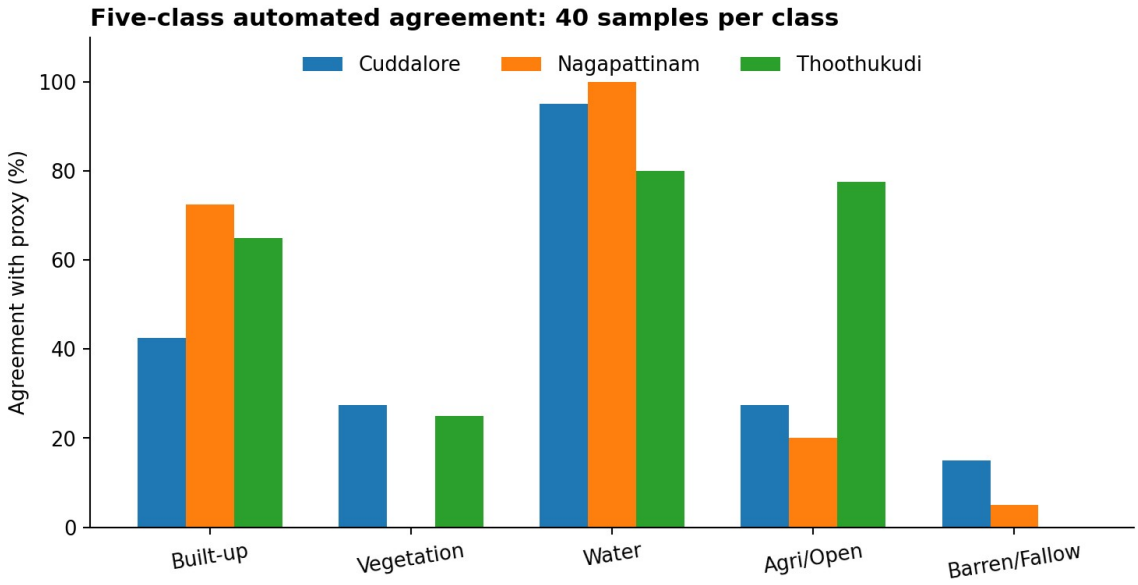


Fig. 10.9 Classwise matches against the automated proxy, 40 points per class and city.

Classwise percentages are counts divided by 40. They are diagnostic cross-model matches, not independently validated producer or user accuracies.

Water agreement is consistently strongest: 38/40 (95%) at Cuddalore, 40/40 (100%) at Nagapattinam and 32/40 (80%) at Thoothukudi. Other classes vary considerably. Built-up matches are 42.5%, 72.5% and 65%; Vegetation 27.5%, 0% and 25%; Agricultural/Open 27.5%, 20% and 77.5%; and Barren/Fallow 15%, 5% and 0%. These patterns support heterogeneity of cross-model agreement rather than a general claim of successful transfer across Tamil Nadu.

> **Interpretation.** Nagapattinam's high aggregate agreement is water-dominated. Non-water agreement and proxy coverage must accompany the headline percentage.

The supported-area weighting can be written as agreement = $\Sigma\, q_k m_k/n_k$, where q_k is the proportion of consensus-supported area assigned to proxy class k and m_k/n_k is its sampled match rate. Non-water agreement removes the Water term and divides by $1 - q\text{Water}$. Consequently, a city can score highly through abundant easy-to-match water while its land classes transfer poorly. Coverage is a separate denominator: an 82.91% supported-subset result cannot be extrapolated to the full pilot without evidence about excluded pixels.

Table 10.8 Exported mapped class areas (km²) across all six experiments.

Experiment	Built-up	Vegetation	Water	Agricultural/ Open	Barren/Fallow
Thanjavur A	1.8732	0.0106	0.3586	13.6733	11.0757
Thanjavur B	2.3790	0.0045	0.2888	10.7553	13.5638
Thanjavur C	6.7055	0.2537	0.4157	11.5379	8.0784
Cuddalore	4.2760	0.4668	5.4008	8.4648	9.2181
Nagapattinam	6.5902	0.0783	10.5754	5.1670	5.4131
Thoothukudi	11.8156	0.0946	4.2900	9.7244	1.8950

10.11 WHAT CAN BE CONCLUDED AND WHAT REMAINS UNCERTAIN

The strongest external finding is the poor manually checked direct transfer to the Thanjavur pilot. More intersecting scenes and greater clear point-month coverage did not prevent it. The monthly and classwise diagnostics indicate a shift in the predictor distributions, but seasonal mixing, mixed pixels, local class composition and reference ambiguity remain competing contributors. B did not improve this fixed model under the chosen restriction. C improved an exploratory area-adjusted estimate but sacrificed Vegetation matches and left accuracy low.

The evidence also sets clear limits. The pilot boundary is provisional; true Barren/Fallow has only four manual references; the same 112 labels informed diagnostic reasoning before B/C comparison; and the paired interval omits spatial and interpretation uncertainty. The three-city results add no independently interpreted external references and cover only the consensus subset of two products at a different native resolution. Neither 82.91% at Nagapattinam nor the +12.76 point Thanjavur gain demonstrates population-wide transferability.

> **Interpretation.** The strongest evidence remains poor direct transfer to manually interpreted Thanjavur. Neither proxy agreement nor a post-hoc preprocessing gain establishes statewide accuracy.

10.12 FINAL DECISION AND REPRODUCIBILITY

The experiments were stopped without further tuning to the 112 labels. A remains the direct-transfer benchmark, B the documented unsuccessful seasonal restriction and C a bounded preprocessing candidate for future testing. A defensible next study would retain these settings and evaluate them on fresh, spatially distributed external reference sites, with class definitions agreed in advance. The benchmark adds a qualification to the project's conclusion: a classifier can perform well within Chennai and still require new external validation before use elsewhere.

The archived handoff preserves original monthly CSVs, the six-band C parameter export, numerical result tables, the A/C audit and the three-city scripts. Some summary CSVs reconstruct values recorded from Earth Engine consoles rather than original export files. B V3 and C V2 reconstructions describe the successful method but are not verified byte-identical final editor snapshots. C was accessible during the final audit; the later supplied GeoTIFF archive now confirms the availability of A/B/C and all three coastal classification outputs. Appendix C records these provenance distinctions. The delivered map files establish export availability; they do not establish that reconstructed scripts are byte-identical to the final editor versions.

Conclusion of the benchmark. Retain A as the benchmark and B as a negative result; carry C forward only as a candidate for evaluation on fresh, spatially distributed references.

CHAPTER 11
DISCUSSION, LIMITATIONS AND CONCLUSION

11.1 FROM THE ORIGINAL PLAN TO THE FINAL OUTCOME

The project set out to produce five historical LULC maps of Chennai with one repeatable Gaussian MLC workflow. It delivered four, followed by a correction of their Built-up class. The three recent maps (2015, 2020, 2025) are comparable and independently validated at 83.7–86.3% five-class overall accuracy, with a persistent confusion between the two open-land classes. The route to 2010 was much harder than planned: automated transfer of 2015 training data produced too few and too spatially concentrated samples and was rejected; manual historical interpretation and a Landsat 5 + 7 stack gave a usable but provisional map (69.33%). That map then implied a 78.94 km² Built-up decline, which pixel-level transitions, GAIA and GHSL cross-checks, dated historical imagery, a reconstruction of both classifiers, likelihood diagnostics and the validation errors all indicated to be classification disagreement rather than verified urban loss. A Random Forest classifier, tested as an alternative, was 2.0–6.0 points less accurate in every recorded comparison. Two confused classes were merged for reporting, 2005 was dropped, and the results were made explorable in a local GIS application.

The final phase showed that the disagreement was largely an omission of Built-up land in the later maps. More training examples (V2A), neighbourhood and segment context (08F) and GAIA-assisted Random Forest (08G-3, 08G-4) each failed to recover it reliably — the last improved 2015 recall but added commission errors and failed badly in 2020. Post-classification fusion of the locked maps with year-specific evidence recovered 4–9 Built-up validation points per year with no new false positives, and gives a reported series of 279.82, 292.24, 296.74 and 342.60 km² mapped Built-up for 2010–2025. A 2010 two-way fusion (304.72 km²; 71.33% on the same points) is retained alongside the original 2010 map, and with it the 2010 → 2015 interval reverses.

The robust findings are therefore those that hold under every documented choice: mapped Built-up increased between 2015 and 2025 in both the original maps (+55.49 km²) and the fusion-adjusted series (+50.36 km²), and between 2010 and 2025 under both 2010 versions (+62.78 or +37.89 km²); Water decreased between 2015 and 2025; and Built-up, Water and Vegetation are mapped with producer's and user's accuracies mostly above 80–90%. The project's contribution is not a perfect historical land-cover series but a documented and critically evaluated workflow whose limitations were investigated, and partly corrected, rather than concealed.

Several lessons follow. Imagery availability and reference-data quality were major limits for historical years. Automated sample selection did not guarantee representative training data; high cross-validation did not guarantee independent accuracy; good overall accuracy did not guarantee a complete Built-up class; changing or retraining the classifier moved errors between classes rather than removing them; merging confused classes raised reported accuracy without improving the model; and both a large mapped decline and a pleasing rising curve had to be investigated before being treated as real.

11.2 LIMITATIONS

1. **Provisional 2010 map** (Section 5.5): lower accuracy, few training sites, and a validation set that was not entirely unseen when the final stack was chosen; the causes of the 2010–2015 anomaly were not individually isolated (Chapter 7).
2. **Sensors and seasons.** 2010 combines Landsat 5 TM and SLC-off Landsat 7 ETM+, later years use OLI, and no cross-sensor harmonisation was applied [8]; full-year medians mix wet and dry seasons, which affects Water, Vegetation and open land.
3. **Reference data.** Labels come from visual interpretation; 2010 sites are not tied to dated imagery; training and validation points are clustered in polygons, so spatial autocorrelation can make accuracy optimistic.
4. **Areas and change.** Validation is balanced rather than probability-based, so areas are mapped areas without error adjustment or confidence intervals [15]; transitions include the errors of both maps [17]; local whole-pixel and Earth Engine areas are not formally reconciled.
5. **Merged class.** Agricultural and vacant/unused land differ in use and meaning, and the merge raises overall accuracy mechanically.
6. **Evidence gaps.** Some records survive only in recovered or secondary form: the explanation of the 528.79 km² diagnostic, the Landsat 7 scene count of the 2010 stack, the intermediate automated 2010 attempts, raw Random Forest console logs, the stage 07W GAIA/GHSL script and output (product versions unconfirmed), the V2A matrix and the 08A–08E site margins; several 08G scripts survive only as later copies. None of the official MLC results depends on them (Appendix A).
7. **Heterogeneous fusion evidence.** Each fusion candidate uses different external products, resolutions, definitions and thresholds (WSF 2015, WorldCover 2020, Dynamic World 2025, GHSL 2010); GAIA ends in 2018 and cannot record demolition. The reported series is therefore not produced by one identical method.

8. **Temporal dependence.** The 2015 candidate requires Built-up in the 2020 candidate, and the 2010 alternative uses the 2015 candidate, so agreement between adjacent years is partly built in.
9. **Reused validation.** The same validation points were used to compare many training, context, RF and fusion variants; all fusion accuracies are exploratory. The 2025 fusion validation uses a new sample drawn from the existing 33 master polygons, not the archived sample or new independent sites.
10. **Two 2010 outcomes. The direction of 2010→2015 change depends on which 2010 version is used; the two-way version was subsequently reconstructed and exported for documentation, but is not substituted into the reported area and transition analysis.**
11. **Application.** It is a local research tool that serves the original maps; fusion candidates and change-analysis features are not yet connected.

Geographical transfer. The Thanjavur boundary is a provisional trace, B/C reuse interpreted references, and only four true Barren/Fallow references are available. Coastal-city proxy agreement covers a selected subset and cannot establish independent regional accuracy.

11.3 CONCLUSION

This project produced LULC maps of a fixed 529.45 km² Chennai Urban Study Area for 2010, 2015, 2020 and 2025 from Landsat surface reflectance, using an explicitly implemented nine-band Gaussian maximum likelihood classifier and independent 300-point validation for every year. The original 2015, 2020 and 2025 maps have five-class overall accuracies of 85.67%, 86.33% and 83.67%, and four-class accuracies of 88.67–96.67% after the two persistently confused open-land classes were merged by post-classification aggregation; the 2010 map is provisional (69.33%; 81.00% four-class). The 78.94 km² mapped Built-up decline between 2010 and 2015 in the original maps is not verified physical urban loss. Investigating it revealed systematic Built-up omission in the later maps, which retraining, context features, segmentation and GAIA-assisted Random Forest did not correct reliably; rule-based post-classification fusion with year-specific external evidence did, at the validation points, raising 2015 and 2020 accuracy to 88.67% and recovering 4–9 Built-up points per year without new false positives. The reported fusion-adjusted series shows mapped Built-up of 279.82, 292.24, 296.74 and 342.60 km²; an equally documented 2010 alternative (304.72 km²) reverses the first interval, so only the overall increase is robust. The final rasters were verified after export, and a local application makes the original series explorable. The project began as a straightforward five-year mapping exercise and became a study of how far a historical map series can be trusted, and

of how it can be corrected without being tuned to an expected answer. The external benchmark extends that lesson beyond time: the frozen Chennai model achieved only 23.19% area-adjusted accuracy in the Thanjavur pilot. Bounded preprocessing raised the exploratory estimate to 35.94%, while a source-month restriction did not help. The coastal-city proxy checks describe cross-model agreement only. Good within-Chennai validation therefore did not establish geographical generalisability.

11.4 FUTURE WORK

- Assess the fusion candidates and both 2010 versions on a new, spatially blocked validation sample that was not used in development, and derive error-adjusted Built-up areas with confidence intervals [15].
- Rebuild 2010 reference data from dated historical imagery at more, better-distributed sites; if the subsequently exported two-way 2010 result is used further, retain its separate sensitivity designation.
- Test seasonal composites and cross-sensor harmonisation for 2010 [8]; extend fusion only where year-matched evidence exists; connect the fusion candidates to the application as separately labelled layers; add change overlays and swipe comparison.

Validate the frozen A/B/C protocols on fresh external reference sites, retain original map-stratum weights for the existing sample, and export authentic spatial comparisons before drawing geographical generalisation claims.

REFERENCES

[1] D. Nandi, R. R. Thakur, D. K. Bera, A. K. Pati, P. Mishra, F. Alshehri and M. Zhran, "A deep learning-geospatial analytics fusion framework for predicting land use land cover dynamics in eastern region of India," *Results in Engineering*, vol. 29, art. 109201, 2026. doi:10.1016/j.rineng.2026.109201.

[2] N. Gorelick, M. Hancher, M. Dixon, S. Ilyushchenko, D. Thau and R. Moore, "Google Earth Engine: Planetary-scale geospatial analysis for everyone," *Remote Sensing of Environment*, vol. 202, pp. 18–27, 2017. doi:10.1016/j.rse.2017.06.031.

[3] C. J. Crawford, D. P. Roy, S. Arab et al., "The 50-year Landsat collection 2 archive," *Science of Remote Sensing*, vol. 8, art. 100103, 2023. doi:10.1016/j.srs.2023.100103.

[4] U.S. Geological Survey, "Landsat Collection 2 Surface Reflectance." https://www.usgs.gov/landsat-missions/landsat-collection-2-surface-reflectance (accessed 20 Oct. 2026).

[5] S. Foga, P. L. Scaramuzza, S. Guo et al., "Cloud detection algorithm comparison and validation for operational Landsat data products," *Remote Sensing of Environment*, vol. 194, pp. 379–390, 2017. doi:10.1016/j.rse.2017.03.026.

[6] B. L. Markham, J. C. Storey, D. L. Williams and J. R. Irons, "Landsat sensor performance: history and current status," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 42, no. 12, pp. 2691–2694, 2004. doi:10.1109/TGRS.2004.840720.

[7] U.S. Geological Survey, "Landsat 7 scan line corrector-off gap-filled product development," in *Proc. 16th William T. Pecora Memorial Remote Sensing Symposium*, Sioux Falls, SD, 2005. https://pubs.usgs.gov/publication/70259127.

[8] D. P. Roy, V. Kovalskyy, H. K. Zhang, E. F. Vermote, L. Yan, S. S. Kumar and A. Egorov, "Characterization of Landsat-7 to Landsat-8 reflective wavelength and normalized difference vegetation index continuity," *Remote Sensing of Environment*, vol. 185, pp. 57–70, 2016. doi:10.1016/j.rse.2015.12.024.

[9] C. J. Tucker, "Red and photographic infrared linear combinations for monitoring vegetation," *Remote Sensing of Environment*, vol. 8, no. 2, pp. 127–150, 1979. doi:10.1016/0034-4257(79)90013-0.

[10] S. K. McFeeters, "The use of the Normalized Difference Water Index (NDWI) in the delineation of open water features," *International Journal of Remote Sensing*, vol. 17, no. 7, pp. 1425–1432, 1996. doi:10.1080/01431169608948714.

[11] Y. Zha, J. Gao and S. Ni, "Use of normalized difference built-up index in automatically mapping urban areas from TM imagery," *International Journal of Remote Sensing*, vol. 24, no. 3, pp. 583–594, 2003. doi:10.1080/01431160304987.

[12] **J. A. Richards**, *Remote Sensing Digital Image Analysis*. Cham, Switzerland: Springer, 2022. doi:10.1007/978-3-030-82327-6.

[13] **R. G. Congalton**, "A review of assessing the accuracy of classifications of remotely sensed data," *Remote Sensing of Environment*, vol. 37, no. 1, pp. 35–46, 1991. doi:10.1016/0034-4257(91)90048-B.

[14] **G. M. Foody**, "Status of land cover classification accuracy assessment," *Remote Sensing of Environment*, vol. 80, no. 1, pp. 185–201, 2002. doi:10.1016/S0034-4257(01)00295-4.

[15] **P. Olofsson, G. M. Foody, M. Herold, S. V. Stehman, C. E. Woodcock and M. A. Wulder**, "Good practices for estimating area and assessing accuracy of land change," *Remote Sensing of Environment*, vol. 148, pp. 42–57, 2014. doi:10.1016/j.rse.2014.02.015.

[16] **R. G. Pontius Jr and M. Millones**, "Death to Kappa: birth of quantity disagreement and allocation disagreement for accuracy assessment," *International Journal of Remote Sensing*, vol. 32, no. 15, pp. 4407–4429, 2011. doi:10.1080/01431161.2011.552923.

[17] **D. Lu, P. Mausel, E. Brondízio and E. Moran**, "Change detection techniques," *International Journal of Remote Sensing*, vol. 25, no. 12, pp. 2365–2401, 2004. doi:10.1080/0143116031000139863.

[18] **Rasterio developers**, Rasterio: geospatial raster I/O for Python. https://rasterio.readthedocs.io.

[19] **FastAPI**, FastAPI web framework documentation. https://fastapi.tiangolo.com.

[20] **Leaflet**, Leaflet: an open-source JavaScript library for interactive maps. https://leafletjs.com.

[21] **React**, React: the library for web and native user interfaces. https://react.dev.

[22] **L. Breiman**, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001. doi:10.1023/A:1010933404324.

[23] **M. Belgiu and L. Drăguţ**, "Random forest in remote sensing: A review of applications and future directions," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 114, pp. 24–31, 2016. doi:10.1016/j.isprsjprs.2016.01.011.

[24] **G. M. Foody**, "Thematic map comparison: evaluating the statistical significance of differences in classification accuracy," *Photogrammetric Engineering & Remote Sensing*, vol. 70, no. 5, pp. 627–633, 2004. doi:10.14358/PERS.70.5.627.

[25] **P. Gong, X. Li, J. Wang, Y. Bai, B. Chen, T. Hu, X. Liu, B. Xu, J. Yang, W. Zhang and Y. Zhou**, "Annual maps of global artificial impervious area (GAIA) between 1985 and 2018," *Remote Sensing of Environment*, vol. 236, art. 111510, 2020. doi:10.1016/j.rse.2019.111510.

[26] **M. Pesaresi and P. Politis**, "GHS-BUILT-S R2023A — GHS built-up surface grid, derived from Sentinel-2 composite and Landsat, multitemporal (1975–2030)," European Commission, Joint Research Centre, 2023. doi:10.2905/9F06F36F-4B11-47EC-ABB0-4F8B7B1D72EA.
https://human-settlement.emergency.copernicus.eu/ghs_buS2023.php.

[27] **M. Marconcini, A. Metz-Marconcini, S. Üreyen, D. Palacios-Lopez, W. Hanke, F. Bachofer, J. Zeidler, T. Esch, N. Gorelick, A. Kakarla, M. Paganini and E. Strano**, "Outlining where humans live, the World Settlement Footprint 2015," *Scientific Data*, vol. 7, art. 242, 2020. doi:10.1038/s41597-020-00580-5.

[28] **D. Zanaga, R. Van De Kerchove, W. De Keersmaecker, N. Souverijns, C. Brockmann et al.**, "ESA WorldCover 10 m 2020 v100," Zenodo, 2021. doi:10.5281/zenodo.5571936.

[29] **C. F. Brown, S. P. Brumby, B. Guzder-Williams et al.**, "Dynamic World, Near real-time global 10 m land use land cover mapping," *Scientific Data*, vol. 9, art. 251, 2022. doi:10.1038/s41597-022-01307-4.

[30] **R. Achanta and S. Süsstrunk**, "Superpixels and polygons using simple non-iterative clustering," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017. doi:10.1109/CVPR.2017.520.

APPENDIX A — REPRODUCIBILITY AND EVIDENCE

Table A.1 Final Earth Engine assets and exported rasters.

Year	Five-class source asset	Final four-class asset	Status
2010	chennai_mlc_lulc_2010_L5L7_manual_exp_v1	chennai_lulc4_2010_L5L7_provisional_v1	Provisional
2015	chennai_mlc_lulc_2015	chennai_lulc4_2015_v1	Accepted
2020	chennai_mlc_lulc_2020	chennai_lulc4_2020_v1	Accepted
2025	chennai_mlc_lulc_2025_from_original_tif	chennai_lulc4_2025_v1	Accepted

Asset prefix `projects/assignment-482806/assets/`. Study area: `chennai_admin_study_area_clean`. Training/validation: `chennai_training_<year>_final` and `chennai_validation_<year>_final` (2015, 2020); 2025 master polygons; `chennai_training_2010_manual_final_v1`, `chennai_training_2010_L5L7_manual_exp_v1`, `chennai_validation_2010_final`; 2010 stack `chennai_predictor_stack_2010_L5L7_exp_v1`. Grid: EPSG:32644, transform [30, 0, 399060, 0, −30, 1463250], 951 × 1,412 cells. The 2025 source asset was uploaded from the original Drive export `Chennai_MLC_LULC_2025.tif` (value 0 = background).

Table A.2 Experimental assets of the Built-up correction (stages 07U2, 08F, 08G).

Asset	Content	Status
chennai_training_polygons_2015_v2_clean9	V2A training polygons (9)	Exists; V2A rejected
chennai_mlc_lulc_2015_V2A_exp_v1	V2A map (planned export)	Export not confirmed
chennai_08f4a_X3_point_snapshot_v1	SNIC predictors at 900 points	Exported (QA)
chennai_08f4c_X3_SNIC_frozen_11bands_segments_v1	12-band frozen SNIC raster	Exported (QA)
chennai_2015_08G3_gaia_assisted_RF_experimental_v1	2015 GAIA-assisted RF map	Exported; not selected
chennai_08g10_2015_temporal_fusion_candidate_v1	2015 fusion candidate	Exported; retained
chennai_08g10_2020_gaia_wc_fusion_candidate_v1	2020 fusion candidate	Exported; retained
chennai_08g10_2025_gaia_dw_fusion_candidate_v1	2025 fusion candidate	Exported; retained
(no asset)	2010 two-way fusion	Not exported; reproducible with 08G-15

Prefix `projects/assignment-482806/assets/`; all on EPSG:32644, transform [30, 0, 399060, 0, −30, 1463250]. Surviving scripts: 08F-2, 08F-4, 08F-4A/B/D; 08G-1 (with a corrected copy), 08G-3, 08G-4, 08G-4A, 08G-4B, 08G-5 (and fixed), 08G-6, 08G-6/7, 08G-8, 08G-9, 08G-10, 08G-11, 08G-12, 08G-13, 08G-15. Stages 08G-2, 08G-3A/B, 08G-4C and 08G-14 are known from the project record only; 08G-10 contains the selected 08G-4C rule. 08G-11 loads the wrong 2010 asset and is superseded by 08G-13; 08G-10 output names are already used and must not be re-exported.

Table A.3 Principal results and their evidence.

Result	Primary evidence	Status
Study area, grid, vector area	Study-area asset and GeoJSON export; raster transforms	Verified
2025 method and parameters	Recovered 2025 master script	Verified
Accuracies, all years and both schemes	Stored confusion matrices; Earth Engine outputs; recomputed	Verified
2010 development history	Transfer scripts and outputs; manual-training and diagnostic outputs; KML audit	Verified; intermediate logs missing
Class areas and transitions	Earth Engine area outputs; row/column sums checked	Verified
2025 raster identity	Area check against archived values; 0-cell comparison with application raster	Verified
Export verification	Post-export comparison of all four rasters	Verified (0 mismatches)
2010 stack accuracy and cross-validation	Original Earth Engine output of the 2010 stack evaluation	Primary

Result	Primary evidence	Status
528.79 km² diagnostic	Value: original export-check output; explanation: later project record	Value primary; explanation secondary
2010→2015 classifier audit (07AB)	Audit script and output; historical-imagery screenshots	Primary; sampled sites
GAIA/GHSL cross-check (07W)	Values recovered from project conversation, 13 Oct 2026; script and console not retained	Recovered; versions unconfirmed
Application features	Repository, integration tests, screenshots	As in Table 9.1
Landsat 7 scenes in 2010 stack	Stack creation/export script not retained	Count unknown
Random Forest comparison (2010–2025)	Recovered transcriptions of original outputs (2025 CV matrix not recovered); 2020/2025 scripts archived	Recovered; recomputed
V2A training experiment (07U2)	Project record and user note; full matrix not retained	Recovered
Context and SNIC experiments (08F)	Stage 08F record; console outputs; exported QA assets	Primary; partial console failures noted
GAIA-assisted RF (08G-3, 08G-4, 08G-4A)	Console outputs; scripts; 2015 RF asset	Primary
Fusion variants and candidates (08G-4B to 08G-10)	Console outputs; scripts; exported candidate assets	Primary; validation reused
Analysis A areas and transitions (08G-13)	Verbatim console paste; script	Primary
2010 lineage and sensitivity test (08G-12, 08G-15)	Console outputs; scripts; two-way map not exported	Primary; unexported
08A–08E site diagnostics and margins	Project record only	Recovered; not re-computed

Primary: original Earth Engine output or file. Recovered: transcribed from project records, with all derived metrics recomputed. Secondary: later explanation without the original output.

APPENDIX B — CONFUSION MATRICES, AREAS AND TRANSITIONS

Table B.1 Gaussian MLC five-class confusion matrices of the official maps (rows = reference, columns = map; bold = correct; *2010 provisional; four-class matrices follow by summing rows and columns 4 and 5).

Year	Reference	Built-up	Veg.	Water	Agri./Open	Barren	PA (%)
2010*	Built-up	**55**	1	0	4	0	91.67
	Vegetation	0	**36**	0	15	9	60.00
	Water	0	0	**59**	1	0	98.33
	Agricultural/Open	7	0	0	**38**	15	63.33
	Barren/Fallow	5	14	1	20	**20**	33.33
	UA (%)	82.09	70.59	98.33	48.72	45.45	OA 69.33
2015	Built-up	**49**	0	0	8	3	81.67
	Vegetation	0	**60**	0	0	0	100.00
	Water	0	0	**40**	9	11	66.67
	Agricultural/Open	0	1	0	**54**	5	90.00
	Barren/Fallow	1	0	1	4	**54**	90.00
	UA (%)	98.00	98.36	97.56	72.00	73.97	OA 85.67
2020	Built-up	**52**	0	0	5	3	86.67
	Vegetation	0	**59**	0	1	0	98.33
	Water	0	0	**60**	0	0	100.00
	Agricultural/Open	0	0	0	**59**	1	98.33
	Barren/Fallow	4	0	0	27	**29**	48.33
	UA (%)	92.86	100.00	100.00	64.13	87.88	OA 86.33
2025	Built-up	**55**	0	0	2	3	91.67
	Vegetation	0	**55**	0	5	0	91.67
	Water	0	0	**60**	0	0	100.00
	Agricultural/Open	0	0	0	**52**	8	86.67
	Barren/Fallow	0	0	0	31	**29**	48.33
	UA (%)	100.00	100.00	100.00	57.78	72.50	OA 83.67

Table B.2 Five-class mapped areas before aggregation (km²).

Class	2010	2015	2020	2025
Built-up	279.82	200.88	225.72	256.37
Vegetation	81.22	33.65	14.11	44.27
Water	21.78	39.73	22.04	20.39
Agricultural/Open	114.46	168.11	177.93	87.31
Barren/Fallow	32.49	87.40	89.98	121.43

Classes 4 and 5 sum to the four-class open-land category in every year.

Table B.3 Random Forest five-class confusion matrices from the exploratory comparison (same layout; †reconstructed paired 2025 samples).

Year	Reference	Built-up	Veg.	Water	Agri./Open	Barren	PA (%)
2010*	Built-up	**56**	1	0	3	0	93.33
	Vegetation	1	**36**	0	13	10	60.00
	Water	0	0	**59**	0	1	98.33
	Agricultural/Open	11	0	0	**37**	12	61.67
	Barren/Fallow	9	15	2	20	**14**	23.33
	UA (%)	72.73	69.23	96.72	50.68	37.84	OA 67.33
2015	Built-up	**43**	0	0	15	2	71.67
	Vegetation	0	**60**	0	0	0	100.00
	Water	0	2	**40**	12	6	66.67
	Agricultural/Open	3	5	0	**50**	2	83.33
	Barren/Fallow	0	0	0	12	**48**	80.00
	UA (%)	93.48	89.55	100.00	56.18	82.76	OA 80.33
2020	Built-up	**33**	0	0	7	20	55.00
	Vegetation	0	**60**	0	0	0	100.00
	Water	0	0	**58**	2	0	96.67
	Agricultural/Open	0	0	0	**59**	1	98.33
	Barren/Fallow	0	0	0	22	**38**	63.33
	UA (%)	100.00	100.00	100.00	65.56	64.41	OA 82.67
2025†	Built-up	**47**	0	0	5	8	78.33
	Vegetation	0	**58**	0	2	0	96.67
	Water	0	0	**60**	0	0	100.00
	Agricultural/Open	0	0	0	**52**	8	86.67
	Barren/Fallow	7	0	0	31	**22**	36.67
	UA (%)	87.04	100.00	100.00	57.78	57.89	OA 79.67

Table B.4 Context, segmentation and GAIA-assisted RF matrices on the 2015 validation set (rows = reference, columns = map; bold = correct).

Model	Reference	Built-up	Veg.	Water	Agri./Open	Barren	PA (%)
X1 3×3	Built-up	**49**	0	0	6	5	81.67
	Vegetation	0	**60**	0	0	0	100.00
	Water	0	0	**38**	12	10	63.33
	Agricultural/Open	0	1	0	**55**	4	91.67
	Barren/Fallow	1	0	0	6	**53**	88.33
	UA (%)	98.00	98.36	100.00	69.62	73.61	OA 85.00
X2 5×5	Built-up	**49**	0	0	6	5	81.67
	Vegetation	0	**60**	0	0	0	100.00
	Water	0	0	**40**	9	11	66.67
	Agricultural/Open	0	0	0	**56**	4	93.33
	Barren/Fallow	3	0	1	4	**52**	86.67
	UA (%)	94.23	100.00	97.56	74.67	72.22	OA 85.67
X3 dynamic	Built-up	**51**	0	0	8	1	85.00
	Vegetation	0	**60**	0	0	0	100.00
	Water	0	0	**41**	7	12	68.33
	Agricultural/Open	0	1	0	**54**	5	90.00
	Barren/Fallow	1	0	1	6	**52**	86.67
	UA (%)	98.08	98.36	97.62	72.00	74.29	OA 86.00
X3 frozen	Built-up	**51**	0	0	8	1	85.00
	Vegetation	0	**60**	0	0	0	100.00
	Water	0	0	**41**	7	12	68.33
	Agricultural/Open	0	1	0	**53**	6	88.33
	Barren/Fallow	1	0	1	5	**53**	88.33
	UA (%)	98.08	98.36	97.62	72.60	73.61	OA 86.00
RF 2015 (08G-3)	Built-up	**55**	0	0	5	0	91.67
	Vegetation	0	**58**	0	2	0	96.67
	Water	0	1	**39**	10	10	65.00
	Agricultural/Open	1	0	0	**57**	2	95.00
	Barren/Fallow	5	0	0	5	**50**	83.33
	UA (%)	90.16	98.31	100.00	72.15	80.65	OA 86.33
RF 2020 (08G-4)	Built-up	**43**	0	0	1	16	71.67
	Vegetation	0	**58**	0	2	0	96.67

Model	Reference	Built-up	Veg.	Water	Agri./Open	Barren	PA (%)
	Water	0	0	**57**	3	0	95.00
	Agricultural/Open	2	0	0	**58**	0	96.67
	Barren/Fallow	29	0	0	10	**21**	35.00
	UA (%)	58.11	100.00	100.00	78.38	56.76	OA 79.00

The RF 2020 matrix is scored on the 2020 validation set. The 08G-4A variant (GAIA by 2018) produced the identical matrix.

Table B.5 Fusion matrices (rows = reference, columns = map; bold = correct).

Map	Reference	Built-up	Veg.	Water	Agri./Open	Barren	PA (%)
2015 broad (rejected)	Built-up	**59**	0	0	1	0	98.33
	Vegetation	0	**60**	0	0	0	100.00
	Water	0	0	**40**	9	11	66.67
	Agricultural/Open	9	1	0	**45**	5	75.00
	Barren/Fallow	3	0	1	4	**52**	86.67
	UA (%)	83.10	98.36	97.56	76.27	76.47	OA 85.33
2015 temporal	Built-up	**58**	0	0	2	0	96.67
	Vegetation	0	**60**	0	0	0	100.00
	Water	0	0	**40**	9	11	66.67
	Agricultural/Open	0	1	0	**54**	5	90.00
	Barren/Fallow	1	0	1	4	**54**	90.00
	UA (%)	98.31	98.36	97.56	78.26	77.14	OA 88.67
2020 fusion	Built-up	**59**	0	0	1	0	98.33
	Vegetation	0	**59**	0	1	0	98.33
	Water	0	0	**60**	0	0	100.00
	Agricultural/Open	0	0	0	**59**	1	98.33
	Barren/Fallow	4	0	0	27	**29**	48.33
	UA (%)	93.65	100.00	100.00	67.05	96.67	OA 88.67
2025 original‡	Built-up	**54**	1	0	1	4	90.00
	Vegetation	0	**53**	0	7	0	88.33
	Water	0	1	**59**	0	0	98.33
	Agricultural/Open	2	0	0	**52**	6	86.67
	Barren/Fallow	1	0	0	27	**32**	53.33
	UA (%)	94.74	96.36	100.00	59.77	76.19	OA 83.33
2025 fusion‡	Built-up	**58**	1	0	0	1	96.67
	Vegetation	0	**53**	0	7	0	88.33

Map	Reference	Built-up	Veg.	Water	Agri./Open	Barren	PA (%)
	Water	0	1	**59**	0	0	98.33
	Agricultural/Open	2	0	0	**52**	6	86.67
	Barren/Fallow	1	0	0	27	**32**	53.33
	UA (%)	95.08	96.36	100.00	60.47	82.05	OA 84.67
2010 add-only	Built-up	**57**	1	0	2	0	95.00
	Vegetation	0	**36**	0	15	9	60.00
	Water	0	0	**59**	1	0	98.33
	Agricultural/Open	8	0	0	**37**	15	61.67
	Barren/Fallow	5	14	1	20	**20**	33.33
	UA (%)	81.43	70.59	98.33	49.33	45.45	OA 69.67
2010 two-way	Built-up	**57**	1	0	2	0	95.00
	Vegetation	0	**36**	0	15	9	60.00
	Water	0	0	**59**	1	0	98.33
	Agricultural/Open	3	0	0	**42**	15	70.00
	Barren/Fallow	4	14	1	21	**20**	33.33
	UA (%)	89.06	70.59	98.33	51.85	45.45	OA 71.33

‡New 300-pixel sample from the 33 validation master polygons. The original 2015 and 2020 matrices are in Table B.1.

Table B.6 Five-class transition matrices of the reported series (km²; rows = earlier year, columns = later year; stage 08G-13).

Period	From \ to	Built-up	Veg.	Water	Agri./Open	Barren	Row total
2010→2015	Built-up	**240.74**	0.74	1.21	27.63	9.50	279.82
	Vegetation	12.38	**20.27**	3.59	19.96	25.02	81.22
	Water	0.87	0.09	**19.18**	1.06	0.58	21.78
	Agricultural/Open	32.47	9.30	14.05	**43.79**	14.85	114.46
	Barren/Fallow	5.79	3.25	1.70	10.71	**11.03**	32.49
	Column total	292.24	33.65	39.73	103.15	60.99	529.77
2015→2020	Built-up	**278.74**	0.04	0.26	7.96	5.24	292.24
	Vegetation	1.45	**10.65**	0.18	18.14	3.24	33.65
	Water	3.07	0.65	**19.92**	13.84	2.25	39.73
	Agricultural/Open	6.95	2.25	1.09	**71.70**	21.16	103.15
	Barren/Fallow	6.53	0.51	0.60	20.79	**32.56**	60.99
	Column total	296.74	14.11	22.04	132.43	64.46	529.77
2020→2025	Built-up	**280.21**	1.05	0.66	5.85	8.96	296.74
	Vegetation	0.54	**10.93**	0.09	1.55	0.99	14.11
	Water	0.92	3.28	**16.60**	0.37	0.87	22.04

Period	From \ to	Built-up	Veg.	Water	Agri./Open	Barren	Row total
	Agricultural/Open	40.63	25.49	2.58	**36.34**	27.39	132.43
	Barren/Fallow	20.31	3.51	0.45	14.08	**26.11**	64.46
	Column total	342.60	44.27	20.39	58.20	64.31	529.77

APPENDIX C — GEOGRAPHICAL TRANSFER EVIDENCE

The frozen source, target labels and all A/B/C estimates refer to 2020. Asset paths below use projects/assignment-482806/assets/. The existing historical map and fusion registers in Appendices A and B are unchanged.

Source: chennai_training_2020_final; chennai_validation_2020_final. Target: tnj_city_trace_pilot_roi_2020; tnj_2020_reference_112_final_v1; thanjavur_mlc_transfer_2020_frozen_chennai. The C candidate was read successfully in 07_TNJ_AB_C_Final_Audit.js. Full asset names and task-status qualifications remain in the handoff GEE_ASSET_REGISTER.csv.

Primary local evidence: TNJ05A2_V3_CHENNAI_2020_MONTHLY_POINTS_original.csv (600 rows), TNJ05A2_V3_THANJAVUR_2020_MONTHLY_POINTS_original.csv (112 rows), TNJ2020_C_V2_ADJUSTMENT_PARAMS_original.csv (six bands), and the original Thanjavur overlay screenshot. Statistical figures reproduce the supplied numeric tables. No map raster is inferred from class areas.

Recorded console results and reconstructed summaries: canonical_results.json, thanjavur_accuracy_by_baseline_stratum.csv, the A/B/C confusion matrices, class-area summaries and three-city agreement/confusion tables. The result tables are not all original Earth Engine CSV exports. Reconstructed B V3 and C V2 source files are explicitly marked as reconstructions; historical B/C drafts are not the final successful editor copies.

Available implementation evidence includes the original frozen Thanjavur classifier, reference-review scripts, 07_TNJ_AB_C_Final_Audit.js, 08_TN_Closest_Spectral_City_Frozen_MLC_Auto_Proxy_2020.js and 09_TN_Frozen_MLC_Nagapattinam_Thoothukudi_Auto_Proxy_2020.js. Map-export scripts are instructions for obtaining authentic class images; task creation alone does not prove export completion.

The reference formula is $O\hat{A} = \Sigma_h W_h c_h / n_h$, with W = (0.0694000106, 0.0003914133, 0.0132850612, 0.5065820489, 0.4103414660) and n = (25, 12, 25, 25, 25). The paired C − A estimate is 12.7590117 percentage points, SE 4.5022855 points, approximate normal 95% CI [3.9345321, 21.5834913]. This interval excludes spatial dependence, reference uncertainty and exploratory model selection.